

# Lecture 1 Motivation for measure theory

1.1

## 1.i) Syllabus + Introduction

## 1.ii) Motivation for Real Analysis

Real analysis is the rigorous foundation of calculus, convergence, and functions.

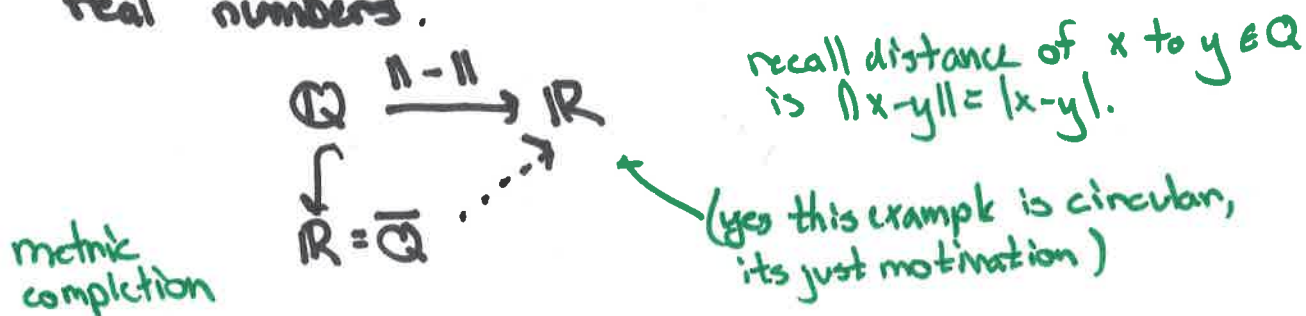
Applications 1.1 : Applications of real analysis include

- 1) Description of physical systems by ODEs and dynamical systems
- 2) Study of partial differential equations (PDEs) describing Electromagnetism, heat, waves, quantum particles, quantum fields, fluids, plasmas, ..
- 3) Analytic Number theory + Automorphic forms
- 4) Representation theory and harmonic analysis
- 5) Probability theory
- 6) Optimization problems in physics, math, economics, etc.
- 7) Study of fractals and chaotic phenomena
- 8) Dynamics and ergodic theory

(Informal) Def 1.2: A space  $X$  is said to be complete if limits of  $x_i \in X$  converge to points of  $X$ .

Ex 1.3: The rational numbers  $\mathbb{Q}$  are an incomplete (metric) space, i.e. not all Cauchy sequences converge.

To fix this is the main motivation for introducing real numbers.



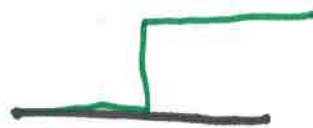
→ in  $\mathbb{R}$  every Cauchy sequence converges.

Motivation 1.4: We can define a metric on the space of continuous functions  $C^0[a,b]$  on an interval  $[a,b] \subseteq \mathbb{R}$  by

$$\|f-g\| = d(f,g) = \int_a^b |f-g| dx \quad (\text{Riemann integral})$$

But this space is certainly not complete!

Ex 1.5:



(not continuous, but is integrable)

Difference of integrals  $\leq \frac{1}{2}bh \rightarrow 0$ .

Ex 1.6: Consider 
$$F(x) = \begin{cases} 1 & x \in \mathbb{Q} \\ 0 & x \in \mathbb{R} \setminus \mathbb{Q} \end{cases}$$

Then  $F(x)$  is not Riemann integrable! But it is a limit of these (more next time).

Question 1.7: Can we find a space  $\bar{X}$  such that

$$C^0[a,b] \subseteq RI[a,b] \xrightarrow{\int_a^b f dx} \mathbb{R}$$



is the metric completion of Riemann integrable functions?

A: Yes! It is  $L^1[a,b]$  the Lebesgue integrable functions

Motivation 1.8: If  $f_n$  is a sequence of functions, when

$$\text{does } \lim_{n \rightarrow \infty} \int_a^b f_n(x) dx = \int_a^b \lim_{n \rightarrow \infty} f_n(x) dx ?$$

This is very annoying w/ Riemann (and RIG may not exist)

It becomes easy with Lebesgue Dominated Convergence Theorem (Lecture 9).

## Motivation 1.9 (Fourier Series)

(1.3)

Fourier Series gives a way of expressing a <sup>(periodic)</sup> function as a sum

$$f(x) = \sum_{n=0}^{\infty} a_n \cos(nx) + b_n \sin(nx).$$

What does it mean for this to converge?  $f_N = \sum_{n=0}^N$  then

$f_N(x) \rightarrow f(x)$  for every  $x$  (pointwise convergence)

$\sup |f_N(x) - f(x)| \rightarrow 0$  (uniform convergence)

$\int |f_N(x) - f(x)| \rightarrow 0$  (Integral convergence).

When are each true? For what sequences  $\{a_n, b_n\}$  is  $f(x)$  continuous? Differentiable.

Studying these questions in the context of PDEs was historically a main motivation for measure theory.

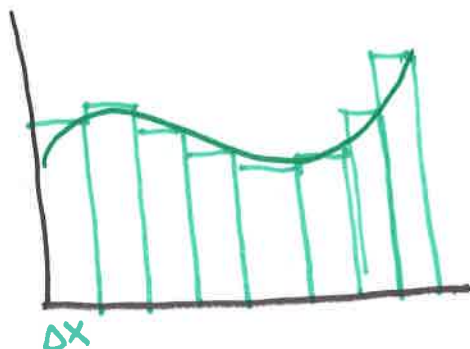


# Lecture 2 $\sigma$ -algebras and measures

2.1

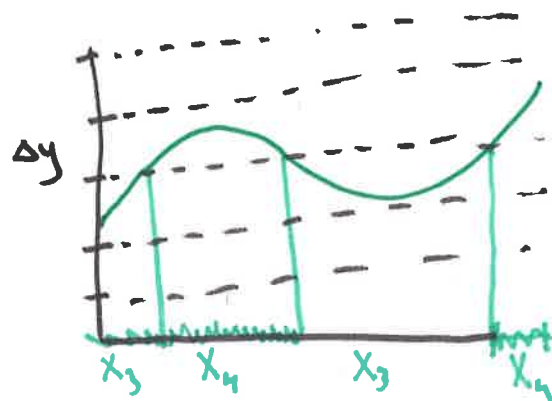
Ex 2.1 : Coin counting example.

## 2.1) Intuition for the Lebesgue Integral



Divide up x-axis, then

$$\int_0^1 f dx = \lim_{\Delta x \rightarrow 0} \sum_{i=1}^N \Delta x_i \cdot f(x_i)$$



$$\int_0^1 f dx = \lim_{\Delta y \rightarrow 0} \sum_{j=1}^N y_j \Delta y \cdot m(X_j)$$

"size or measure of  $X_j$ "

## Goal 2.2 : Define a measure

$$m : \{A \mid A \subseteq \mathbb{R}^n \text{ is a subset}\} \rightarrow \mathbb{R} \cup \{\infty\}$$

that satisfies "reasonable" properties.

- 1)  $m([a, b]) = b - a$  (normalization)
- 2)  $m(A \cup B) = m(A) + m(B) - m(A \cap B)$  (well-behaved under  $\cup, \cap$ )
- 3)  $m(A + x) = m\{a + x \mid a \in A\} = m(A)$  (translation invariance)

Rem 2.3 :  $\nexists m$  satisfying the above for all subsets

$$\{\text{finite unions of intervals}\} \subsetneq \{\text{Borel sets}\} \subsetneq \{\text{measurable subsets}\} \subsetneq \{\text{All subsets}\}$$

$$\downarrow m$$

$$\mathbb{R} \cup \{\infty\}$$

Ex 2.4 : Consider  $f: [0,1] \rightarrow \mathbb{R}$

[2.2]

$$f(x) = \begin{cases} 1 & x \in \mathbb{Q} \cap [0,1] \\ 0 & \text{else} \end{cases}$$



Claim :  $f(x)$  is not Riemann integrable

$$\left| \sum_{i=1}^N \inf_{x \in [x_{i-1}, x_i]} f(x) \cdot \Delta x - \sum_{i=1}^N \sup_{x \in [x_{i-1}, x_i]} f(x) \cdot \Delta x \right| = 1$$

For all  $\Delta x > 0$ .

But in a Lebesgue sense

$$\int_0^1 f(x) dx = 1 \cdot \underbrace{m(\mathbb{Q} \cap [0,1])}_{=0}$$

$\forall \varepsilon > 0$ ,  $m(\mathbb{Q} \cap [0,1]) \leq \varepsilon$ . Let  $\{q_n\}_{n \in \mathbb{N}}$  be an enumeration of  $\mathbb{Q}$ .

Let  $I_n = [q_n - \frac{\varepsilon}{2^{n+1}}, q_n + \frac{\varepsilon}{2^{n+1}}]$ . Then  $\mathbb{Q} \cap [0,1] \subseteq \bigcup I_n$

but  $m(\bigcup I_n) \leq \sum_{n=1}^{\infty} m(I_n) = m(\cap \dots) \leq \sum_{n=1}^{\infty} m(I_n) = \varepsilon \sum_{n=1}^{\infty} \frac{1}{2^n} = \varepsilon \cdot 1 = \varepsilon$ .

2.ii) Open and Borel sets,  $\sigma$ -algebras.

Def 2.5 : An open subset  $U \subseteq \mathbb{R}$  is one such that  $\forall x \in U, \exists r_x$  st  $B_{r_x}(x) \subseteq U$ .

A closed subset  $C$  is one so that  $C^c = \{x \in \mathbb{R} \mid x \notin C\}$  is open.

Facts 2.6 : 1) Arbitrary unions of open sets are open (inf of closed are closed)  
2) finite intersection of open sets is open ( $\bigcup$  of closed sets is closed).

Lemma 2.7 : Every open  $U \subseteq \mathbb{R}$  may be written

$$U = \bigcup_{n=1}^{\infty} (a_n, b_n)$$

where  $a_n, b_n = \pm\infty$  is allowed.

Proof :  $\forall x \in U$ , let  $I_x = \left( \inf_{a \in U} a, \sup_{b \in U} b \right)$   
st  $(a, x) \subseteq U$  st  $(x, b) \subseteq U$

$I_x \subseteq U$   
 $I_x$  is open, and  $x \in I_x$ , so  $U = \bigcup I_x$ .

Note  $I_x, I_y$  distinct or equal. If  $\text{not } x \neq y$ , then  $I_x \cup I_y$  ~~open~~ is an interval contradicting maximality unless  $I_x = I_y$ . And each  $I_x$  is open so contains some  $q_i \in \mathbb{Q}$ . Since the collection of distinct intervals contains distinct rationals, ~~it~~ it is countable  $\square$

Lemma 2.8: Every open  $U \subseteq \mathbb{R}^d$  is a union

$$U = \bigcup_{n=1}^{\infty} [a_n, b_n] \times \dots \times [a_n, b_n]$$

that have disjoint interiors.

Proof: Exercise (see pages 7-8).

Let  $P(X)$  denote the power set  $P(X) = \{A \mid A \subseteq X\}$ .

Definition 2.9: A  $\sigma$ -algebra on  $X$  is a subset  $\Sigma \subseteq P(X)$

such that

i)  $X \in \Sigma$

ii)  $A \in \Sigma \Rightarrow A^c \in \Sigma$  (complements)

iii)  $A_i \in \Sigma \quad i=1, \dots \Rightarrow \bigcup_{i=1}^{\infty} A_i \in \Sigma$  (countable unions).

(same for intersections by ii)  $\cap A_i = (\cup A_i^c)^c$ .

Def 2.10: If  $X$  is a metric space, the  $\sigma$ -algebra

$$\mathcal{B}(X) = \{ \text{smallest } \sigma\text{-algebra containing open sets} \}$$

$$= \bigcap_{\substack{\Sigma \\ \text{open} \in \Sigma}} \Sigma \quad \leftarrow P(X) \text{ is a } \sigma\text{-algebra so non-empty.}$$

is called the Borel sets of  $X$ .

Rem 2.11:

$\left\{ \begin{array}{l} \text{open/closed} \\ \text{sets} \end{array} \right\} \subsetneq \left\{ \begin{array}{l} \text{Borel} \\ \text{subsets} \end{array} \right\}$

$\uparrow$   
 $[0, 1/2) = \bigcup \text{closed}$

$\nwarrow$  has cardinality of  $\mathbb{R}$   
 $\nearrow$  bigger cardinality  
 $\nearrow$  "most sets are not Borel".  
 $\neq P(\mathbb{R})$



Rem 2.12 : Every  $x \in \mathbb{R}$  has a continued fraction expression

(2.4)

$$x = a_0 + \frac{1}{a_1 + \frac{1}{a_2 + \frac{1}{\dots}}}$$

wl  $a_0 \in \mathbb{Z}$  and  $a_i \in \mathbb{N}, \mathbb{Z}^{\geq 0}$

$K = \left\{ x \in \mathbb{R} \mid x \in \mathbb{Q} \text{ and } (a_0, a_1, \dots) \text{ has a subsequence } a_{i_k} \right. \\ \left. \text{st } \{a_{i_k} \mid a_{i_{k+1}}\} \right\}$

is not a Borel set.

2.iii) Measures :

Def 2.13 : Let  $(X, \Sigma)$  be a metric space with  $\sigma$ -algebra.

A measure

$$\mu : \Sigma \longrightarrow \mathbb{R} \cup \{\infty\}$$

is a function such that

i)  $\mu(E) \geq 0 \quad \forall E \in \Sigma$

ii)  $\mu(\emptyset) = 0$

iii) If  $\{E_i\}_{i=1}^{\infty}$  are a countable disjoint collection

$$\mu\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} \mu(E_i).$$

Def 2.14 : A measure space is a triple  $(X, \Sigma, \mu)$

of a metric space and  $\sigma$ -algebra with a measure.

It is complete if  $E \subseteq A \in \Sigma$  and then  $\mu(A) = 0 \Rightarrow \mu(E) = 0$ .

Theorem 2.15 : There exists a measure and  $\sigma$ -algebra

$\mathcal{M}(\mathbb{R}^n)$

$\mathcal{M}(\mathbb{R}^n)$

making  $(\mathbb{R}^n, \mathcal{M}(\mathbb{R}^n), m)$  into a measure space with the addition properties

i)  $m([a, b]) = b - a$

ii)  $m$  is complete

iii) If  $A \in \mathcal{M}(\mathbb{R}^n)$ ,  $A+x \in \mathcal{M}(\mathbb{R}^n)$  and  $m(A) = m(A+x)$ .

iv)  $\mathcal{B}(\mathbb{R}^n) \subseteq \mathcal{M}(\mathbb{R}^n)$

$m$  is called the Lebesgue measure, and  $\mathcal{M}(\mathbb{R}^n)$  the measurable sets.

### Lecture 3 Outer Measure + Lebesgue Measure

3.1

Recall: We wish to construct a measure ("Lebesgue measure") and a  $\sigma$ -algebra  $\mathcal{M}(\mathbb{R}^n)$  such that

- i)  $m(Q) = |Q|$  if  $Q = [a_1, b_1] \times \dots \times [a_n, b_n]$  is a cube
- ii)  $m$  is complete and translation invariant
- iii)  $\mathcal{B}(\mathbb{R}^n) \subseteq \mathcal{M}(\mathbb{R}^n)$

Ex 3.1: It is easy to construct an arbitrary measure on  $\Sigma = \mathcal{B}(\mathbb{R})$

Set 
$$\mu(E) = \begin{cases} 1 & \text{if } 0 \in E \\ 0 & \text{else.} \end{cases}$$

Check  $\mu(E)$  satisfies the properties of a measure.  
This is called an "atomic" or "Dirac" measure.

Plan 3.2: First define "outer measure" on  $\mathcal{P}(\mathbb{R}^n)$

outer measure on  $\mathcal{P}(\mathbb{R}^n)$   $\xrightarrow{\text{when defined.}}$  Lebesgue measure on  $\mathcal{M}(\mathbb{R}^n)$

Def 3.3: Let  $E \subseteq \mathbb{R}^n$  be a subset. Let

$$m^*(E) = \inf \left\{ \sum_{j=1}^{\infty} |Q_j| \mid E \subseteq \bigcup_{j=1}^{\infty} Q_j \text{ is a covering by cubes} \right\}$$

Ex 3.4:  $m^*(pt) = 0$  since  $pt = x_0$ ,  $Q = \prod [x_0^i - \frac{\varepsilon}{2}, x_0^i + \frac{\varepsilon}{2}]$  has volume  $\varepsilon > 0$  for arbitrary  $\varepsilon$ .

Ex 3.5:  $m^*(Q) = |Q|$  for a cube  $Q$ . Obvious  $Q$  covers itself so  $m^*(Q) \leq |Q|$ . Must show  $|Q| \leq \sum_{j=1}^{\infty} |Q_j|$  for  $Q \subseteq \bigcup Q_j$ .

For each  $j$ , let  $Q_j \subseteq S_j$  open w/  $|S_j| \leq (1+\varepsilon)|Q_j|$  for  $\varepsilon > 0$ .

$Q_j \subseteq \bigcup_{i=1}^N S_j$  by compactness,

$$|Q| \leq \sum_{j=1}^{\infty} |S_j| \leq (1+\varepsilon) \sum_{j=1}^{\infty} |Q_j| \leq (1+\varepsilon) \sum_{n=1}^{\infty} |Q_n|, \text{ take } \varepsilon \rightarrow 0$$



Ex 3.6 :  $Q$  an open cube,  $|Q| = m^*(Q)$  still holds. (3.2)  
 $Q$  covers, so  $m^*(Q) \leq |Q|$ . There is a closed  $Q_0 \subseteq Q$   
 with volume  $|Q_0| \geq (1-\varepsilon)|Q| \forall \varepsilon$ , and

$$(1-\varepsilon)|Q| \leq |Q_0| \leq m(Q)$$

since any covering of  $Q$  covers  $Q_0$ . Now take  $\varepsilon \rightarrow 0$ .

Ex 3.7 :  $m^*(\mathbb{R}^n) = \infty$ , obviously.

### 3.ii) Properties of the Outer Measure

Prop 3.8 (Monotonicity) If  $E_1 \subseteq E_2$ , then  $m^*(E_1) \leq m^*(E_2)$ .

Proof: Coverings of  $E_2$  also cover  $E_1$ .

Prop 3.9 (countable sub-additivity) If  $E = \bigcup_{j=1}^{\infty} E_j$ , then  $m^*(E) \leq \sum_{j=1}^{\infty} m^*(E_j)$ .

Proof: Can show for  $m^*(E_j) < \infty$ , else holds trivially.

Let  $Q_k^j$  be a collection of cubes covering  $E_j$  with

$$\begin{aligned} \sum_{k=1}^{\infty} |Q_k^j| &\leq m^*(E_j) + \frac{\varepsilon}{2^j} \\ m^*(E) &\leq \sum_{j,k=1}^{\infty} |Q_k^j| \leq \sum_{j=1}^{\infty} \sum_{k=1}^{\infty} |Q_k^j| \\ &\leq \sum_{j=1}^{\infty} m^*(E_j) + \varepsilon \quad \text{take lim } \varepsilon \rightarrow 0 \quad \square. \end{aligned}$$

Prop 3.10:  $m^*(E) = \inf_{\substack{U \subseteq \mathbb{R}^n \\ \text{open } E \subseteq U}} m^*(U)$ .

Proof:  $m^*(E) \leq m^*(U) \forall E \subseteq U$  by Prop 3.8. Let  $Q_j$  be a covering so

$$\sum |Q_j| \leq m^*(E) + \varepsilon/2.$$

Take  $U_j =$  open cube containing  $Q_j$  w/ area  $\leq |Q_j| + \frac{\varepsilon}{2^{j+1}}$ .

$$m^*(\bigcup U_j) \leq \sum m^*(U_j) \leq \sum |Q_j| + \frac{\varepsilon}{2^{j+1}} \leq \sum |Q_j| + \frac{\varepsilon}{2} \leq m^*(E) + \varepsilon \quad \square$$

### Prop 3.11 (Translation Invariance)

(3.3)

Suppose  $A$  is a set and  $A+y = \{u+y \mid u \in A\}$  for a fixed  $y \in \mathbb{R}^n$ .

Then  $m^*(A) = m^*(A+y)$ .

Proof: If  $Q$  is a cube, then  $Q+y$  is also, so a covering  $Q_j$  of  $A$  gives  $m^*(A+y) \leq \sum_{j=1}^{\infty} |Q_j+y| = \sum_{j=1}^{\infty} |Q_j| \leq m^*(A) + \varepsilon$ .

For reverse, note  $(A+y)-y = A$ . □

Corollary 3.12: If  $E \subseteq \mathbb{R}^n$  is countable,  $m^*(E) = 0$ .

Proof:  $E = \bigcup_{j=1}^{\infty} E_j$  where  $E_j = \text{pt.}$  Follows from Prop 3.9.

### 3.iii) Measurable Sets

Def 3.13: A subset  $E \subseteq \mathbb{R}^n$  is said to be measurable if for any  $A \subseteq \mathbb{R}^n$ ,

$$m^*(A) = m^*(A \cap E) + m^*(A \cap E^c).$$

ie  $E$  cuts any  $A$  into two pieces. (Note  $\leq$  holds by subadditivity)

Ex 3.14: If  $I \subseteq \mathbb{R}$  is an open interval,  $I$  is measurable.

Proof: Let  $A \subseteq \mathbb{R}$  and let  $A \subseteq \bigcup_{j=1}^{\infty} Q_j$  be a covering

Let  $Q_j' = I \cap Q_j \Rightarrow Q_j'$  cover  $A \cap I$

$Q_j'' = I^c \cap Q_j \Rightarrow Q_j''$  cover  $A \cap I^c$

$$\begin{aligned} m^*(I \cap A) + m^*(I^c \cap A) &\leq \sum_{j=1}^{\infty} |Q_j'| + \sum_{j=1}^{\infty} |Q_j''| + \varepsilon \\ &= \sum_{j=1}^{\infty} |Q_j| + \varepsilon \leq m^*(A) + 2\varepsilon. \end{aligned}$$

and now take  $\lim \varepsilon \rightarrow 0$ .

Ex 3.15: Same for closed intervals.

Ex 3.15.5: Same for  $\mathbb{R}^n$  but subdivide



Prop 3.16 : If  $E_j$  are disjoint, measurable, then  $\bigcup_{j=1}^{\infty} E_j$  is measurable and

$$m^*(A \cap \bigcup_{j=1}^{\infty} E_j) = \sum m^*(A \cap E_j).$$

3.4

Proof: If  $N=1$ , this is vacuous. Take  $A \in \mathbb{R}^n$  then set  $A' = A \cap (E_1 \cup E_2)$

$$m^+(A') = m^+(A' \cap E_1) + m^+(A' \cap E_1^c) \quad (E_1 \text{ measurable})$$
$$= m^+(A \cap E_1) + m^+(A \cap E_2)$$

now use induction.

Thm 3.17 : The measurable sets form a  $\sigma$ -algebra, and  $m^* : \mathcal{M}(\mathbb{R}^n) \rightarrow [0, \infty]$  is a measure.

Proof : i)  $\mathbb{R}^n$  is measurable, since  $m^*(A) = m^*(A \cap \mathbb{R}^n) + \underbrace{0}_{\text{def.}}$   
 ii)  $E$  measurable  $\Rightarrow E^c$  measurable by symmetry of def.  
 iii) Suppose  $E_i$  are measurable, then wts  $\bigcup_{i=1}^{\infty} E_i$  is.

Step 1: True for finite unions/intersections. Suppose  $E, F$  measurable

$$\begin{aligned} m^*(A) &= m^*(A \cap E) + m^*(A \cap E^c) \\ &= m^*(A \cap E \cap F) + m^*(A \cap \underline{E \cap F^c}) + m^*(A \cap \underline{E^c \cap F}) \\ &= m^*(A \cap E \cap F) + m^*(A \cap (E \cap F)^c) \quad \begin{matrix} \text{by Prop 16.} \\ U = E \cap F \end{matrix} \end{aligned}$$

Step 2 : Suppose  $E_i$  measurable, then

$$E'_i = E_i - \bigcup_{j=1}^{i-1} E_j = E_i \cap \left( \bigcup_{j=1}^{i-1} E_j \right)^c$$

is measurable, and  $\bigcup E'_i = \bigcup E_i$ , so suffices to assume disjoint. Now let  $F_N = \bigcup_{i=1}^N E_i$  (measurable)

$A \in \mathbb{R}^n$  arbitrary.

$$\begin{aligned} m^*(A) &= m^*(A \cap \bigcup_{i=1}^{\infty} E_i) + m^*(A \cap F_n^c) \\ &= \sum_{i=1}^{\infty} m^*(E_i \cap A) + m^*(A \cap F_n^c) \\ &\geq \sum_{i=1}^{\infty} m^*(E_i \cap A) + m^*(A \cap E^c) \quad \text{by monotonicity} \end{aligned}$$

Take lim

$$\begin{aligned} &\geq \sum_{i=1}^{\infty} m^*(E_i \cap A) + m^*(A \cap E^c) \\ &\geq m^*(E \cap A) + m^*(E^c \cap A) \text{ by countable subadditivity.} \end{aligned}$$



and  $\mu(E) \geq 0$ ,  $\mu(\emptyset) = 0$ . If  $E = \bigcup E_j$  disjoint,

3.5

$$\mu(E) \leq \sum_{j=1}^{\infty} m^*(E_j) \quad (\text{additivity}).$$

$$\sum_{j=1}^{\infty} m^*(E_j) \leq \sum_{j=1}^{\infty} \sum_{k=1}^{\infty} |Q_{j,k}| \quad \text{covering}$$

$\therefore$  take  $A = E$  in step 2.

Def 3.18 :  $\mathcal{M}(\mathbb{R}^n)$  are called the measurable sets,  
and  $m^*|_{\mathcal{M}(\mathbb{R}^n)}$  is called the Lebesgue measure.

## Lecture 4 Properties of Lebesgue Measure + Counterexamples.

[4.1]

Recall last time we showed  $(\mathbb{R}^n, \mathcal{M}(\mathbb{R}^n), m)$  is a measure space.

Thm 2.15 :  $(\mathbb{R}^n, \mathcal{M}(\mathbb{R}^n), m)$  is a measure space, and additionally satisfies

(1) Normalization  $m(Q) = |Q|$

(2) Complete  $E \in \mathcal{M}(\mathbb{R}^n)$  then  $m(E) = 0 \Rightarrow A \subseteq E$  measurable

(3) translation invariant  $E \in \mathcal{M}(\mathbb{R}^n) \Rightarrow E+y \in \mathcal{M}(\mathbb{R}^n)$  w/  $m(A) = 0$ .  
w/  $m(E) = m(E+y)$

(4) Borel complete  $\mathcal{B}(\mathbb{R}^n) \subseteq \mathcal{M}(\mathbb{R}^n)$ .

Proof : (1) For open/closed  $Q$ ,  $m(Q) = m^*(Q) = |Q|$ .

(2) Suppose  $E$  has <sup>outer</sup> measure 0. Then  $A \cap E \subseteq E$ ,  $A \cap E^c \subseteq A$   
so by monotonicity

$$m^*(A) \geq m^*(A \cap E^c)$$

$$\geq m^*(A \cap E^c) + m^*(A \cap E) \stackrel{0}{=} m^*(A) \Rightarrow E \text{ is measurable.}$$

If  $A \subseteq E$ , then  $m^*(A) = 0$  so also measurable.

(3)  $A \cap (E+x) = (A-x) \cap E$  ~~so~~, and  $m^*$  is translation inv.

(4) open sets  $\in \mathcal{M}(\mathbb{R}^n)$  and it's a  $\sigma$ -algebra,  $\square$

### 4.i) Continuity of Measure

Prop 4.1 : Suppose  $E = \bigcap_{i=1}^{\infty} E_i$  with  $E_1 \supseteq E_2 \supseteq \dots$  and  $m(E_1) < \infty$ .

$$\text{Then } m(E) = \lim_{N \rightarrow \infty} m(E_N).$$

Proof :  $E_1 = E \cup (E_1 \setminus E_2) \cup (E_2 \setminus E_3) \cup \dots$

$$m(E_1) = m(E) + \sum_{n=1}^{\infty} m(E_n \setminus E_{n+1}) = m(E) + m(E_1) - \lim_{i \rightarrow \infty} m(E_i)$$

$$\Rightarrow m(E) = \lim m(E_i) \quad \square$$

Rem 4.2 : Informally we can write

$$m(\lim E_i) = \lim m(E_i)$$

which looks more like standard continuity.

Lemma 4.2 (Borel - Cantelli) Suppose  $E_i$  measurable,  $\sum_{n=1}^{\infty} m(E_n) < \infty$ . [4.2]

Then  $m\{x \mid x \in E_n \text{ for } \infty\text{-many } n\} = 0$ .

Proof: Call this set  $A$ .  $A \subseteq \bigcup_{n=N}^{\infty} E_n$  for any  $N$ .

$$m(A) \leq m\left(\bigcup_{n=N}^{\infty} E_n\right) \leq \sum_{n=N}^{\infty} m(E_n) \rightarrow 0$$

by convergence.

#### 4.ii) Some funky sets.

Ex 4.3 (Cantor Set)  $K \subseteq [0,1]$  a set uncountable but  $m(K)=0$ .

$$K_0 = [0,1]$$

$$K_1 = K_0 \setminus (\text{middle third})$$

$$K_2 = K_1 \setminus (\text{middle thirds})$$



⋮

$$K = \bigcap_{n=0}^{\infty} K_n$$

$$\text{Clearly } m(K_n) = \frac{2}{3} m(K_{n-1}) = \left(\frac{2}{3}\right)^n$$

$$m(K) = \lim_{n \rightarrow \infty} m(K_n) = \lim_{n \rightarrow \infty} \left(\frac{2}{3}\right)^n = 0.$$

Claim:  $K$  is uncountable.  $K = \{x \in [0,1] \mid x = 0.x_1x_2x_3 \dots \text{ 'in base 3'}$   
(bijective w/  $\mathbb{R}$ )  $\wedge x_i = 0, 2$ .

Suppose we can enumerate  $K_1, K_2, \dots \Rightarrow$  Cantor's diagonalization  $\square$ .

Corollary 4.4: There exist measurable sets that are not Borel.  
In fact, "almost all" are such.

Proof:  $m(K)=0$ , so any  $A \subseteq K$  is measurable with measure 0.

$\Rightarrow \mathcal{P}(K) \subseteq \mathcal{M}(\mathbb{R})$ , but  $\mathcal{P}(K)$  has cardinality  $2^{\aleph_0} \geq$   
Borel sets have cardinality  $\aleph_1$ .



# Ex 4.5 (Normal Numbers)

[4.3]

Def 4.6:  $x \in [0,1]$  is normal if any finite sequence of digits occurs infinitely often.

Lemma 4.7:  $m\{x \in [0,1] \mid x \text{ normal}\} = 1$ .

Proof: Let  $E_i = \{x \in [0,1] \mid i^{\text{th}} \text{ digit is } 1\}$ . Clearly  $m(E_i) = \frac{1}{10}$ .

$A = \{x \mid 1 \text{ occurs inf often}\}$

$$A^c = \{x \mid \exists N \text{ st } x \notin \bigcup_{n=N}^{\infty} E_n\} = \bigcup_{N=1}^{\infty} \bigcap_{n=N}^{\infty} E_n^c$$

$$m(A^c) = m\left(\bigcup_{N=1}^{\infty} \bigcap_{n=N}^{\infty} E_n^c\right) \leq \sum_{N=1}^{\infty} m\left(\bigcap_{n=N}^{\infty} E_n^c\right) = 0.$$

= 0 by Borel-Cantelli.

$$\Rightarrow m(A) = 1.$$

$E_i' = \{x \in [0,1] \mid i^{\text{th}} - i \cdot 10^{\text{th}} \dots\}$  same proof.

Conjecture 4.8:  $\pi, \sqrt{2} - 1$  are normal.

Corollary 4.9: It is not known if  $\pi$  contains Hamlet in some decimal form.

Ex 4.10: There exists a non-measurable set.

Let  $N = \{ \text{a lift of } \mathbb{R} \downarrow \mathbb{R}/\mathbb{Q} \text{ ie a choice of representatives of cosets } x \sim y \text{ if } x - y \in \mathbb{Q} \}$ .

this invokes the axiom of choice!

Note each  $x \in \mathbb{R}$  can be written  $x = a + q$  uniquely w/  $a \in N, q \in \mathbb{Q}$ .

Lemma 4.11:  $N$  is not measurable.

Proof: Suppose it is. Set  $S = [0,1] \cap N$ .

$$B_{m,s} = \{s + a \mid s \in S, a \in [m, m+1]\}$$

$\bigcup_{s \in S} B_{m,s} \subseteq [m, m+2]$ , and all disjoint, and translates, ~~so  $\sum m(B_{m,s})$~~

Therefore

$$\sum_{s \in S} m(B_{m,s}) \leq 2 \Rightarrow m(B_{m,s}) = 0 \text{ b/c all equal.}$$

In particular,  $m(N \cap [m, m+1]) = m(B_{m,0}) = 0 \Rightarrow m(N) = 0$ .

But

$$\infty = m(\mathbb{R}) = m\left(\bigcup_{q \in \mathbb{Q}} N + q\right) = \sum 0 = 0 \quad \rightarrow \leftarrow \quad \square$$

Corollary 4.12 : If  $E \subseteq \mathbb{R}$  has positive measure,  $\exists$  a  $N \subseteq E$  non-measurable.

Remark 4.13

Thm 4.13 (Solovay 1970) : A non-measurable <sup>Lebesgue</sup> set cannot be constructed without the Axiom of Choice.

Rem 4.14 : If  $A$  is measurable  $A - A = \{x - y \mid x, y \in A\}$  need not be measurable!

Note also  $K - K = [-1, 1]$  despite  $m(K) = 0$ .

Open Problem 4.15 : If  $x \in K$ , then can  $x$  be algebraic?  
and  $x \in \mathbb{Q}$

## Lecture 5 Measurable Functions

[5.1]

Def 5.1 : A property  $P$  is said to hold "almost everywhere" if it holds on a set of full measure.

### 5.1) Littlewood's First Principle

Principle 5.2 : "Every measurable set is closed to a finite union of intervals"

Def 5.3 : Given two set  $A, B$  the symmetric difference  
 $A \Delta B = (A \setminus B) \cup (B \setminus A)$

Prop 5.4 : If  $m(E) < \infty$ , then  $\forall \varepsilon > 0$ ,  
 $\exists J$  a finite union of intervals st  
 $m(J \Delta E) < \varepsilon$ .



Proof : Note that  $m(B \setminus A) = m(B) - m(A)$   
since  $m(B) = m(B \cap A) + m(B \cap A^c)$   
 $= m(A) + m(B \setminus A)$

Take  $E \subseteq \bigcup Q_n$  such that  
 $\sum |Q_n| \leq m(E) + \varepsilon/2$ .

Now take  $N$  large so that

$$\sum_{n=N+1}^{\infty} |Q_n| < \frac{\varepsilon}{2}$$

$$m(E \setminus \bigcup_{n=1}^N Q_n) \leq m(\bigcup_{n=1}^{\infty} Q_n \setminus \bigcup_{n=1}^N Q_n) < \varepsilon/2$$

$$m(\bigcup_{n=1}^N Q_n \setminus E) \leq m(\bigcup_{n=1}^{\infty} Q_n \setminus E) < \varepsilon/2 \quad \square$$

Def 5.5 : A  $G_\delta$  set is a countable intersection of opens <sup>[5.2]</sup>  
an  $F_\sigma$  set is a countable union of closed

Note both are Borel.

Prop 5.6 : A set  $E \subseteq \mathbb{R}^n$  is measurable : ff  $\exists$

$$F_\sigma \subseteq E \subseteq G_\delta$$

such that  $m(E \setminus F) = m(G \setminus E) = 0$ .

Proof : If the statement holds,  $E$  is measurable  
since  $F, G$  are and all measure 0 sets are.

Let  $U_n$  be open with  $m(U_n \setminus E) \leq \frac{1}{n}$ ,  
and set  $G_\delta = \bigcap U_n$ , and  $m(G \setminus E) \leq \frac{1}{n} \forall n$ .

Let  $V_n$  be open w/  $m(V_n \setminus E^c) \leq \frac{1}{n}$ .

$$F_\sigma = \bigcup V_n^c$$

suffices.

5.ii) Measurable Functions (We will work only on  $\mathbb{R}$ )  
for now

Prop 5.7 : TFAE for  $f: \mathbb{R} \rightarrow \mathbb{R}$

1)  $f^{-1}(a, \infty]$  is measurable  $\forall a \in \mathbb{R}$ .

2)  $f^{-1}(U)$  is measurable  $\forall U \subseteq \mathbb{R}$  open

3)  $f^{-1}(B)$  is measurable  $\forall B \subseteq \mathbb{R}$  Borel.

Proof :  $3 \Rightarrow 2 \Rightarrow 1$  a fortiori.

Let  $\Sigma(f) = \{ A \in \mathcal{P}(\mathbb{R}) \mid f^{-1}(A) \text{ is measurable} \}$

It is easy to check  $\Sigma(f)$  is a  $\sigma$ -algebra, and  $(a, \infty)$   
generates  $\mathcal{B}$  so  $\mathcal{B}(\mathbb{R}) \subseteq \Sigma(f)$



Prop 5.8:  $\mathcal{M}(\mathbb{R})$  is an algebra containing  $\mathcal{C}^0(\mathbb{R})$ .

Proof: Obviously  $\mathcal{C}^0 \subseteq \mathcal{M}$  by 2) of Prop 5.7.

1)  $f \in \mathcal{M}(\mathbb{R})$  then  $\alpha f \in \mathcal{M}(\mathbb{R})$  for  $\alpha \in \mathbb{R}$  is clear, because  $f^{-1}(a, b) = \alpha f^{-1}(\frac{a}{\alpha}, \frac{b}{\alpha})$ .

2) Suppose  $f, g \in \mathcal{M}(\mathbb{R})$ . Suppose  $x \in (f+g)^{-1}(a, \infty)$ .

Let  $\frac{2}{q} \in \mathbb{Q}$ ,  $a - g(x)$  be a rational, such a  $p/q$  exists iff  $f(x) + g(x) > a$ .

$$(f+g)^{-1}(a, \infty) = \bigcup_{\substack{p/q \in \mathbb{Q} \\ f, g \in \mathcal{M}(\mathbb{R})}} \{x \mid f(x) > p/q\} \cap \{x \mid \frac{2}{q} + g(x) > a\}$$

↑ countable union of measurable.

3) If  $f, g \in \mathcal{M}(\mathbb{R})$  then  $f \cdot g \in \mathcal{M}(\mathbb{R})$ .

$\frac{1}{2}[(f+g)^2 - f^2 - g^2] = f \cdot g$  so by 1), 2) enough to show for

$$f = g. \quad (f^2)^{-1}(a, \infty) = f^{-1}(\sqrt{a}, \infty) \cup f^{-1}(-\infty, -\sqrt{a}). \quad \square$$

Warning 5.9:  $\mathcal{M}(\mathbb{R})$  is not closed under composition!

but  $\mathcal{C}^0(\mathbb{R}) \times \mathcal{M}(\mathbb{R}) \rightarrow \mathcal{M}(\mathbb{R})$

$(h, f) \rightarrow h \circ f$  is measurable.

Theorem 5.10: Suppose  $f_n$  is a sequence of measurable functions such that  $f_n \rightarrow f$  pointwise, then  $f \in \mathcal{M}(\mathbb{R})$ .

Proof: For  $c \in \mathbb{R}$ , write

$$\begin{aligned} f^{-1}(c, \infty) &= \{x \mid \exists k \in \mathbb{N} \text{ st } \exists N \in \mathbb{N} \text{ st } \forall n \geq N, f_n(x) > c + \frac{1}{k}\} \\ &= \bigcup_{k \in \mathbb{N}} \bigcup_{N \in \mathbb{N}} \bigcap_{n \geq N} \{x \mid f_n(x) > c + \frac{1}{k}\}. \end{aligned}$$

Countable  $\cup$  and  $\cap$  of measurable

□.

Def 5.11 : The Indicator function of a measurable set  $A$  is

$$\chi_A = \begin{cases} 1 & x \in A \\ 0 & \text{else} \end{cases}$$

It is clearly measurable.

Def 5.12 : A function is called simple if it is a finite sum

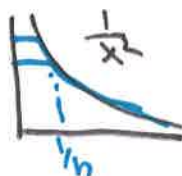
$$f = \sum_{i=1}^N \alpha_i \chi_{E_i} \quad E_i \text{ measurable.}$$

and a step function if  $E_i$  are disjoint intervals.

Def 5.13 :  $f_n$  converges to  $f$  in measure if  $\forall \varepsilon > 0$ .

$$\lim_{n \rightarrow \infty} m\{x \mid |f_n(x) - f(x)| > \varepsilon\} = 0.$$

Ex 5.14 :



$$f_n = \min(n^2, \frac{1}{x^2})$$

then  $f_n \rightarrow f$  in measure

but  $\sup |f_n - f| = \infty$  so not uniformly

Thm 5.15 Suppose  $f: [a, b] \rightarrow \mathbb{R}$  is measurable.  $\exists f_n \rightarrow f$  in measure such that  $f_n$  are

- i) bounded and measurable
- ii) Simple
- iii) Step
- iv) Continuous.

Proof i) let  $f_n = \begin{cases} f(x) & |f(x)| \leq n \\ 0 & \text{else} \end{cases}$

then  $|f_n - f| = 0$  outside  $E_n = \{x \mid |f(x)| > n\}$ .

Since  $\bigcap E_n = \emptyset$  and  $m(E_n) < \infty$ ,  $m(E_n) \rightarrow 0$  by continuity of measure.

ii) By diagonalization can assume  $f$  bounded. Say  $|f(x)| \leq M$ .

$$\text{Let } I_{nk} = [-M + \frac{k}{n}, -M + \frac{(k+1)}{n}]$$

$$E_k = \{x \mid f(x) \in I_{nk}\}$$

Take  $f_n = \sum_{k=1}^{2Mn} (-M + \frac{k}{n}) \chi_{E_k}$   $|f_n - f| \leq \frac{1}{n}$ . Now send  $n \rightarrow \infty$ .

iii) Approximating  $\chi_{E_k}$  by step is same as approximating  $E$  by opens.



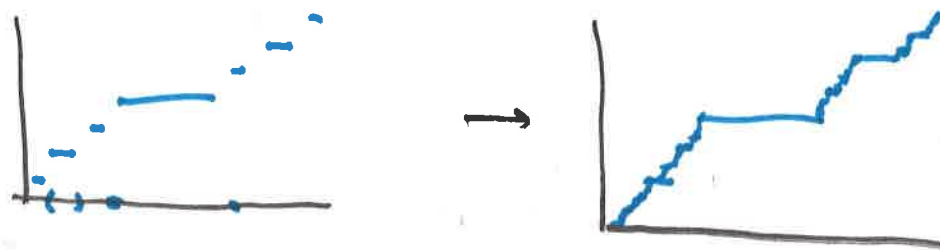
# Lecture 6 Measurable Functions II: Littlewood's 2<sup>nd</sup> - 3<sup>rd</sup> Principles.

6.1

## 6.i) The Cantor Function / Devil's Staircase

Ex 6.1: Let  $c(x): [0,1] \rightarrow [0,1]$  be defined by

$$c(x) = \begin{cases} \frac{1}{2} & x \in (\frac{1}{3}, \frac{2}{3}) \\ \frac{1}{4}, \frac{3}{4} & x \in (\frac{1}{9}, \frac{2}{9}) \text{ or } (\frac{7}{9}, \frac{8}{9}) \end{cases} \quad \text{on } K^c = \sup \{C_H \mid t \in K_n^c \cap [0,x]\} \text{ on } K$$



Lemma 6.2:  $C(x): [0,1] \rightarrow [0,1]$  is continuous (hence measurable) and monotonically <sup>non-dec.</sup> increasing, but  $C'(x) = 0$  a.e.

Proof: Clearly continuous on  $K^c$ . If  $x \in K$ , then  $x$  lies between two intervals of  $K_n^c$  w/  $K^c = \bigcup_n K_n^c$ , for <sup>any</sup>  $n$ , and these have  $|C(x) - C(y)| \leq \frac{1}{2^n}$ .  
Monotonically non-decreasing is obvious from sup.  
 $C'(x)$  exists and  $= 0$  on  $K^c$  and  $m(K^c) = 1$ .

Corollary 6.3: (Even though it's not precisely defined yet)

$$\int_0^1 C'(x) dx = 0 \cdot m(K^c) + ? \cdot m(K) = 0$$

So in particular, the FTC fails for  $C$ ,

$$0 = \int_0^1 C'(x) dx \neq C(1) - C(0) = 1.$$

Corollary 6.4:  $\exists \varphi$  a homeomorphism  $\varphi: [0,1] \rightarrow [0,2]$  that sends a set of measure 0 to a set of positive measure, and vice-versa.

Proof:  $\varphi(x) = x + C(x)$  is <sup>(strictly)</sup> monotonically increasing so a homeomorphism  $K \rightarrow \text{measure } 1$ .  
 $\frac{1}{2}\varphi^{-1}(2x)$  has opposite property.

## 6:ii) Littlewood's principles II

6.2

Principle 6.5 (Littlewood I: Measurable Sets): Any measurable set is nearly a finite union of open intervals

Principle 6.6 (Littlewood II: Measurable Functions): Any measurable function is nearly continuous

Principle 6.7 (Littlewood III: Convergence): Any convergent sequence is nearly uniformly convergent.

Here's a formal version of II:

Thm 6.8 (Lusin) Let  $f: \mathbb{R} \rightarrow \mathbb{R}$  be measurable. Then given  $\varepsilon > 0$ ,  $\exists g: \mathbb{R} \rightarrow \mathbb{R}$  continuous st.  $f = g$  outside a set of measure  $< \varepsilon$ .

Recall we can construct  $f_n \rightarrow f$  where  $f: [a, b] \rightarrow \mathbb{R}$   
 $f_n$  bounded, simple, step,  $C^0$ .

Proof: 1) It suffices to consider  $f: [n, n+1] \rightarrow \mathbb{R}$ , then use the " $\frac{\varepsilon}{2^n}$ " trick.

2) Since  $f_n \rightarrow f$  in measure,  $f_n$  bounded, can assume  $f$  bounded.

3)  $g_n \in C^0$  st  $g_n \rightarrow f$  in measure  
 $\Rightarrow \exists g_1$  st  $m(\underbrace{|g_1 - f|}_{h_1} > \frac{1}{2}) < \frac{\varepsilon}{2}$ .

$$f = g_1 + h_1$$

Now  $\exists g_2$  st  $g_2 \rightarrow g_1$  in measure  $\Rightarrow \exists g_2$  st  $m(|g_2 - g_1| > \frac{1}{2^2}) < \frac{\varepsilon}{4}$   
Inductively

$$f = g_1 + \dots + g_n + h_n$$

$g_i$  continuous,  $|h_N| \leq \frac{1}{2^N}$  except on  $E = \bigcup_{j=1}^N E_j$   
with  $m(E_j) \leq \frac{\varepsilon}{2^j}$ ,  $h_N = 0$  on  $E^c$ .

4) Let  $\sup |f_N(x)| \leq \sup |h_{N-1}| \leq \frac{1}{2^{N-1}}$ ,  $g = \sum g_i$  converges uniformly  
and  $f = g$  outside  $m(\bigcup E_j) = \varepsilon$ .



### 6.iii) Convergence

6.3

There are three types of convergence discussed so far.

$$\left\{ \begin{array}{l} f_n \rightarrow f \\ \text{in measure} \end{array} \right\} \overset{\text{implies}}{\subseteq} \left\{ \begin{array}{l} f_n \rightarrow f \\ \text{pointwise} \\ \text{(a.e.)} \end{array} \right\} \overset{\text{implies}}{\subseteq} \left\{ \begin{array}{l} f_n \rightarrow f \\ \text{uniformly} \end{array} \right\}$$

$m(|f_n - f| > \varepsilon) \rightarrow 0 \quad \forall \varepsilon$ 
 $f_n(x) \rightarrow f \quad \forall x$ 
 $\sup_x |f_n(x) - f(x)| \rightarrow 0$   
 = Convergence in  $\ell^\infty$ .

Prop 6.9: Suppose  $f_n \rightarrow f$  ~~in measure~~ pointwise a.e. on  $[a, b]$

then  $f_n \rightarrow f$  in measure

Proof: Let  $\varepsilon > 0$ .  $E_n = \{|f_n - f| > \varepsilon\}$   ~~$\bigcap_{n=1}^{\infty} E_n = \emptyset$  by pointwise a.e.~~

~~$\lim_{n \rightarrow \infty} m\left(\bigcap_{k=n}^{\infty} E_k\right) = m\left(\bigcap_{n=1}^{\infty} E_n\right) = 0$ , so~~

$B_N = \bigcup_{n=N}^{\infty} E_n$  so  $B_N \supseteq B_{N+1} \supseteq \dots$

$\lim_{n \rightarrow \infty} m(E_n) \leq \lim_{N \rightarrow \infty} m(B_N) \overset{\text{cont. of measure}}{=} m\left(\bigcap_{N=1}^{\infty} B_N\right)$

$= m\left\{x \mid \exists n \geq N \text{ w/ } |f_n - f| > \varepsilon\right\}$

$= 0$  by pointwise a.e.

Ex 6.10: Note convergence  $f_n \rightarrow f$  in measure for  $f_n$  bounded cannot happen on  $\mathbb{R}$  (on any inf. measur. set)

$f(x) = x$  is a counter example, However, any such  $f$  is still a lim in measure of  $f$  cont.

Ex 6.11: Prop 6.9 also fails on  $\mathbb{R}$ . b/c  $m(B_1) = \infty$  is possible

$f_n = \sin x \quad \forall [-n, n] \rightarrow \sin x$  pointwise but

$m(|f_n - f| > \frac{1}{2}) = \infty \quad \forall n.$

Ex 6.11 :  $\exists f_n \rightarrow f$  in measure but not pointwise.

Let  $I_n =$  interval of length  $\frac{1}{\sqrt{n}}$ .

"limp function"



so  $\bigcup_{n=1}^{\infty} I_n = [0,1] \forall N$ . so  $\sum |I_n| = \infty$ .

Then  $f_n = \chi_{I_n} \rightarrow 0$  in measure, but  $f_n \rightarrow 0$  pointwise nowhere.

Thm 6.12 :  $f_n \rightarrow f$  in measure on  $\mathbb{R}$ ,  $\exists$  a subseq converging pointwise a.e.

Proof :  $E_n = \{|f_n - f| > \frac{1}{2^n}\}$ , take  $n_k$  such that  $\{m(E_{n_k}) < \frac{1}{k^2}$ . Then  $f_n \rightarrow f$  outside  $\bigcup_{k=1}^{\infty} E_{n_k}$  but this  $\rightarrow 0$  w/k because  $\sum m(E_{n_k}) < \infty$ .

6.14 : Littlewood's 3<sup>rd</sup> Principle

Precise version:

Thm 6.13 (Egoroff): Let  $f_n \rightarrow f$  pointwise a.e. on  $[a,b]$ .

$\forall \varepsilon > 0$ ,  $\exists E_{\varepsilon} \subseteq [a,b]$  st.  $m([a,b] \setminus E_{\varepsilon}) < \varepsilon$  and  $f_n \rightarrow f$  uniformly on  $E_{\varepsilon}$ .

Proof : Set  $\delta_n = \sup_{i \geq n} |f_i(x) - f(x)|$

Note  $\delta_n \geq \delta_{n+1}$  and  $\delta_n \rightarrow 0$  pointwise a.e.

$\exists n_k$  st.  $\delta_{n_k} \rightarrow 0$  in measure, so  $\forall k \exists$

$E_k$  st.  $m(E_k) \leq \frac{\varepsilon}{2^k}$  and  $|\delta_{n_k}| \leq \frac{1}{k}$  on  $E_k^c$ .

Then  $\delta_{n_k} \rightarrow 0$  uniformly on  $E_{\varepsilon}^c \cap E_k^c$  and

$$m(E_{\varepsilon}) \geq m([a,b]) - \sum E_k \geq (b-a) - \sum \frac{\varepsilon}{2^k} \geq b-a-\varepsilon.$$

But  $\delta_{n_k}$  decreasing so conv on subseq  $\Rightarrow$  conv of whole seq  $\square$

## Lecture 7 | Lebesgue Integration I: bounded functions

(7.1)

Assumption 7.1 : Throughout this lecture, assume  $E$  measurable w/  $m(E) < \infty$  and all measurable functions lie in  $\mathcal{M}_B(E) = \{ f \in \mathcal{M}(E) \mid \exists M \text{ w/ } |f| \leq M \}$ .

Def 7.2 : Define the (Lebesgue) Integral of a function  $f \in \mathcal{M}_B(E)$

w/  $m(E) < \infty$ , denote

$$\int_E dx : \mathcal{M}_B(E) \rightarrow \overline{\mathbb{R}}$$

or  $\int_E dm(x)$   
↑  
Lebesgue measure.

By

$$\int_E f dx := \sup_{\substack{\varphi \text{ simple} \\ \varphi \leq f}} \left\{ \sum_{i=1}^N a_i m(E_i) \mid \varphi = \sum_{i=1}^N a_i \chi_{E_i} \right\}$$

Def 7.3 :  $f \in \mathcal{M}_B(E)$  is said to be (absolutely) Lebesgue integrable if  $\int_E |f| dx < \infty$

Rem 7.4 : If  $E' \subseteq E$ , then one can extend

$$- : \mathcal{M}_B(E') \rightarrow \mathcal{M}_B(E)$$

and

$$f \mapsto \bar{f} = \begin{cases} f & x \in E' \\ 0 & \text{else} \end{cases}$$
$$\int_{E'} f dx = \int_E \bar{f} dx.$$

Prop 7.5 :  $\int_E dx : \mathcal{M}_B(E) \rightarrow \overline{\mathbb{R}}$  satisfies

1)  $\int_E af + bg dx = a \int_E f dx + b \int_E g dx$  (linearity)

2)  $\int_E \chi_A dx = m(A)$  for  $A \subseteq E$  measurable (normalization)

3)  $f \leq g \Rightarrow \int_E f dx \leq \int_E g dx$  (monotonicity)

4)  $\left| \int_E f(x) dx \right| \leq \int_E |f(x)| dx$

5)  $\int_E |f(x)| dx \leq m(E) \sup |f(x)| < \infty.$

Rem 7.6: A simple function  $\varphi: E \rightarrow \mathbb{R}$  has a canonical representation <sup>(7.2)</sup>

$$\varphi = \sum_{k=1}^N \lambda_k \chi_{E_k}$$

where  $E_k = \varphi^{-1}(\lambda_k)$  for all  $k$ . Other representations arise from subdividing or repetition. We assume all simple functions are in this canonical representation. (See Royden 4.2.1)

Prop 7.7: If  $f \in \mathcal{M}_B(E)$  then

Note

$\int_E \varphi dx = \sum a_k m(E_k)$   
is immediate

$$\sup_{\varphi \leq f} \int_E \varphi dx = \int_E f dx = \inf_{\varphi \geq f} \int_E \varphi dx$$

agree and are finite. In fact, if  $f$  bounded, this is iff.

Proof:  $\Rightarrow$  For any choice  $\varphi \leq f \leq \psi$ , so suffices to show  $|\inf - \sup| < \varepsilon$  for any  $\varepsilon$ . Given  $\varepsilon$ , let  $|f| \leq M$ , and divide  $[-M, M] = \bigcup I_n = \bigcup [a_n, b_n]$  where  $|I_n| < \varepsilon$ .

Set  $E_n = f^{-1}(I_n)$ . Then

$$\sum_{n=1}^N a_n m(E_n) \leq \inf_{\varphi \leq f} \sum \alpha_k m(E_k) \quad \text{and on the other hand}$$

$$\inf_{\varphi \leq f} \sum \beta_k m(E_k) \leq \sum_{n=1}^N b_n m(E_n) \leq m(E) \cdot \varepsilon + \sum_{n=1}^N a_n m(E_n)$$

so  $|\sup - \inf| \leq m(E) \cdot \varepsilon$ .

$\Leftarrow$  Suppose  $\sup = \inf$ . Let  $\varphi_n, \psi_n$  be with  $\frac{1}{2^n}$  of these so  $\int_E |\varphi_n - \psi_n| dx \leq \frac{1}{n}$ . monotonic, bounded above

Since  $\varphi_n \leq f \leq \psi_n$ ,  $\forall n$   $\lim_{n \rightarrow \infty} \varphi_n \leq f \leq \lim_{n \rightarrow \infty} \psi_n$

Thus  $\varphi = \psi$  a.e.  $\Rightarrow f = \varphi = \psi$  a.e.  $\Rightarrow f$  measurable since  $\varphi, \psi$  are.

Claim:  $\varphi_n \rightarrow \psi_n$  in measure. i.e.  $\forall \varepsilon > 0$   $\lim_{n \rightarrow \infty} \frac{1}{\varepsilon} m(|\varphi_n - \psi_n| > \varepsilon) = 0$ .

Pick  $\delta > 0$ .  $N$  st  $\frac{1}{N} < \varepsilon \delta$ . Then

$$m(|\varphi_n - \psi_n| > \varepsilon) \cdot \varepsilon \leq \int_E |\varphi_n - \psi_n| \leq \frac{1}{N} \leq \varepsilon \delta$$

$m(\quad) \leq \delta$  (now subseq conv. but monotone)  $\square$



Proof (of Prop 7.5).

(7.3)

$$1) \int_E f dx + \int_E g dx \leq \sup_{\psi \leq f, \psi \leq g} \int \psi \leq \sup_{\xi \leq f+g} \int \xi dx = \int_E f+g.$$

Using inf get reverse.  
 $\int_E \alpha f dx = \alpha \int_E f dx$  is obvious.

$$2) \text{ If } f = \chi_A \text{ then } \psi \leq \chi_A \Rightarrow \psi \leq 0 \text{ on } A^c, \psi \leq 1 \text{ on } A, \text{ so}$$
$$\int_E \psi dx \leq m(A), \text{ but } \psi = \chi_A \text{ itself gives } =.$$

$$3) \text{ If } f \leq g, \sup_{\psi \leq f} \leq \sup_{\psi \leq g} \text{ is the sup over a strictly larger set.}$$

$$4) -|f| \leq f \leq |f| \text{ so by 3, } -\int_E |f| dx \leq \int_E f dx \leq \int_E |f| dx$$

$$5) \text{ by inf def, } \inf \leq m(E) \cdot \sup |f|.$$

$$\sup \geq m(E) \cdot (-\sup |f|)$$

Prop 7.8 : If  $A, B \in \mathcal{E}$  disjoint, measurable

$$\int_{A \cup B} f dx = \int_A f dx + \int_B f dx$$

Proof : By linearity

$$\begin{aligned} \int_{A \cup B} f dx &= \int_E f \chi_{A \cup B} dx = \int_E f (\chi_A + \chi_B) \\ &= \int_E f \chi_A dx + \int_E f \chi_B dx = \int_A f dx + \int_B f dx. \end{aligned}$$

Thm 7.9 (Bounded convergence) Suppose  $f_n \in M_b(E)$  are uniformly 7.4  
 bounded  $|f_n| \leq M$ ,  $f_n \rightarrow f$  pointwise a.e. Then

$$\lim_{n \rightarrow \infty} \int_E f_n dx = \int_E \lim f_n dx = \int_E f dx$$

Proof: By Littlewood III,  $\exists A \subseteq E$  with  $m(E \setminus A) < \frac{\varepsilon}{2M}$  st  
 $\sup_{x \in A} |f_n(x) - f| \rightarrow 0$ .

Now let  $\varepsilon > 0$ .  $n$  large enough so  $\frac{\varepsilon}{2} \cdot m(E) \leq \int_E f_n - f$ .

$$\begin{aligned} \left| \int_E f dx - \int_E f_n dx \right| &\leq \left| \int_A f_n - f dx + \int_{A^c} f_n - f \right| \\ &\leq \int_A |f_n - f| dx + \int_{A^c} |f_n - f| dx \\ &\leq m(A) \sup_{x \in A} |f_n - f| + m(A^c) 2M \\ &\leq m(E) \cdot \frac{\varepsilon}{2} m(E) + \frac{\varepsilon}{4M} \cdot 2M \leq \varepsilon. \end{aligned}$$

### 7.ii) Riemann Integrals

Def 7.10: Recall  $f: [a, b] \rightarrow \mathbb{R}$  is Riemann integrable if  
 $\lim_{n \rightarrow \infty} \sum_{n=1}^N \frac{b-a}{n} \cdot \inf_{x \in I_n} f = \lim_{n \rightarrow \infty} \sum_{n=1}^{\infty} \frac{b-a}{n} \sup_{x \in I_n} f$

where  $I_n = [a + \frac{b-a}{N}(n-1), a + \frac{b-a}{N} \cdot n]$ .

Thm 7.11: If  $f$  is Riemann integrable, then it is Lebesgue integrable, and they agree.

Proof:  $\frac{b-a}{n} \cdot \inf_{I_n} f = m(I_n) \cdot \inf_{I_n} f$  is a special case of a simple function,  
 likewise for sup □

## Lecture 8 Lebesgue Integration II: ~~functions~~ functions on $\mathbb{R}$ . [8.1]

Throughout this lecture we assume  $E = \mathbb{R}$  or has (possibly) infinite measure, and  $f \in \mathcal{M}_+(E)$  is non-negative but not (necessarily) bounded (above).

8.i) Non-Negative functions <sup>at first</sup>

Def 8.2: The support of a function  $f \in \mathcal{M}(\mathbb{R})$  is

$$\text{supp } \varphi = \{x \mid \varphi(x) \neq 0\}$$

which is measurable. It has finite support if  $m(\text{supp } \varphi) < \infty$ .

Recall if  $f \in \mathcal{M}(E)$  then it may be extended by 0  $f'' = f \chi_E$ .

Def 8.3: Define the Lebesgue Integral

$$\int_E dx : \mathcal{M}_+(E) \rightarrow \overline{\mathbb{R}}$$

by

$$\int_E f dx = \sup_h \left\{ \int_E h \mid h \in \mathcal{M}_B(E) \text{ has finite support, } 0 \leq h \leq f \right\}$$

Rem 8.4: By the way,  $f \in \mathcal{M}(E)$  is tacitly allowed to take value  $\pm\infty$  on a set of measure 0, thus e.g.  $\frac{1}{x} \in \mathcal{M}(\mathbb{R})$  while technically perhaps we should write  $\frac{1}{x} \in \mathcal{M}(\mathbb{R}, \{0\})$ .

Prop 8.5:  $\int : \mathcal{M}_+(E) \rightarrow \overline{\mathbb{R}}$  still satisfies

1) Linearity

2)  $\int_E \chi_A dx = m(A)$

3)  $f \leq g \Rightarrow \int_E f dx \leq \int_E g dx$

4)  $\left| \int_E f(x) dx \right| \leq \int_E |f(x)|$

Proof: 3), 4) same as before. 2) same for  $m(A) < \infty$ ,  $m(A) = \infty$

1)  $\int f + \int g \leq \int f + g$  as before

$\sup \rightarrow \infty$   
is obvious.

$\geq$  used inf characterization, which we no longer have.

Let  $h \leq f+g$  be bounded w/ finite support. Write

$$h = h_1 + h_2 \quad \text{w/ } h_1 = \min(h, f) \quad h_2 = h - h_1.$$

Note that  $h_1 \leq f$  by construction

[8.2]

$h_2 \leq g$  (if  $h_1 = f$ , then  $h_2 = 0 \leq g$   
if  $h_1 = f$ , then  $h_2 = h - h_1 \leq f + g - h_1 \leq g$ )

Thus

$$\begin{aligned} \int_E h dx &= \int_E h_1 dx + \int_E h_2 dx && \text{(by linearity for bounded, finite supp)} \\ &\leq \int_E f dx + \int_E g dx \end{aligned}$$

$$\Rightarrow \sup \int_E h dx = \int_E f + g dx \leq \int_E f dx + \int_E g dx$$

$$\int_E \alpha f dx = \alpha \int_E f dx \text{ still immediate}$$

Prop 8.6: If  $A, B$  disjoint

$$\int_{A \cup B} f dx = \int_A f dx + \int_B f dx$$

continues to hold

Proof:

$$\begin{aligned} \int_{A \cup B} f dx &= \sup_{\substack{h \text{ bd} \\ f \leq h}} \int_{A \cup B} h = \sup_{\substack{h \text{ bd} \\ f \leq h}} \int_A h dx + \int_B h dx \\ &\leq \int_A f dx + \int_B f dx \end{aligned}$$

but also  $\geq$  up to  $\varepsilon$  by taking  $h = h_1 \chi_A + h_2 \chi_B$  for  $h_i$  within  $\varepsilon/2$ .  $\square$

8.ii) General Functions: Now drop the assumption  $f \geq 0$ .

Def 8.7: Define  $f \in \mathcal{M}(E)$  as Lebesgue integrable if

$$\int_E |f| dx < \infty.$$

Set  $f^\pm = \max\{0, \pm f\}$ , both measurable, and  $\int_{\{f \geq 0\}} f^+ dx + \int_{\{f < 0\}} f^- dx = \int |f| dx < \infty$   
so both are integrable.

Def 8.8: If  $f \in \mathcal{M}(E)$  is Lebesgue integrable, define

$$\int_E f dx = \int_E f^+ dx - \int_E f^- dx.$$



Prop 8.7 : Properties (1)-(4) continue to hold.

2) is same since  $\chi_A \geq 0$ .

3) Follows from linearity because  $f \leq g \Rightarrow g - f \geq 0$  so  $\int g - \int f \geq 0$ .

4) Follows from 3.

1) Let  $f, g \in \mathcal{M}(\mathbb{R})$ . Write  $f = f^+ - f^-$   
 $g = g^+ - g^-$  (all are positive)  
 $f + g = h^+ - h^-$

$$h^+ - h^- = f + g = f^+ - f^- + g^+ - g^-$$

$$h^+ + f^- + g^- = f^+ + g^+ + h^-$$

By linearity for positive functions

$$\int h^+ dx + \int f^- dx + \int g^- dx = \int f^+ dx + \int g^+ dx + \int h^- dx$$

$$\begin{aligned} \int f + g &= \int h^+ - \int h^- = \int f^+ - \int f^- + \int g^+ - \int g^- \\ &= \int f + \int g. \end{aligned}$$

Scalar mult again clean.

8.iii) Some Inequalities : Assume  $f \geq 0$ .

[8.4]

Lemma 8.9 : Let  $L_{f,t} = \{x \in E \mid f(x) \geq t\}$ ,  $t \geq 0$ .

$$\chi_{L_{f,t}} = \begin{cases} 1 & x \in L_{f,t} \\ 0 & \text{else.} \end{cases}$$

Then

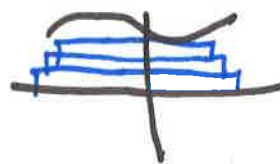
$$f(x) = \int_0^\infty \chi_{L_{f,t}}(x) dt.$$

Proof : Observe

$$\chi_{L_{f,t}}(x) = \begin{cases} 1 & \text{if } f(x) \geq t \\ 0 & \text{if } f(x) < t \end{cases} = \chi_{[0, f(x)]}(t).$$

$$\int_0^\infty \chi_{L_{f,t}}(x) dt = \int_0^\infty \chi_{[0, f(x)]}(t) dx = \int_0^{f(x)} 1 dt = f(x).$$

Corollary 8.10 (Layer Cake Representation)



$$\int_0^\infty f dx = \int_0^\infty m(\{x \mid f(x) \geq t\}) dt$$

Proof :

$$\begin{aligned} \int_0^\infty f(x) dx &= \int_0^\infty \int_0^\infty \chi_{[0, f(x)]}(t) dt dx \\ &= \int_0^\infty \int_0^\infty \chi_{L_{f,t}}(x) dt dx \\ &= \int_0^\infty \int_0^\infty \chi_{L_{f,t}}(x) dx dt \\ &= \int_0^\infty m(L_{f,t}) dt \end{aligned}$$

Pretend we can switch order, we will prove this later.

Thm 8.11 : (Markov's Inequality) If  $f \geq 0$ ,

$$m(\{x \mid f \geq a\}) \leq \frac{\int_E f dx}{a}$$

Proof :  $a \cdot m(\{x \mid f \geq a\}) \leq \int f(x) dx$  is obvious.

Thm 8.12 (Chebyshev's Inequality) If  $\int |f|^p dx < \infty$  for  $p > 0$ ,

$$m(\{x \mid f \geq a\}) \leq \frac{1}{a^p} \int_E |f|^p dx$$

Proof : Markov applied to  $|f|^p$

□

## Lecture 9 Lebesgue Integration III: the convergence theorems.

9.1

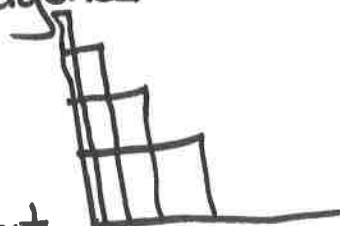
Question 9.1: Suppose  $f_n \rightarrow f$  pointwise a.e. when can we conclude

$$\lim_{n \rightarrow \infty} \int_E f_n dx = \int_E \lim_{n \rightarrow \infty} f_n dx ?$$

Ex 9.2: Two phenomena can ruin convergence

1) Mass escapes vertically

$$f_n = n \cdot \chi_{[0, 1/n]}$$

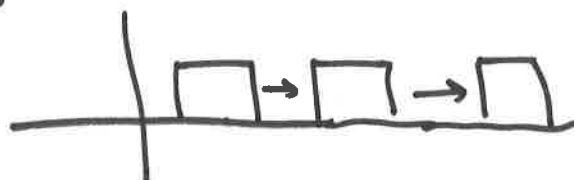


Then  $f_n \rightarrow 0$  pointwise a.e., but

$$\lim_{n \rightarrow \infty} \int_0^1 f_n dx = \lim_{n \rightarrow \infty} 1 = 1 \neq 0 = \int_0^1 f dx.$$

2) Mass escapes horizontally

$$f_n = \chi_{[n, n+1]}$$



Then  $f_n \rightarrow 0$  pointwise,

$$\lim_{n \rightarrow \infty} \int_{\mathbb{R}} f_n dx = \lim_{n \rightarrow \infty} 1 \neq 0 = \int_{\mathbb{R}} f dx.$$

Rem 9.3: Recall convergence holds if  $m(E) < \infty$  and  $|f_n| \leq M$  as it rules out both these behaviors, i.e. Ex 9.2 is "all" that can go wrong.

### 9.2 Fatou's Lemma

Lemma 9.4 (Fatou's Lemma) Suppose  $f_n \geq 0$  are non-negative measurable functions on  $E$ . If  $f_n \rightarrow f$  pointwise a.e.,

$$\int_E f \leq \liminf_n \int_E f_n.$$

Proof: Note  $f$  is measurable since it is pointwise a.e. limit,  $f \geq 0$ .  
Let  $h$  be bounded, finite support with  $0 \leq h \leq f$ .

Define

$$h_n = \min \{h, f_n\} \quad \text{on } E \supseteq \text{supp } h.$$

9.2

Then  
and

$$0 \leq h_n \leq h \leq M := \text{Sup } h, \text{ so is bounded,}$$
$$\text{supp } (h_n) \subseteq \text{supp } h.$$

By Bounded Convergence Thm (7.9),

$$\lim_{n \rightarrow \infty} \int_E h_n = \lim_{n \rightarrow \infty} \int_{\text{supp } h} h_n = \int_{\text{supp } h} h = \int_E h.$$

But  $h_n \leq f_n \quad \forall n$  so

$$\int_E h = \lim_{n \rightarrow \infty} \int_E h_n \leq \liminf_{n \rightarrow \infty} \int_E f_n$$

If  $\liminf = \infty$  trivial,  
if  $\liminf < \infty$  take  $N$  st  
 $\inf_{n \geq N} = \text{Inf} + \epsilon/2$   
increase  $n$  so that  $\lim \int h_n$  within  $\epsilon/2$

Now take  $\sup_{h \leq f, \text{supp } h \text{ finite}}$

□

Rem 9.5 : Ex 9.2 shows inequality may be strict w/ right side including the "escaped" mass.

## 9.ii Monotone Convergence :

Theorem 9.6 (Monotone Convergence) Suppose  $f_n$  is an increasing sequence of non-negative functions on  $E$ , and  $f_n \rightarrow f$  pointwise a.e. Then

$$\lim_{n \rightarrow \infty} \int_E f_n dx = \int_E \lim_{n \rightarrow \infty} f_n dx.$$

Proof : By Fatou,

$$\int_E f \leq \liminf_{n \rightarrow \infty} \int_E f_n dx$$

But

$$\leq \limsup_{n \rightarrow \infty} \int_E f_n dx \leq \limsup_{n \rightarrow \infty} \int_E f dx = \int_E f dx.$$

Therefore  $\liminf = \limsup = \int_E f dx.$

□



### 9.iii) Dominated Convergence

[9.3]

Def 9.7: A measurable function  $g$  is said to dominate a sequence  $f_n$  if  $|f_n| \leq g$  on  $E$  for all  $n$ .

Lemma 9.8 (Modulus of Integrability) Suppose  $f \geq 0$  is integrable. Then  $\forall \varepsilon > 0$ ,  $\exists \delta$  such that  $m(A) < \delta \Rightarrow \int_A f dx < \varepsilon$ .

Proof: Let  $\varepsilon > 0$ . For  $m$  sufficiently large,  $\exists f_m$  with  $|f_m| \leq M$  and

$$\int_E |f - f_m| dx < \frac{\varepsilon}{2}.$$

Now let  $A \subseteq E$  be measurable w/  $m(A) < \delta := \frac{\varepsilon}{2M}$ .

Then

$$\begin{aligned} \left| \int_A f dx \right| &\leq \int_A |f| dx \leq \int_A |f - f_m + f_m| dx \\ &\leq \int_A |f - f_m| dx + \int_A |f_m| dx \\ &\leq \int_E |f - f_m| dx + m(A) \cdot M \\ &\leq \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \quad \square \end{aligned}$$

Corollary 9.9:  $f: \mathbb{R} \rightarrow \mathbb{R}$  integrable,  
 $F(x) = \int_{-\infty}^x f(t) dt$   
is uniformly continuous.

Thm 9.10 (Lebesgue Dominated Convergence) Suppose  $f_n \rightarrow f$  pointwise a.c. with  $|f_n|, |f|$  dominated by  $g$  with  $\int_E g dx < \infty$ . Then  $\lim_{n \rightarrow \infty} \int_E f_n dx = \int_E \lim_{n \rightarrow \infty} f_n dx$

Proof: By Lemma 9.8, given  $\varepsilon > 0 \exists \delta$  such that  $A \subseteq E$  with  $m(A) < \delta \Rightarrow \int_A g dx < \varepsilon/2$ .

Also  $\exists h \leq g$  supported on  $[-m, m]$  st  $\int g - h dx < \frac{\varepsilon}{2}$ .

By Littlewood 3,  $\exists A$  with  $m(A) < \delta$  s.t.  $f_n \rightarrow f$  uniformly outside  $A$ .

Then

$$\begin{aligned}
 \left| \int_E f_n - f \, dx \right| &\leq \int_E |f_n - f| \, dx \\
 &\leq \int_{E \setminus [-m, m] \cap E} |f_n - f| \, dx + \int_{[-m, m] \cap E} |f_n - f| \, dx + \int_A |f_n - f| \, dx
 \end{aligned}$$

Note  $m$  is defined by  $g$  so ind. of  $n$ !

$$\begin{aligned}
 &\leq 2 \int_{E \setminus [-m, m] \cap E} |g| \, dx + \int_{[-m, m] \cap E} |f_n - f| \, dx + 2 \int_A |g| \, dx \\
 &\leq 2 \cdot \frac{\varepsilon}{2} + 2m \cdot \sup_{\text{uniform}} |f_n - f| + 2 \cdot \frac{\varepsilon}{2} \\
 &\leq \varepsilon + \varepsilon \quad \text{for } n \text{ large} \quad \square.
 \end{aligned}$$

#### 9. iv) Examples

Ex 9.11 : Evaluate  $\lim_{k \rightarrow \infty} \int_0^1 \frac{k[\sin(kx) + \sin(\frac{1}{k}x)]}{\tan(\frac{\pi}{2k}x) + K^{3/2}\sqrt{x}} \, dx$ .

$$\left| \frac{k[\sin(kx) + \sin(\frac{1}{k}x)]}{\tan(\frac{\pi}{2k}x) + K^{3/2}\sqrt{x}} \right| \leq \frac{2k}{\text{positive} + K^{3/2}\sqrt{x}} \leq \frac{1}{\sqrt{kx}} \leq \frac{1}{\sqrt{x}}$$

and  $\int_0^1 \frac{1}{\sqrt{x}} \, dx = 2x^{1/2} \Big|_0^1 = 2 < \infty$ . So

$$\lim_{k \rightarrow \infty} \int_0^1 \frac{k[\sin(kx) + \sin(\frac{1}{k}x)]}{\tan(\frac{\pi}{2k}x) + K^{3/2}\sqrt{x}} \, dx = \int_0^1 \lim_{k \rightarrow \infty} \frac{k[\sin(kx) + \sin(\frac{1}{k}x)]}{\tan(\frac{\pi}{2k}x) + K^{3/2}\sqrt{x}} \, dx = 0.$$

Ex 9.12 : Suppose  $f(t, x)$  is  $C^1$  a.e. and  $|f(t, x)| \leq m$ . Then

$$\frac{d}{dt} \int_{\mathbb{R}} f(t, x) \, dx = \int_{\mathbb{R}} f'(x, t) \, dx$$

Proof :  $|f|$  dominates the difference quotients  $\frac{f(t+h) - f(t)}{h}$ .

Corollary 9.13 : If  $E_1 \subseteq E_2 \subseteq \dots$  and  $\int_E |f| < \infty$ , then

$$\int_{\bigcup E_n} f = \lim_{n \rightarrow \infty} \int_{E_n} f$$

and

$$\int_{\bigcap E_n} f = \lim_{n \rightarrow \infty} \int_{E_n} f. \quad (\text{Apply LDC to } \chi_{E_n} f.)$$

## Lecture 10 Differentiation and Integration I: monotone functions 10.1

Question 10.1: Under what assumptions of  $f$  does the FTC

$$\frac{d}{dx} \int_a^x f(t) dt = f(x)$$

hold? Note we saw that it fails for  $\sin(x)$  the Cantor function.

Ex 10.2: There exists a continuous but nowhere differentiable function. Let  $a_n = 10^{-n}$ ,  $b_n = 10^{2n}$ .

Consider

$$f = \sum_{n=1}^{\infty} a_n \sin(b_n x) \quad \text{on } [0, 2\pi].$$

$f$  is a uniform limit of continuous, so continuous, but

$$f'(x) = \sum_{n=1}^{\infty} a_n b_n \sin(b_n x) = \sum_{n=1}^{\infty} 10^n \sin(b_n x).$$

would have to hold. Can show  $\frac{f(x+h) - f(x)}{h} \rightarrow \infty$  everywhere.

### 10.2) Monotone Functions

Def 10.3: A function  $f: [a, b] \rightarrow \mathbb{R}$  is said to be monotone if  $x \leq y \Rightarrow f(x) \leq f(y)$ .

Prop 10.4: Let  $f: [a, b] \rightarrow \mathbb{R}$  be monotone. Then  $f$  is continuous away from a countable (hence measure 0) set.

Proof: Suffices to assume  $[a, b]$  finite and take unions.

$$\text{Let } f^-(x_0) = \sup \{ f(x) \mid a < x < x_0 \}$$

$$f^+(x_0) = \inf \{ f(x) \mid x_0 < x < b \}.$$

Discontinuity iff  $f^- < f^+$  is strict. If so  $[f^-(x_0), f^+(x_0)] \subseteq [f(a), f(b)]$ .

$\forall n \in \mathbb{N}$ , only finitely many have size  $> \frac{1}{n}$ . So

$$\text{discontinuities} = \bigcup_{n \in \mathbb{N}} \{\text{finite}\}.$$

□



Ex 10.5 : If  $C \subseteq [a, b]$  is countable  $\exists f: [a, b] \rightarrow \mathbb{R}$  monotonic 10.2  
with discontinuities at  $C$ .

Enumerate  $C = \{q_1, q_2, \dots\}$

$$f(x) = \sum_{i: q_i \leq x} \frac{1}{2^i}$$

Thm 10.6 (Monotone Differentiability) If  $f: [a, b] \rightarrow \mathbb{R}$  is monotone it is differentiable almost everywhere.

## 10.ii) Covering Lemmas

Lemma 10.7 : Let  $K \subseteq \mathbb{R}^n$  be compact. If  $\mathcal{U} = \bigcup_{i \in I} D_i$  is a cover by balls,  $\exists i \in 1, \dots, N$  st.  
 $K \subseteq \bigcup_{i=1}^N 3D_i$

Proof : Let  $D_i$  for  $1 \dots N$  be a finite subcover ordered by decreasing radii.  $D_i = B(x_i, r_i)$   $r_1 \geq r_2 \geq \dots$ .  
Starting from  $i=1$  throw out each  $D_i$  if it is not disjoint from those already selected. Suppose  $x \in K$ . Either  $x$  is in chosen  $D_i$  or another  $D_j$ . In the latter case  $\exists i$  so  $x \in D_i \cap D_j$  w/  $i \geq j$ , but then  
 $x \in 3D_i$  

Def 10.8 : A collection  $\mathcal{U} = \{D_i\}$  of balls is a Vitali covering of  $K$  if  $\forall x, \forall \varepsilon > 0$ ,  $\exists i \in I$  such that  $x \in D_i$  and  $r_i < \varepsilon$ .

Lemma 10.9 (Vitali Covering Lemma) Let  $E \subseteq \mathbb{R}^n$  be a set of finite positive measure, and let  $\mathcal{U}$  be a Vitali covering. Then  $\exists$  finite collection  $D_i \in \mathcal{U}$  st.  
 $m(E \Delta \bigcup_{i=1}^N D_i) < \varepsilon$ .



Lemma 10.10 : Let  $\mathcal{U}$  be a Vitali covering of  $K$  compact. Then 10.3

$$\forall N \in \mathbb{N} \exists D_i \text{ disjoint so}$$

$$K \subseteq \bigcup_{i=1}^N \bar{D}_i \cup \bigcup_{i=N+1}^{\infty} 3D_i$$

Proof : Assume  $\mathcal{U} = \bigcup_{n \in \mathbb{N}} B(x_n, \frac{1}{n})$  for  $x_n$  finite, by compactness.

Proceed w/ algorithm as before on (now countable sequence).

Either  $x \in \bigcup_{i=1}^N \bar{D}_i$  or it lies in a ball thrown out, and radii  $r_i \geq r_j$ , so same logic.  $\square$

Proof (of Vitali Covering Lemma) We may choose

$$K \subseteq E \subseteq \mathcal{U} \quad \text{and } m(\mathcal{U} \setminus K) < \varepsilon.$$

$\nearrow$  closed, finite measure  $\Rightarrow$  compact  $\nwarrow$  open

By 10.10,  $K \subseteq \bigcup_{i=1}^N \bar{D}_i \cup \bigcup_{i=N+1}^{\infty} 3D_i \quad \forall N,$

By throwing out balls of large radii, may assume each  $D \in \mathcal{U}$  lies in  $\mathcal{U}$ .

Since  $D_i$  disjoint,  $\sum_{i=1}^{\infty} m(D_i) < m(\mathcal{U}) < m(E) + \varepsilon.$

Let  $N$  be large enough so that  $\sum_{i=N+1}^{\infty} 3m(D_i) < \varepsilon.$

$$m(E \setminus \bigcup_{i=1}^N D_i) \leq m(\mathcal{U} \setminus \bigcup_{i=1}^N D_i) \leq m(\mathcal{U} \setminus K) + m(K \setminus \bigcup_{i=1}^N \bar{D}_i) \leq m(\mathcal{U} \setminus K) + m(K \setminus \bigcup_{i=1}^N \bar{D}_i)$$

$$m(\bigcup_{i=1}^N D_i \setminus E) \leq \varepsilon + m(\bigcup_{i=1}^N D_i \setminus K) \leq 2\varepsilon \text{ since } \mathcal{U} \text{ is a Vitali covering.}$$

Proof (of monotone differentiability)

Suffices to assume  $[a, b]$  finite, and  $f$  continuous (bc discords are measure 0)

For  $x \in [a, b]$ , let  $I_{\delta_1, \delta_2} = [x - \delta_1, x + \delta_2].$

Set  $D_{\delta_1, \delta_2}(x) = \frac{f(x + \delta_1) - f(x - \delta_2)}{x + \delta_1 - (x - \delta_2)}.$

Clearly  $\lim_{\substack{\delta_1 \rightarrow 0 \\ \delta_2 \rightarrow 0}} D_{\delta_1, \delta_2}(x)$  exists  $\Rightarrow f'(x)$  exists by monotonicity by continuity.

Thus if  $f'$  does not exist  $\lim_{\delta, \delta_2 \rightarrow 0} D_\delta(x)$  does not exist, (10.4)  
and  $\exists$  intervals  $I, J$  so that

$$D_I(x) \neq D_J(x) \Rightarrow \exists \text{ a rational in between.}$$

Let  $E_{r,s}$  be  $\{x \mid \nexists r, s \in \mathbb{Q} \text{ w/ } D_I(x) < r < s < D_J(x)\}$   
 $\exists I, J$  arbitrarily small.

Thus if  $f'(x)$  DNE,  $x \in E_{r,s}$  for some  $r, s$ .

Claim:  $m(E_{r,s}) = 0$  (hence  $m(\cup E_{r,s}) = m\{x \mid f' \text{ DNE}\} = 0$ ).

Idea: Intuitively,  $f' > s$ , and  $f' < r$  both hold, so we show  
 $s \cdot m(A) \leq m(f(E)) \leq r m(A) \rightarrow$  since  $r \neq s$   
if  $m(A) \neq 0$ .

Let  $\varepsilon > 0$ . By Vitali's Lemma  $\exists \bigcup_{n=1}^N I_n$  disjoint st  
 $m(E_{r,s} \Delta \bigcup_{n=1}^N I_n) < \varepsilon$ .

In fact, can assume  $D_{I_n}(x) < r$ . (Take  $\mathcal{U}$  to be the covering of  $E_{r,s}$  by the intervals  $I$  of size  $\frac{1}{n}$ ).

$$\text{Thus } \sum |f(I_n)| \leq r \sum |I_n|$$

Now apply Vitali to  $E_{r,s} \cap \bigcup_{n=1}^m I_n$  using covering by  $\mathcal{J}_s$ .

$$\exists \bigcup_{m=1}^M J_m \text{ st } m(E_{r,s} \cap \bigcup_{n=1}^N I_n \Delta \bigcup_{m=1}^M J_m) < \varepsilon \text{ and } D_{J_m}(x) > s.$$

$$\Rightarrow \sum |f(J_m)| \geq s \sum |J_m|$$

$\rightarrow$  monotonicity  $\bigcup J_m \subseteq \bigcup I_n$  and disjoint

$$s m(\bigcup J_m) \leq \sum |J_m| \leq \sum |f(J_m)| \leq \sum |f(I_n)| \leq r \sum |I_n| = r m(\bigcup I_n)$$

$$|m(E_{r,s}) - m(\bigcup I_n)| \leq \varepsilon \Rightarrow \lim \varepsilon \rightarrow 0$$

$$|m(E_{r,s}) - m(\bigcup J_m)| \leq 2\varepsilon \Rightarrow s m(E_{r,s}) \leq r m(E_{r,s}) \rightarrow \text{unless } m(E_{r,s}) = 0$$

# Lecture 11 Integration and Differentiation II: absolute continuity 11.1

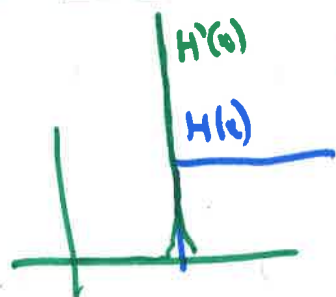
Recall if  $F: [a, b] \rightarrow \mathbb{R}$  is measurable and monotone, then  $F' = f$  exists a.e.

Ex 11.1: As before if  $C(x): [0, 1] \rightarrow \mathbb{R}$  is the Cantor function,  $C' = f$  indeed exists and is 0 a.e.  
Hence

$$0 = \int_0^1 C'(x) \neq C(1) - C(0) = 1.$$

But  $\leq$  is true.

Intuition 11.2: Consider  $H(x) = \begin{cases} 0 & x \leq 1/2 \\ 1 & x > 1/2 \end{cases}$

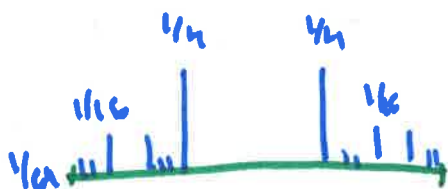


Then intuitively  $H'(x) = \delta(x - 1/2)$  is the derivative

$$\int_a^b H'(x) = \begin{cases} 1 & \text{if } a \leq 1/2 \leq b \\ 0 & \text{else.} \end{cases}$$

For the Cantor function, increase comes

from  $\sum_{C \in C} \frac{1}{4^k} \delta_{C_k}$   $k = \text{layer of Cantor set.}$



Thm 11.3: If  $f: [a, b] \rightarrow \mathbb{R}$  is monotone increasing

$$\int_a^b f'(x) dx \leq f(b) - f(a).$$

Proof: Let  $f_n(x) = \frac{f(x + 1/n) - f(x)}{1/n}$ . By  $f'$  existing a.e.  $f$  extend  $f$  by  $f(a), f(b)$  const.

$f_n(x) \rightarrow f'(x)$  pointwise a.e.

Thus

$$\int f' \leq \liminf \int f_n \leq \limsup \int f_n$$

$$\int_a^b f_n = \frac{1}{n} \int_{a+1/n}^{b+1/n} f(x) dx - \frac{1}{n} \int_a^b f(x) dx = \frac{1}{n} \int_b^{b+1/n} f(x) dx - \frac{1}{n} \int_{a+1/n}^{a+1/n} f(x) dx \leq f(b) - f(a)$$

by monotone.



## 11.ii) Absolute Continuity + Bounded Variation

11.2

Q 11.4 : What is the condition for equality? We have to rule out the  $\delta$ -function jumps. We will define

$$\begin{aligned} BV[a,b] &\supseteq AC[a,b] \\ &= \{f \text{ has bounded variation}\} = \{f \text{ absolutely continuous}\} \end{aligned}$$

Def 11.5 : A function has bounded variation if

$$\|f\|_{BV} := \sup_{P \in [a,b]} \sum_{i=0}^n |f(a_{i+1}) - f(a_i)|$$

is finite, where the sup is over all finite partitions

$$a = a_0 \leq a_1 < a_2 < \dots < a_n = b.$$

Prop 11.6 : A function  $f$  has bounded variation iff

$$f = g(x) - h(x)$$

for  $g, h$  monotone increasing.

Proof : Suppose  $f = g(x) - h(x)$ . Then

$$\begin{aligned} \|f\|_{BV} &= \sum_{k=0}^n |g(x_{k+1}) - g(x_k) + h(x_{k+1}) - h(x_k)| \\ &\leq \sum_{k=0}^n |g(x_{k+1}) - g(x_k)| + |h(x_{k+1}) - h(x_k)| \\ &\leq g(b) - g(a) + h(b) - h(a) < \infty \end{aligned}$$

Now suppose  $\|f\|_{BV} < \infty$ . Write  $y = y^+ - y^-$  where  $y^\pm \geq 0$ .  
 $y = y^+ - y^-$  for  $y^\pm \geq 0$ .

$$f_\pm(x) = \sup_P \sum_{k=0}^n [f(a_{k+1}) - f(a_k)]_\pm$$

where  $P$  partitions  $[a, x]$ .



Then  $f_{\pm}$  are monotone and bounded since  $\|f\|_{BV} < \infty$ . 11.3

Claim  $f = f_+ - f_- + f(a)$ .

Note  $f_{\pm}$  are increasing as we refine partition (by triangle inequality)  
so can find  $P$  subdivision for which both are within  $\varepsilon/2$  of sup.

Then

$$\begin{aligned} & \sum_{i=0}^{n-1} [f(a_{i+1}) - f(a_i)]_+ - [f(a_{i+1}) - f(a_i)]_- \\ &= \sum_{i=0}^{n-1} f(a_{i+1}) - f(a_i) = f(a_n) - f(a_0) = f(x) - f(a) + \varepsilon. \end{aligned}$$

Corollary 11.7: If  $f \in BV[a, b]$  then  $f'$  exists a.e.  
and  $f' \in L^1[a, b]$

Proof: apply monotone differentiability to  $g, h$ . □

Def 11.8: A function  $F: [a, b] \rightarrow \mathbb{R}$  is absolutely continuous  
if  $\forall \varepsilon > 0, \exists \delta > 0$  st. for any  $\sqcup I_n$  w/

$$\sum |I_n| < \delta, \quad \sum_{i=1}^n |f(a_i) - f(b_i)| < \varepsilon.$$

For  $I_j = (a_j, b_j)$ .

Prop 11.9: Suppose  $f: [a, b] \rightarrow \mathbb{R}$  is integrable. Then  
 $F(x) = \int_a^x f \, dt$  is absolutely continuous.

Proof:  $\forall \varepsilon > 0 \exists \delta$  s.t. if  $\mu(A) < \delta$  then  $\int_A |f| \, dx < \varepsilon$ .

(approximate by bounded)

Ex 11.10 : Absolute continuity is designed to precisely rule out the cantor function. If  $f$  has discontinuity jumps

11.4



then  $\bigcup I_n$  can be taken  $< \varepsilon$  while  $|f(b) - f(a)| > \text{const}$

Eg.  $2^n$  intervals of length  $3^{-n}$  so that

$$\sum |f(I_n)| = 1 \text{ but } |I_n| = \left(\frac{2}{3}\right)^n \rightarrow 0.$$

Prop 11.11 : Recall Lipschitz means  $\exists M$  st

$$|f(b) - f(a)| \leq M|b - a|,$$

One has  $\swarrow$  Lipschitz

$$C^0[a, b] \subset AC[a, b] \subset BV[a, b]$$

Proof : Let  $\varepsilon > 0$ , and  $\delta$  be given by A.C. Then

$$\|f\|_{BV} \leq \varepsilon \frac{(b-a)}{\delta}$$

For Lipschitz, take  $\delta = \frac{\varepsilon}{M}$

Corollary 11.12 : If  $f \in AC[a, b]$  then  $f'$  exists a.e. D

### 11.iii) The FTOC

Theorem 11.13 (FTOC with absolute continuity)

1) Suppose  $f \in L^1[a, b]$ . Then  $F = \int_a^x f \, dt$  is A.C. and

$$\frac{d}{dx} F = f \quad (\text{a.e. hence in } L^1).$$

2) Suppose  $f \in AC[a, b]$ . Then

$$\int_a^x f'(t) \, dt = f(x) - f(a).$$

Proof (next time)

# Lecture 12 Integration and Differentiation III: generalized FTC.

Recall we wish to show

Thm 11.13 (generalized FTC)

1) If  $f \in L^1[a, b]$ , then  $F(x) = \int_a^x f \, dt \in AC[a, b]$

and  $F'(x) = f$  a.e.

2) If  $f \in AC[a, b]$ , then  $f'$  exists a.e. and

$$\int_a^x f'(t) \, dt = f(x) - f(a).$$

Rem 12.1: Phrased a different way,

$$\begin{array}{c} D = \frac{d}{dx} \\ AC[a, b] \xrightarrow{\quad} L^1[a, b] \\ \uparrow I \\ \mathbb{R} \end{array}$$

$I = \int_a^x \, dt$

is an isomorphism

## 12.1) The lemmas

Intuition 12.1: We will show  $D \circ I$  and  $I \circ D$  give the identity.

$\Rightarrow D \circ I$  first prove for Lipschitz, then take limits  
 $I \circ D$  shown to be injective.

Lemma 12.3: If  $I(f) = 0$  for  $f \in L^1[a, b]$ , then  $f = 0$ , i.e.  
 $I$  is injective.

Proof: If  $f \neq 0$  in  $L^1$ , then  $\exists E \subseteq [a, b]$  s.t.  $f \neq 0$  on  $E$   
 $\wedge m(E) > 0$ . Take  $F \subseteq E$  closed of positive measure.

$F$  is a union of intervals, take one,  $J$ .  $\int_J f \neq 0$  so

$J = [c, d]$  cannot have  $\int_c^d f \, dt = \int_a^d f \, dt = 0$  □



Lemma 12.4 (Lipschitz Case) Suppose  $f \in L^\infty[a, b]$ . Then  $I(f)$  is Lipschitz, and  $D \circ I(f) = f$  a.e. 12.2

Rem 12.5: This shows the theorem on the subspaces

$$C^0[a, b] \simeq L^\infty[a, b]$$

$$\cap AC[a, b] \simeq \cap L^1[a, b]$$

Proof: Let  $|f| \leq M = \|f\|_{L^\infty}$ . Then  $F = I(f)$  satisfies

$$|F(x+h) - F(x)| = \left| \int_x^{x+h} f(t) dt \right| \leq M \cdot h$$

is Lipschitz. Let  $F_n = \frac{1}{n} [F(x + 1/n) - F(x)] \leq M$

and  $F_n \rightarrow F'$  pointwise a.e. so by dominated convergence.

$$\begin{aligned} \int_a^x F'(t) dt &= \lim_{n \rightarrow \infty} \int_a^x n [F(t + 1/n) - F(t)] dt \\ &= \lim_{n \rightarrow \infty} \int_{a+1/n}^{x+1/n} n F(t) dt - \int_a^{x+1/n} n F(t) dt \\ &= \lim_{n \rightarrow \infty} \int_x^{x+1/n} n F(t) dt - \int_a^{x+1/n} n F(t) dt = F(x) - F(a) \end{aligned}$$

Thus  $\int_a^x F'(t) - f(t) dt = 0 \quad \forall x$ , so  $F' = f$  a.e. by 12.3.  $= \int_a^x f(t) dt$

Lemma 12.6: Let  $X, Y$  be sets and  $X \xrightarrow{D} Y$  maps such that  $\xrightarrow{D} Y \xrightarrow{I} X$  If  $D$  is injective, then  $I \circ D = \text{id}_X$ .

Proof:  $D(I(D)x) = Dx$ , hence  $I \circ D x = x$  by injectivity.



## 12.ii) The proof

12.3

Proof  $I \circ D = \text{id}$  (general case) Suppose  $f \in L^1[a, b]$

Write  $f = f^+ - f^-$ , so enough to assume  $f \geq 0$ .

Let  $f_n = \min(n, f)$ , hence  $f_n \rightarrow f$  pointwise and is monotone.

Also  $F_n = I(f_n)$  is monotone and  $F_n \rightarrow F$  pointwise.

By previous lemma,

$$F_n'(x) = f_n \quad \text{a.e.}$$

Therefore

$$\begin{aligned} \int_a^x F' dt &\geq \int_a^x f_n dt = F_n(x) - F_n(a) \\ \int_a^x F' dt &\geq \lim_{n \rightarrow \infty} F_n(x) - F_n(a) \\ &= F(x) - F(a) = \int_a^x f dt = I(f). \end{aligned}$$

$$\int_a^x F' dt \leq F(x) - F(a) = I(f) \quad \text{b.c. } F \text{ is monotone.}$$

Hence  $I(F') = I(f)$ , and  $F' = f$  a.e. by injectivity.

Proof that D is injective). Suppose  $F \in AC[a, b]$  and  $F'(x) = 0$  a.e. then  $F$  is constant.

Let  $\varepsilon > 0$ ,  $\delta$  as in def of a.e., may assume  $\delta < \varepsilon$ .

Let  $B = \left\{ U \subseteq [a, b] \text{ open st } \left| \frac{F(u)}{|u|} \right| < \frac{\delta}{n} \right\}$  for all  $n$ .

Since  $F'(x) = 0$ , and this is open, it is a covering of the set where  $F'$  exists.

Then let  $I_1, \dots, I_n$  be intervals st.  $m([a, b] - I) < \delta$ .

$$\text{Then } |F(x) - F(a)| \leq \varepsilon + \sum_{I_n} |F(I_n)|$$

$$\leq \varepsilon + \sum \delta |I_n| \leq \varepsilon (1 + (b-a)) \rightarrow 0. \quad \square$$

## 12.4 ii) : Rectifiable Curves

Let  $\gamma = (x(t), y(t))$  be such that

$$\|\gamma\|_{BV} < \infty.$$

Def 12.7 : A curve satisfying the above is said to be rectifiable.

Its length is

$$L(\gamma) = \int_a^b |x'(t)^2 + y'(t)^2|^{1/2} dt < \infty.$$

Def 12.8 : Let  $K_\delta = \{(x, y) \mid \inf_{t \in [a, b]} d((x, y), \gamma(t)) < \delta\}$ .

The 1-dim Minkowski content of  $\gamma$  is  $\lim_{\delta \rightarrow 0} \frac{m(K_\delta)}{2\delta}.$

Ex 12.9 : The Koch snowflake is not rectifiable.

• A space filling curve has  $\infty$  Minkowski content.

Rem 12.10 : Minkowski content leads to notion of "Hausdorff measure".

This leads to "Geometric measure theory" and is key for studying singular limiting behavior of PDEs/minimal surfaces.

Thm 12.11 (Isoperimetric Inequality)

Let  $\Omega \subset \mathbb{R}^2$  be measurable, with  $\partial\Omega$  rectifiable.

Then  $4\pi m(\Omega) \leq L(\partial\Omega)^2,$

with equality iff it's a ~~disk~~ disk.

Let  $X$  be a metric space.

Def 13.1 : An extension measure on  $X$  is a function  $\mu^*: 2^X \rightarrow [0, \infty]$  with

- 1)  $\mu^*(\emptyset) = 0$
- 2)  $E_1 \subseteq E_2 \Rightarrow \mu^*(E_1) \leq \mu^*(E_2)$
- 3)  $\mu^*\left(\bigcup_{j=1}^{\infty} E_j\right) \leq \sum_{j=1}^{\infty} \mu^*(E_j)$ .

\* A set  $E \subseteq X$  is said to be measurable if  $\mu^*(A) = \mu^*(A \cap E) + \mu^*(A \cap E^c)$  for all  $A \subseteq X$ .

Prop 13.2 :  $\mathcal{M}(X) = \{E \subseteq X \mid E \text{ is measurable}\}$  forms a  $\sigma$ -algebra, and  $\mu^*|_{\mathcal{M}(X)} : \mathcal{M}(X) \rightarrow \bar{\mathbb{R}}^{\geq 0}$  is a measure.

Proof : Nothing about previous proof used  $\mathbb{R}$ !

Def 13.3 : A pre-measure on an algebra  $\mathcal{A} \subseteq 2^X$  (closed under finite  $\cup, \cap$ ) is a function  $\mu_0 : \mathcal{A} \rightarrow [0, \infty]$  such that

- 1)  $\mu_0(\emptyset) = 0$
- 2)  $\mu_0\left(\bigsqcup_{k=1}^{\infty} E_k\right) = \sum_{k=1}^{\infty} \mu_0(E_k)$

the outer measure completing a premeasure is

$$\mu^*(E) = \inf \left\{ \sum_{j=1}^{\infty} \mu_0(E_j) \mid E \subseteq \bigcup E_j, E_j \in \mathcal{A} \right\}$$

Prop 13.4 :  $\mu^*$  as above is an extension measure, and  $\mathcal{A} \subseteq \mathcal{M}(X)$  in the induced measure.

Thm 13.5 : If  $\mathcal{A} \subseteq 2^X$  is a (covering) algebra <sup>has countable cover of  $\mu_0$  sets.</sup> and  $\mu_0$  a pre-measure, then  $\exists!$  measure  $\mu$  extending  $\mu_0$ .

Proof : Existence immediate from 13.2 + 13.4. For uniqueness Suppose  $\nu, \mu$  are two measures extending  $\mu_0$ .



$$\text{If } F \subset \bigcup E_j \text{ w/ } E_j \in \mathcal{A},$$

$$\nu(F) \leq \sum_{j=1}^{\infty} \nu(E_j) = \sum_{j=1}^{\infty} \mu_0(E_j)$$

$$\nu(F) \leq \inf ( \quad ) = \mu(F).$$

and vice-versa. □

Proof of 13.4 :  $\mu$  is clearly an outer measure

b/c

$$1) \mu(\emptyset) = 0, \quad 2) E_1 \subset E_2 \text{ means } \mu_0(E_1 \cup E_2 \setminus E_1) = \mu(E_1) + \underbrace{\mu(E_2 \setminus E_1)}_{\geq 0}$$

b/c strictly bigger set.

$$3) \mu^*\left(\bigcup_{j=1}^{\infty} E_j\right) = \mu^*\left(\bigcup_{j=1}^{\infty} (E_j \setminus \bigcup_{k=1}^{j-1} E_k)\right) \leq \sum_{j=1}^{\infty} \mu(E_j \setminus \bigcup_{k=1}^{j-1} E_k) \leq \sum_{j=1}^{\infty} \mu(E_j).$$

On  $\mathcal{A}$ ,  $\mu(E) \leq \mu_0(E)$  since it covers itself

and if  $E \subset \bigcup_{j=1}^{\infty} E_j$  then

$$\mu_0(E) = \sum (E_j \setminus \bigcup_{k=1}^{j-1} E_k) \leq \sum \mu_0(E_k) \text{ and take inf.}$$

$\mathcal{A} \subset \mathcal{M}(\mathbb{R})$  is similar. □

### 13.ii) Lebesgue Measure on $\mathbb{R}^n$

Def 13.6 : Given  $(X, \Sigma, \mu)$ ,  $(Y, \mathcal{M}, \nu)$

the product pre-measure is the measure

$$\lambda(A \times B) = \mu(A) \nu(B)$$

on the algebra of products of measurable sets.

Claim 13.7 : this is a pre-measure.

$$\bullet \lambda(\emptyset) = 0 \quad \checkmark$$

$\bullet$  If  $A \times B = \bigsqcup A_j \times B_j$  all products of measurable,  
then  $(x, y)$  belongs to a single  ~~$A_j \times B_j$~~   $A \times B_j$  for  $B_j$  depends  
on  $y$ .  
hence  $B = \bigsqcup B_j$ .



Then  $\chi_A(x_1)\mu_2(B) = \sum_{j=1}^{\infty} \chi_{A_j}(x_1)\mu_2(B_j)$  for  $x_1 \in A_j$ .

13.3

Integrating

$$\begin{aligned} \mu(A)\mu(B) &= \lim \int \chi_A(x_1)\mu_2(B) dx_1 = \lim_{N \rightarrow \infty} \int \sum_{j=1}^N \chi_{A_j}(x_1)\mu_2(B_j) dx_1 \\ &= \lim_{N \rightarrow \infty} \sum_{j=1}^N \mu(A_j)\mu(B_j). \quad \square \end{aligned}$$

Def 13.8: The product measure on  $X \times Y$  is the one induced by the product pre-measure.

Thm 13.9: The product of Lebesgue measures on  $\mathbb{R}^k \times \mathbb{R}^m$  is the Lebesgue measure on  $\mathbb{R}^{m+k}$ .

Proof:  $k=m=1$ . Suffices to show that  $\lambda = m^*$  product = Lebesgue outer

$A \subset \mathbb{Q}; B \subset \mathbb{Q}$  on  $A \times B$  measurable.  
 $\Rightarrow m(A \times B) \leq \sum_{Q_j, P_k} m(Q_j \times P_k) = \sum_{Q_j, P_k} m(Q_j)m(P_k)$  they agree on cubes!  
 $\leq \sum m(Q)m(P) \leq m(A)m(B) + \varepsilon$ .

$\Leftarrow$  If  $Q_k$  cover  $A \times B$ , then  $Q_k = P_j \times M_i$ . Therefore

if  $(x, y) \in A \times B$ ,  $x \in P_j$ ,  $y \in M_i$  same  $i$  so  
 $A \subset \cup P_j$   $B \subset \cup M_i$

$$m(A)m(B) \leq \sum_{j,i} m(P_j)m(M_i) = \sum m(Q_k) \leq m(A \times B) + \varepsilon. \quad \square$$

### 13.iii) Fubini's Theorem

Thm 13.10 (Fubini I) Suppose that  $\int |f| dV < \infty$  on  $\mathbb{R}^m \times \mathbb{R}^n$  where  $dV$  is the product Lebesgue measure. Then for a.e.  $x \in \mathbb{R}^m, y \in \mathbb{R}^n$   
 •  $f(x, \cdot)$  is integrable on  $\mathbb{R}^n$ ,  $f(\cdot, y)$  on  $\mathbb{R}^m$ ; and

$$\int_{\mathbb{R}^{m+n}} f dV = \int_{\mathbb{R}^m} \left( \int_{\mathbb{R}^n} f(x, y) dy \right) dx = \int_{\mathbb{R}^n} \left( \int_{\mathbb{R}^m} f(x, y) dx \right) dy$$

Rem 13.11: Requires verifying  $\int |f| dV < \infty$ .

13.4

Thm 13.11 (Fubini II / Tonelli)

Suppose  $|f| \geq 0$  on  $\mathbb{R}^{m+n}$  is measurable. Then  $f(x, -)$ ,  $f(-, y)$  are ~~integrable~~ measurable for a.e.  $x, y$  resp and

$$\int_{\mathbb{R}^{m+n}} |f| dV = \int_{\mathbb{R}^n} \left( \int_{\mathbb{R}^m} |f| dy \right) dx = \int_{\mathbb{R}^n} \left( \int_{\mathbb{R}^m} f(x, y) dx \right) dy$$

In particular, the hypotheses of Fubini's thm hold.

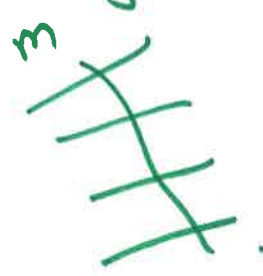
Proof: Stein-Shakarchi pg 76.

Corollary 13.12: If  $E \subseteq \mathbb{R}^{m+n}$ , then  $E_x \cap \{x \times \mathbb{R}^n\}$  is measurable for almost every  $x$ , and

$$m(E) = \int_{\mathbb{R}^m} m(E_x) dx = \int_{\mathbb{R}^n} m(E_y) dy.$$

Proof: Tonelli on  $\chi_E$ .

Corollary 13.13 (informal) If  $M \subseteq \mathbb{R}^n$  is a submanifold, w/  $\text{codim } M \geq 1$ , then  $M$  has measure 0.

Proof: After a change of variable,  $\text{Vol}(M) = \int_{\substack{\mathbb{R}^k \times \{0\} \\ \subseteq \mathbb{R}^n}} |\det \psi| dx dy$   

$$= \int_{\mathbb{R}^k} \chi_M |\det \psi| dx dy$$

↓

$$= 0 \text{ for a.e. } y \in \mathbb{R}^{n-k}.$$

0

Lecture 14 Radon-Nikodym Theorem

Let  $(X, \mathcal{M}, \mu), (X, \mathcal{M}, \nu)$  be two different measures on the same space.

Def 14.1:  $\nu$  is said to be absolutely continuous with respect to  $\mu$  if

$$E \in \mathcal{M}(X), \mu(E) = 0 \Rightarrow \nu(E) = 0.$$

Ex 14.2: Let  $f \geq 0$  be continuous, bounded.

$$\text{Set } \nu(E) = \int_E f \, d\mu.$$

Clearly  $\mu(E) = 0 \Rightarrow \nu(E) = 0$ .

Same holds for  $f$  measurable (choose bounded approximation).

Lemma 14.3:  $\nu$  is a.c. with respect to  $\mu$  iff

$$\forall \varepsilon > 0, \exists \delta \text{ such that}$$

$$\mu(E) < \delta \Rightarrow \nu(E) < \varepsilon.$$

Proof:  $\Rightarrow$  a.c. is obvious.

$\Leftarrow$  Suppose not. Then  $\exists E_n$  w/  $\mu(E_n) \leq \frac{1}{2^n}$  but  $\nu(E_n) \geq \varepsilon_0$ .

Let  $A_n = \bigcup_{k=n}^{\infty} E_k$ , so  $A_1 \supseteq A_2 \supseteq \dots$

Then  $\mu(A_n) \leq \frac{1}{2^{n-1}}$   $\nu(A_n) \geq \varepsilon_0$

Set  $A = \bigcap_{n=1}^{\infty} A_n$ . Then  $\mu(A) \leq \mu(A_n) \forall n$   
 $= 0.$

But by cont. of measure  $\nu(A) = \lim \nu(A_n) \geq \varepsilon_0 \rightarrow \Leftarrow$ .



Def 14.4 : Two measures are said to be mutually singular if

[14.2]

$\exists A, B$  disjoint s.t.

$$\nu(E) = \nu(A \cap E) \quad \mu(E) = \mu(B \cap E)$$

for all  $E \in \mathcal{M}$

Fact 14.5.  $L^2(\mathbb{R}, \mu)$  is a complete space in any measure.

Thm 14.6 (Radon-Nikodym)

Suppose  $\mu$  is ( $\sigma$ -finite) and  $\nu$  is absolutely continuous wrt  $\mu$ . Then  $\exists f \in L^1_{loc}(\mathbb{R}^n; \mu)$  such that

$$\nu(E) = \int_E f \, d\mu.$$

Rem 14.7 :  $f \in L^1_{loc}(X; \mu)$  means  $f \in L^1(E; \mu)$  for any set of finite measure.

Ex 14.8 : If  $\nu = 2\mu$ , then clearly  $f = 2$  is not in  $L^1(\mathbb{R}^n)$ .

Def 14.9 : On  $\mathbb{R}$ ,  $\nu(E) = \int_E g \, dx$  with  $g \in C^1$ , then

~~$\nu(E) = \int_E g \, dx$~~  Set  $G(x) = \nu[0, x]$ .

Then  $G(x) = \int_0^x g \, dx$  so  $G' = g$  by FTC.

$f$  is in general called the Radon-Nikodym derivative denoted  $\frac{d\nu}{d\mu}$ .



Proof: First assume  $X$  is a cube! Finite measures for both! [14.3]  
 Let  $\nu = \mu + \nu$ . Let  $\varphi \in L^2(\mathbb{R}^n; \nu)^*$  be given by

$$\varphi(f) = \int_{\mathbb{R}^n} f d\nu.$$

Clearly  $\|\varphi(f)\|_{\mu} \leq \int_{\mathbb{R}^n} f d\nu \leq \|f\|_{L^2(\nu)}(X)$  so bounded.

By Riesz rep,  $\exists g$  st.

$$\int_{\mathbb{R}^n} f d\nu = \int_{\mathbb{R}^n} f g d\mu.$$

Claim:  $0 \leq g < 1$  a.e. w.r.t  $\mu$ .

First, take  $f = \chi_E$  for some  $E$  so,

$$\int_{\mathbb{R}^n} g d\mu = \int_{\mathbb{R}^n} \chi_E d\nu = \nu(E) < \mu(E) + \nu(E).$$

< because if  $\exists E$  w/  $\mu(E) > 0$ , then equality implies  $\mu(E) = 0$   
 $\rightarrow$  absolute cont.  $\square$

Now take  $f = \chi_E (1 + g + \dots + g^n)$ . Since

$$\int_{\mathbb{R}^n} f d\nu = \int_{\mathbb{R}^n} f g (d\nu + d\mu)$$

$$\int_{\mathbb{R}^n} f (1 - g) d\nu = \int_{\mathbb{R}^n} f g d\mu$$

then

$$\int_{\mathbb{R}^n} f (1 - g) d\nu = \int_{\mathbb{R}^n} \chi_E \frac{1 - g^{n+1}}{1 - g} (1 - g) d\nu = \int_{\mathbb{R}^n} \chi_E (1 - g^{n+1}) d\nu = \int_{\mathbb{R}^n} g (1 + g + \dots + g^n) d\mu$$

Now

$1 - g^{n+1} \rightarrow 1$  pointwise, D.C.T.  $\nu(E)$

$g + g^2 + \dots \rightarrow \frac{g}{1 - g}$

$\Rightarrow$

$$\int_{\mathbb{R}^n} \chi_E d\nu = \int_{\mathbb{R}^n} f d\mu \quad f = \frac{g}{1 - g}.$$

For general case, cover

$$X = \bigcup E_j$$

with  $\mu(E_j) < \infty$ ,  $\nu(E_j) < \infty$  disjoint. Take  $f = \sum f_j$ .  $\square$ .

### Thm 14.10 (Lebesgue Decomposition)

Let  $(X, \mathcal{M}, \mu) \subset (\mathbb{R}^n, \mathcal{M}, m)$  be a  $\sigma$ -finite measure space. Let  $\nu$  on  $(X, \mathcal{M}, \nu)$  be another  $\sigma$ -finite measure. Then,  $\exists!$  decomposition

$$\nu = \nu_a + \nu_s$$

st.  $\nu_a$  is absolutely continuous wrt  $\mu$ ,  $\nu_s$  is mutually sing.

Proof: In above proof, "atomic" part  $\nu_s$  arises from  $\{g=1\}$ . Take  $B = \{x \in X \mid g(x)=1\}$ .

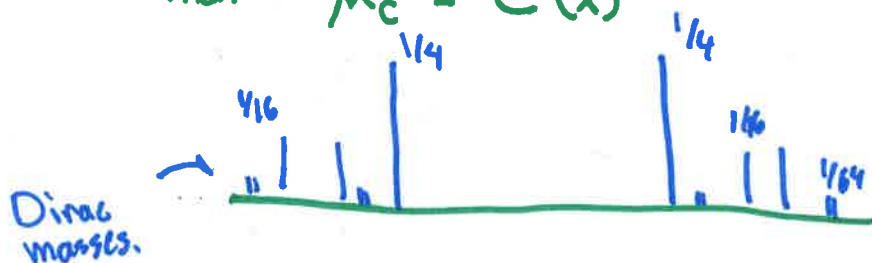
Then  $\nu_s(E) = \nu(E \cap B)$ .

$\mu(E) \in \nu(E \cap B^c)$  mutually sing.

Ex 14.11: Let  $\mu_c$  be the measure

$$\mu_c([0, x]) = C(x) \text{ the cantor function}$$

Then  $\mu_c = "C'(x)"$



$$x \in C$$

$$x = x_0 x_1 x_2 \dots$$

$\mu_c$  is mutually sing. w/ Lebesgue b/c  $m(C) = 0$ .

# Lecture 15 General Measures III: Interlude on probability

15.1

Def 15.1 : A probability space  $(X, \mathcal{F}, P)$  is a measure space such that  $m(X) = P(X) = 1$ .  $P$  is a probability measure.

Ex 15.2 : Any discrete space  $S$  is a probability space

where  $\mathcal{F} = 2^S$ , and  $P(x_i) = p_i$  s.t.  $\sum p_i = 1$ .

c.g. a 52-card deck w/  $p_i = \frac{1}{52}$ .

Ex 15.3 : For  $X = \mathbb{R}$ , the normal distribution

$$P = \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx$$

is a probability space for  $\mathcal{M} = \mathcal{F}$  the Borel measurable sets.

$$P(E) = \int_E \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx.$$

More generally  $X = \mathbb{R}^n$ ,  $\mu \in \mathbb{R}^n$ ,  $\Sigma$  positive def  $n \times n$

$$P = \frac{1}{(2\pi)^{n/2}} \frac{1}{\det \Sigma} e^{-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)}.$$

## 15.1) Terminology Dictionary

### Probability theory

Def 15.4 : The following standard terminologies are equivalent

Measure theory  $(X, \Sigma, m)$

Probability theory  $(\Omega, \mathcal{F}, P)$

$x \in X$  point  $\Rightarrow$

$x \sim P$  a sample  
 $x \in \Omega$  ~~an event~~

$E \subseteq X$  measurable  $\Rightarrow$

$E \subseteq \Omega$  an event

$m(E)$  measure  $\Rightarrow$

$P(E)$  probability of  $E$ .

$f: X \rightarrow \mathbb{R}$  measurable function  $\Rightarrow$

$X: \Omega \rightarrow \mathbb{R}$  random-variable



Ex 15.5: A random variable is an assignment of a number to each outcome. Eg  $\Omega = \{P, N, D, Q\}$  coins,  
 $V: \Omega \rightarrow \mathbb{R}$  value  
 $\Omega = \text{cards}$   
 $R: \Omega \rightarrow \mathbb{R}$  value ( $J=11, Q=12 \dots$ )

Note  $X_E$  is a R.V.  $Y_E$  measurable.

Def 15.6: The expectation value or average of a random variable is  

$$IE(f) = \int_{\Omega} f dP$$

eg.  $IE(f) = \sum_{x_i \in \Omega} f(x_i) P_i$      $IE(f) = \int_{\mathbb{R}} f(x) \frac{e^{-x^2/2}}{\sqrt{2\pi}} dx$ .

Lemma 15.7: If  $(X, \Sigma, \mu)$  is a ~~probability~~ <sup>measure</sup> space,  
 $Y \in X$  measurable, then  
 $(Y, \{Y \cap E \mid E \in \Sigma\}, \mu(E \cap Y))$  is a measure space.

Proof: Obvious.

Def 15.8: If  $(\Omega, \mathcal{F}, P)$  is a probability space,  $E \in \mathcal{F}$  and event.  
 $(E, \mathcal{F} \cap E, \tilde{P} = \frac{P(- \cap E)}{P(E)})$  is a probability space  
 called the Conditional probability space.

$P(A | E) = \tilde{P}(A) = \frac{P(A \cap E)}{P(E)}$     "the probability of A given E"  
 the conditional probability

Corollary 15.9 (Bayes Rule)  $P(A) P(A|E) = P(E \cap A) \cdot P(A)$ .

Proof: Both are definitionally  $P(A \cap E)$ .

Rem 15.10: Radon-Nikodym theorem is used in defining conditional expectations.



## 15.ii) Some Inequalities

15.3

Thm 15.11 : The following hold

1) Suppose  $X: \Omega \rightarrow \mathbb{R}$  is a positive R.V.

$$P(X \geq a) \leq \frac{E(X)}{a} \quad (\text{Markov Inequality})$$

2)  $X$  any R.V.

$$P(|X - E(X)| > a) \leq \frac{\text{Var}(X)}{a^2} \quad (\text{Chebyshev's Inequality})$$

$$\text{Var} = E(|X - E(X)|^2).$$

Proof : this is just translating notation.

## 15.iii) Large numbers and Central Limits

"The average of larger numbers of samples converges to the expectation value."

Def 15.12 : Let  $\mathbb{I}: (X, \Sigma, \mu) \rightarrow (Y, \Sigma', -)$  a measurable mapping, the pushforward measure is defined by

$$\mathbb{I}_* \mu (B) = \mu(\mathbb{I}^{-1}(B)). \in \Sigma'.$$

! Pushforward of R.V. is a measure containing Borel sets.

Def 15.13 : Two random variables  $X_1, X_2$  are said to be Identically distributed if

$$(X_1)_* P = (X_2)_* P. \text{ on } \mathbb{R}.$$

Def 15.14 : Two random variables  $X, Y$  are said to be independent if

$$P(X \leq a) \cdot P(Y \leq b) = P(X \leq a \text{ and } Y \leq b) \text{ on } \Omega \times \Omega.$$

i.e. those measurable functions on  $\Omega \times \Omega$  of the form  $f(x)g(y)$ ,

Thm 15.15 (Law of Large Numbers) Suppose that  $X_1, X_2, \dots$  are a sequence of random variables that are identically distributed, and independent, w/

$$E(X_i) = \mu \quad \forall i.$$

Then for  $\bar{X} = \frac{1}{N} \sum_{i=1}^N X_i$

1) (Weak Law)

$$\bar{X}_N \longrightarrow \mu \quad (\text{the constant function})$$

in probability (in measure)

2) (Strong Law)

$$\bar{X}_N \longrightarrow \mu$$

almost surely (pointwise a.e.).

Proof: Weak law assuming  $\text{Var}(X_i) = \sigma^2 \quad \forall i$ .

$$\begin{aligned} \text{Var}(\bar{X}_N) &= \text{Var}\left(\frac{1}{n}(X_1 + \dots + X_N)\right) \quad \checkmark \text{ Ind.} \\ &= \frac{1}{n^2} \text{Var}(X_1 + \dots + X_N) = \frac{1}{n^2} \sum \text{Var}(X_i) = \frac{\sigma^2}{n} \end{aligned}$$

Then

$$P(|\bar{X}_N - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{n\varepsilon^2} \quad \text{by Chebyshev.}$$

$$\Rightarrow P(|\bar{X}_N - \mu| < \varepsilon) = 1 - \frac{\sigma^2}{n\varepsilon^2} \xrightarrow{n \rightarrow \infty} 1 \quad \forall \text{ fixed } \varepsilon. \quad \square$$

Thm 15.16 (Central Limit Theorem) Suppose  $X_i$  are independent, un $\mathbb{R}^d$  identically distributed,  $E[X_i] = \mu$  and  $\text{Var}(X_i) = \sigma^2 < \infty$ .

Then

$$\sqrt{N}(\bar{X}_N - \mu) \longrightarrow \frac{1}{\sqrt{2\pi\sigma^2}} e^{-x^2/2\sigma^2} = N(0, \sigma^2)$$

in distribution  $(P(\cdot \leq a) \longrightarrow P(N \leq a) \quad \forall a \in \mathbb{R}).$

Rem 15.17: By the way,  $P(Y \leq a) = \int_{-\infty}^a g \, dx$  where  $g$  is the R-N density wrt Lebesgue measure.

## Lecture 16 Banach Spaces I: $L^p$ , and the fundamental inequalities. 16.1

Ex 16.1 : Consider the equivalence relation on  $E$  measurable,  
 $f \sim g$  if  $f = g$  a.e.  $\Leftrightarrow \int_E |f - g| dx = 0$ .

Def 16.2 : Let  $(L^1(E), \| \cdot \|_1)$  denote the space of equivalence classes with the metric

$$d(f, g) = \|f - g\|_1 := \int_E |f - g| dx$$

Lemma 16.3 :  $L^1(E)$  is a metric space.

Proof :  $\|f - g\|_1 = 0$  iff  $f \sim g$  so  $f = g$  in  $L^1$  by definition.

•  $\|f - g\|_1 = \|g - f\|_1$  is clear.

•  $\forall h$ , by triangle inequality on  $\mathbb{R}$ ,

$$\|f - g\|_1 = \int_E |f - g| \leq \int_E |f - h| + |h - g| \leq \int_E |f - h| + \int_E |h - g| \leq \|f - h\|_1 + \|h - g\|_1$$

Prop 16.4 :  $L^1(E)$  is complete, i.e., if  $f_n \in L^1(E)$  is Cauchy, then  $f_n \rightarrow f$  in  $L^1(E)$  and  $\|f_n - f\|_1 \rightarrow 0$ .

Proof : Note that if  $|g_n| \geq 0$ ,  $\sum_{n=1}^{\infty} \int |g_n| < \infty$  then monotone convergence implies (apply to  $G_N = \sum_{n=1}^N |g_n|$ )

$$\sum_{n=1}^{\infty} \int |g_n| = \lim_{N \rightarrow \infty} \int G_N = \int \sum_{n=1}^{\infty} |g_n|. \quad \text{In particular, } \sum_{n=1}^{\infty} |g_n| \text{ is integrable.}$$

Now let  $f_{n_k}$  be a subsequence so  $\|f_{n_k} - f_{n_{k+1}}\|_1 \leq \frac{1}{2^k}$ .

$$f = f_{n_1} + \sum_{k=1}^{\infty} f_{n_{k+1}} - f_{n_k}$$

$$g = |f_{n_1}| + \sum_{k=1}^{\infty} |f_{n_{k+1}} - f_{n_k}|$$

By the above,  $\int g < \infty$ , since  $\sum_{k=1}^{\infty} \frac{1}{2^k} < \infty$ , so  $f \in L^1$  integrable. Then claim.  
 $f_{n_k} \rightarrow f$  a.e.

$$m(|f_{n_k} - f_{n_k}| > \frac{1}{\sqrt{2^k}}) \leq \frac{1}{\sqrt{2^k}} \int |f_{n_{k+1}} - f_{n_k}| \leq \frac{1}{\sqrt{2^k}} \text{ by Markov.}$$

so  $\sum m(|f_{n_{k+1}} - f_{n_k}|) < \infty \Rightarrow m(x | x \text{ in infinitely many}) = 0$   
 $m(x | f_{n_k} \text{ doesn't converge}) = 0$   
 by Borel - Cantelli.

Then  $f_{n_k} \rightarrow f$  in  $L^1$  by dominated convergence. Now choose  $N$  large  
 $\|f_n - f\| \leq \|f_{n_k} - f_n\| + \|f_{n_k} - f\| < \frac{\epsilon}{2} + \frac{\epsilon}{2} \quad \blacksquare$

### 16.ii) Banach Spaces

Def 16.5: A norm on a vector space is a function  
 $\|\cdot\|: X \rightarrow \mathbb{R}^{\geq 0}$   
 such that.  $(X, \|\cdot\|)$  is a normed vector space.

- i)  $\|\alpha x\| = |\alpha| \|x\|$
  - ii)  $\|x\| = 0 \iff x = 0$
  - iii)  $\|x+y\| \leq \|x\| + \|y\|$ .
- Note  $d_X(x,y) = \|x-y\|$  is therefore a metric.

Def 16.6: A normed vector space is a Banach Space if  
 $X$  is complete as a metric space.

Ex 16.7:  $\ell^1 \cong \mathbb{R}^{\infty}$  is the space of sequences  $a = (a_0, a_1, \dots)$   
 with  $\|a\|_{\ell^1} = \sum_{i=1}^{\infty} |a_i|$   
 completeness follows from completeness of  $\mathbb{R}$ .

Ex 16.8  $\ell^2 = \{a_0, a_1, \dots\}$   
 $\|a\|_{\ell^2} = \left( \sum_{i=1}^{\infty} |a_i|^2 \right)^{1/2}$   
 $\|a\|_{\ell^p} = \left( \sum_{i=1}^{\infty} |a_i|^p \right)^{1/p}$   
 $\|a\|_{\ell^\infty} = \sup_i |a_i|$   
 (all complete, different spaces)



Lemma 16.1 : If  $Y \subseteq X$  is a vector subspace closed in the induced topology then  $Y$  is also a Banach space. 16.3

Ex 16.10 : Let  $K \subseteq \mathbb{R}$  be compact, then

$C^0(K) := \{f \mid f \text{ continuous, } \|f - g\|_\infty = \sup_x |f(x) - g(x)|\}$   
is a Banach space.

$C^\alpha(K) = \{f \mid f \text{ Hölder continuous, } 0 < \alpha \leq 1\}$   
 $\|f\|_{C^\alpha} = \sup |f(x)| + \sup_{x,y} \frac{|f(x) - f(y)|}{|x - y|^\alpha}$   
 $\alpha = 1$  is Lipschitz continuous.

Def 16.11 : Let  $E$  be a measurable set then the  $L^p$  spaces

$$L^p(E) = \{f: E \rightarrow \mathbb{R} \mid f \text{ measurable, } \|f\|_{L^p} < \infty\}$$

where  $\|f\|_{L^p} = \left( \int_E |f|^p dx \right)^{1/p}$ , is the  $L^p$ -norm

For  $1 \leq p < \infty$ , and

$$\|f\|_{L^\infty} = \inf \{M \mid |f| \leq M \text{ a.e.}\}.$$

16.iii) Minkowski, Young, and Hölder Inequalities.

Thm 16.12 (Minkowski) If  $1 \leq p < \infty$ ,  $\|f + g\|_{L^p} \leq \|f\|_{L^p} + \|g\|_{L^p}$   
w/ equality iff  $f = \alpha g$ .

Proof : It suffices to show the unit ball is convex, ~~le~~ by linearity  
may assume  $\|f\| + \|g\| = 1$ . Then  $\bar{f} = \frac{f}{\|f\|_{L^p}}$ ,  $\bar{g} = \frac{g}{\|g\|_{L^p}}$

so  $f + g = \alpha \bar{f} + \beta \bar{g}$  .  $\bar{f}, \bar{g} \in B$ , and  $\alpha + \beta = 1$ ,

so can show  $\|f + g\| = \|\alpha \bar{f} + \beta \bar{g}\| \leq 1 = \|f\| + \|g\|$   
 $\downarrow$   
 $\in B$ .

Thus  $\forall t \in [0, 1]$  need  $\|tf + (1-t)g\|_{L^p} \leq 1$ . or  $\|tf + (1-t)g\|_{L^p}^p \leq 1$

$$\int |tf + (1-t)g|^p \leq \int t|f|^p + (1-t)|g|^p \leq 1 \quad (x^p \text{ is convex})$$

Thm 16.13 (Riesz-Fischer) For  $1 \leq p \leq \infty$ ,  $L^p$  is complete, and 16.4  
a Banach space.

Proof: The triangle inequality is precisely Minkowski's inequality.

For  $1 < p < \infty$ , the proof mimics  $p=1$ .

Suppose  $f_n$  is Cauchy, set  $\|f_{n_k} - f_{n_{k+1}}\| \leq \frac{1}{2^k}$

$$f = f_{n_1} + (f_{n_2} - f_{n_1}) + \dots$$

$$g = |f_{n_1}| + |f_{n_2} - f_{n_1}| + \dots$$

then  $\lim_{n \rightarrow \infty} \|G_n\|_{L^p} = \left( \int \left( \sum |g_n| \right)^p \right)^{1/p}$

$$\leq \left( \int \left( \sum |g_n|^p \right)^{1/p} \right)^{1/p}$$

~~convex~~  
Minkowski's obvious for  $p = \infty$ .  
(apply Minkowski N times)

$$\leq \sum_{n=1}^N \left( \int |g_n|^p \right)^{1/p}$$

$$\leq \infty.$$

so  ~~$f_n \in L^p$~~   $g \in L^p \Rightarrow f \in L^p$ . Rest same w/ Chebychev, and  $|g|^p$  in dominated convergence.  $\square$

Lemma 16.14: If  $m(A) < \infty$  and  $f \in L^p, L^\infty$  for all  $p$ ,

$$\lim_{p \rightarrow \infty} \|f\|_{L^p} = \|f\|_{L^\infty}$$

Proof: If  $\chi$  is a step function

$$\|\chi\|_{L^p} = m(A)^{1/p} \rightarrow 1 = \|\chi\|_{L^\infty}$$

Now take limits.  $\square$

## Lecture 17 Banach Spaces II : ~~Density~~. Density, Young-Hölder 17.18

Def 17.1 : A Banach space has a dense subset (subspace)

$$S \subseteq X$$

if  $\forall x \in X, \varepsilon > 0 \exists s \in S$  with  $\|s - x\| < \varepsilon$ .

$X$  is separable if  $\exists$  a countable, dense subset.

Prop 17.2 : For  $1 \leq p < \infty$  simple functions are dense.

For  $1 \leq p < \infty$  step, continuous, smooth functions are dense.

Proof : For  $L^\infty$ , take  $f$ , so  $M = \|f\|_{L^\infty}$ . Divide  $[-M, M]$  up and take  $\sum_{n=1}^N (M - \frac{M}{N}) \cdot \chi_{f^{-1}[(M - \frac{M}{N}, M - \frac{M}{N+1})]}$

For  $p \neq \infty$ , let  $f_M = \min_{\text{if } \infty} [M, f] \cdot \chi_{[-M, M]}$ .

Since  $\|f_M - f\| \rightarrow 0$  pointwise, dominated by  $|f|^p$

$$\int |f_M - f|^p \rightarrow 0.$$

If  $S \subseteq T \subseteq X$  are dense inclusions,  $S \subseteq X$  is dense.

For  $f_M$  use same trick as  $L^\infty$  (since  $f_M \in L^1 \cap L^p \cap L^\infty$  and compact support)

$\text{smooth} \subseteq \text{cont} \subseteq \text{step} \subseteq \text{simple}$ .

Prop 17.3 :  $L^p$  is separable for  $p \neq \infty$ .

Proof : Step functions  $\sum_{i=1}^N q_i \chi_{[p_i, t_i]}$   $q_i, p_i, t_i \in \mathbb{Q}$   
are dense in step functions.

### 17.ii Young and Hölder Inequalities

Thm 17.4 : Suppose  $\frac{1}{p} + \frac{1}{q} = 1$ . Then

$$\|fg\|_{L^1(E)} \leq \frac{\|f\|_{L^p(E)}^p}{p} + \frac{\|g\|_{L^q}^q}{q}.$$

Proof: It suffices to show

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q} \quad (\text{note } p=q=2 \text{ is } 2|ab| \leq a^2 + b^2 \text{ is clear})$$

Since  $\ln$  is concave,  $t = \frac{1}{p}$   $1-t = \frac{1}{q}$

$$\begin{aligned} \ln(ta^p + (1-t)b^q) &\geq \frac{1}{p} \ln(a^p) + \frac{1}{q} \ln(b^q) \\ &= \ln(a) + \ln(b) = \ln(ab). \end{aligned}$$

Take exp.

Corollary 17.5 (Young w/  $\varepsilon$ ) or "Peter-Paul" or "Absorption"

$\forall \varepsilon > 0$

$$\|fg\|_{L^1} \leq \frac{\varepsilon^p \|f\|_{L^p}^p}{p} + \frac{\|f\|_{L^q}^q}{q \varepsilon^q}$$

Proof: Apply Young to  $fg = \varepsilon f \cdot \frac{g}{\varepsilon}$ .

Rem 17.6: The above is, in my opinion, the most useful fact in analysis.

Thm 17.7 (Hölder's Inequality) Suppose  $\frac{1}{p} + \frac{1}{q} = 1$ . Then (including  $p=q=\infty$ )

$$\|fg\|_{L^1} \leq \|f\|_{L^p} \|g\|_{L^q}$$

Proof: Set  $\bar{f} = \frac{f}{\|f\|_{L^p}}$   $\bar{g} = \frac{g}{\|g\|_{L^q}}$ .

$$\text{Then } \|\bar{f}\bar{g}\|_{L^1} \leq \frac{\|\bar{f}\|_{L^p}^p}{p} + \frac{\|\bar{g}\|_{L^q}^q}{q} = \frac{1}{p} + \frac{1}{q} = 1. \text{ By Young.}$$

$$\Rightarrow \|fg\| \leq \|f\|_{L^p} \|g\|_{L^q}$$

$$\text{If } q = \infty, \quad \int |fg| \leq \sup f \cdot \int |g|.$$

□



17.iii) Examples17.3Ex 17.8 : Suppose  $m(E)$  is finite.

Then  $q \geq p$  :

$$\int |f|^q \leq \left( \int |f|^{p \cdot \frac{q}{p}} \right)^{1/q} \left( \int 1^{q \cdot \frac{p}{q}} \right)^{1/p}$$

$$\frac{1}{q} + \frac{1}{q/p} = 1.$$

$$\leq \|f\|_{L^p} m(E)^{1/q}$$

so finite  $L^q$  implies finite  $L^p$  so  $L^q \subseteq L^p$ .

Thus

~~$L^1(E) \subseteq L^2(E) \subseteq L^p(E) \subseteq L^q(E) \subseteq L^\infty(E)$~~

are strict inclusions.

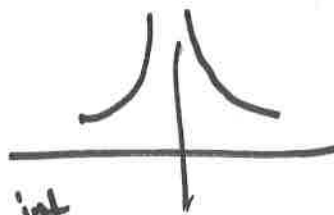
Intuition 17.1 : Large  $p$  makes mass escaping vertically more problematic.  
Small  $p$  horizontally.

Ex 17.10 :  $E = \mathbb{R}$ .  $p \leq q$ . Then  $\exists f \in L^p \setminus L^q$  and vice versa.

Take

$$f = \frac{1}{x^{1/q}} \chi_{[-1,1]}.$$

$$|f|^q = \frac{1}{x} \text{ not int, } |f|^p = \frac{1}{x^{p/q}} \text{ int.}$$



Take

$$f = \frac{1}{x^{1/p}} \chi_{[1,\infty)}$$

$$|f|^q = \frac{1}{x^{q/p}} \text{ integrable}$$

$$|f|^p = \frac{1}{x} \text{ not integrable.}$$



Rem 17.11: Recall every metric space has a completion  $X \hookrightarrow \bar{X} = \{\text{equiv class of Cauchy sequences}\}$  17.4

The density shows that

$(C^0[a,b], \|\cdot\|_{L^p}) \hookrightarrow L^p([a,b])$   
is this completion. So all we've done in some sense is very precisely characterize this completion.

### 17.iv) Dual Spaces

Def 17.12

A linear functional  $\varphi: X \rightarrow \mathbb{R}$  is said to be bounded if

$$|\varphi(x)| \leq C \|x\| \quad \forall x \in X.$$

Def 17.13: Suppose  $X$  is a Banach space, then

$$X^* = \{\varphi \mid \varphi: X \rightarrow \mathbb{R} \text{ bounded}\}, \quad \|\varphi\|_{X^*} = \sup_{\|x\|=1} |\varphi(x)|.$$

Prop 17.14:  $X^*$  is a Banach space.

Proof: Linear functionals are linear, and triangle inequality on  $\mathbb{R}$  shows norm.

Suppose  $\varphi_i$  is Cauchy, so that

$$\|\varphi_i - \varphi_j\|_{X^*} = \sup_{\|x\|=1} |\varphi_i(x) - \varphi_j(x)| \rightarrow 0.$$

Then  $\varphi_i(x)$  is Cauchy  $\forall x$ , set  $\varphi(x) = \lim_{i \rightarrow \infty} \varphi_i(x)$ . It is a simple matter to show  $\varphi(x)$  is linear.  $\square$

Lemma 17.15: There is a natural inclusion  $X \hookrightarrow X^{**}$ .

If it is an isomorphism,  $X$  is called Reflexive.

Proof: Elements of  $X^{**} = \{\xi: X^* \rightarrow \mathbb{R} \mid \text{bounded linear}\}$ .

$x \mapsto \xi_x$  defined by  $\xi_x(\varphi) := \varphi(x)$ .  
Then obviously linear,  
 $\|\xi_x\| \leq \|\varphi\| \|x\|$  so bounded.  $\square$

# Lecture 18 Banach Spaces III: Riesz Representation and Weak Topology 18.1

Recall each Banach space has a dual

$$X^* = \{ \varphi \mid \|\varphi(x)\|_X \leq \|\varphi\|_{X^*} \|x\| < \infty \}.$$

Question 18.1: What is the dual of  $L^p(E)$ .

Ex 18.2: There is an inclusion  $L^q(E) \hookrightarrow (L^p(E))^*$  for  $\frac{1}{p} + \frac{1}{q} = 1$ .

$$f \longmapsto \underbrace{\int_E f \, dx}_{\varphi_f}.$$

It is bounded because by Hölder,

$$\|\varphi_f(g)\|_{\mathbb{R}} \leq \|fg\|_{L^1} \leq \|f\|_{L^q} \|g\|_p$$

$$\text{so } \|\varphi_f\|_{(L^p)^*} = \|f\|_{L^q}.$$

Theorem 18.3 (Riesz Representation) If  $\frac{1}{p} + \frac{1}{q} = 1$  w/  $1 < p, q < \infty$ ,

then

$$L^q(E) \xrightarrow{\sim} (L^p(E))^*$$

$$f \longmapsto \varphi_f$$

is an isometry.

! We will prove for  $E = \mathbb{R}$  or  $[a, b]$

Corollary 18.4:  $L^p$  is reflexive.

Proof: Given  $\varphi \in (L^p(E))^*$  define a candidate

$$F(x) = \begin{cases} \varphi(\chi_{[0, x]}) & x > 0 \\ -\varphi(\chi_{[x, 0]}) & x < 0 \end{cases}$$

Claim  $F$  is absolutely cont.

$$\begin{aligned} \sum_{i=1}^N |F(b_i) - F(a_i)| &= \left| \sum_{i=1}^N \varphi(\chi_{[a_i, b_i]}) \right| \\ &\leq \varphi\left(\sum_{i=1}^N \chi_{[a_i, b_i]}\right) \leq C_\varphi \left\| \sum_{i=1}^N \chi_{[a_i, b_i]} \right\|_{L^p} \\ &\leq C_\varphi \delta^{1/p}. \end{aligned}$$

Thus  $\forall [n, n+1] \exists f$  st  $f(x) = F'(x)$  a.e.

18.2

Then

$$\varphi_f(\chi_{[a,b]}) = \int f \chi_{[a,b]} dx = \int_a^b f dx = F(b) - F(a) = \varphi(\chi_{[a,b]})$$

agree on step functions + continuity / density.

It remains to show  $f \in L^q(\mathbb{R})$ . Suppose first  $f \geq 0$ .

Let  $f_m = \min(m, f) \chi_{[-m, m]}$ .

$$\int_{\mathbb{R}} |f_m|^q \leq \int_{\mathbb{R}} |f_m|^{q-1} |f_m| = \varphi_f(|f_m|^{q-1})$$

$$\begin{aligned} &\leq C_\varphi \|f_m^{q-1}\|_{L^p} \\ &\leq C_\varphi \left( \int |f_m|^{(q-1)p} \right)^{1/p} \\ &\leq C_\varphi \left( \int |f_m|^q \right)^{1/p} \end{aligned}$$

$$\frac{1}{p} + \frac{1}{q} = 1 \Rightarrow p+q = pq$$

$$\text{so } pq - p = q$$

$$\frac{q}{q(1-\frac{1}{p})} = q \cdot \frac{1}{1-\frac{1}{p}} = 1$$

$$\text{so } \|f_m\|_{L^q} \leq C_\varphi$$

Then  $M \rightarrow \infty$  by monotone convergence,  $\|f\|_{L^q} \leq C_\varphi$ .

For  $f$  general use  $f = f^+ \chi^+ - f^- \chi^-$  trick.

Thm 18.5 (Reisz Rep  $p = \infty$ )  $L^1(\mathbb{R})^* = L^\infty(\mathbb{R})$ , but not vice versa.

Proof:  $F(x)$  as before,

~~extend  $\varphi$  to  $L^\infty$~~  extend  $\varphi$ .

$$\|F(b) - F(a)\|_{\mathbb{R}} \leq \|\varphi\|_* \|\chi_{[a,b]}\|_{L^1} = (b-a) \|\varphi\|_*$$

so absolutely cont as before,

$$f = F' \text{ a.e.}$$

$$\text{and } \|f\|_{L^\infty} \leq \|\varphi\|_*$$

$$\text{Again } \varphi_f(\chi_{[a,b]}) = \int_a^b F' = F(b) - F(a) = \varphi(\chi_{[a,b]})$$

agree, so equal by density.



# 18.4) The Weak Topologies

18.3

Def 18.6 a sequence  $f_n \in X$  to converge weakly to  $f$   
 $f_n \rightarrow f$

if  $\varphi(f_n) \rightarrow \varphi(f)$  in  $\mathbb{R} \quad \forall \varphi \in X^*$ .

Ex 18.7:  $f_n = \chi_{[n, n+1]} \rightarrow 0$  in  $L^2(\mathbb{R})$  but not strongly. By Riesz  $(L^2)^* \cong L^2$

$$\text{So } \int_{\mathbb{R}} \chi_{[n, n+1]} g \leq \|g\|_{L^2[n, n+1]}$$

$$\text{but } \sum_{n=1}^{\infty} \|g\|_{L^2[n, n+1]} < \infty \text{ so } \rightarrow 0.$$

$$\text{but } \|f_n\|_{L^2} = 1 \quad \forall n.$$

Prop 18.8: If  $f_n \rightarrow f$  in  $L^p(E)$   $1 \leq p < \infty$   
 then  $\exists M$  st  $\|f_n\|_{L^p} < M$ .

Proof: Suppose not, By passing to a subsequence,  
 May assume  $f = 0$ .  $\|f_n\| \geq n \cdot 3^n$ .

Renormalize so

$$g_n = \frac{n \cdot 3^n}{\|f_n\|} \cdot f$$

bounded by  $[0, 1]$   
 so sub converges.

$$\text{Claim } \|g_n\| = n \cdot 3^n \text{ and } g_n \rightarrow g \text{ weakly } \varphi(g_n) = \frac{n \cdot 3^n}{\|f_n\|} \varphi(f_n)$$

$$\text{Now let } \varepsilon_1 = \frac{1}{3}$$

$$\varepsilon_{n+1} = \frac{1}{3^{n+1}} \cdot \text{sign} \left( \int_E \left[ \sum_{k=1}^n \varepsilon_k (f_k)^* \right] f_n \right)$$

$$f_n^* = |f_n|^{\frac{p}{p-1}} \text{sgn}(p) |f_n|^{\frac{p}{p-1}}$$

$$\left( \int f_n^* f_n \right)^{\frac{1}{p}} = \|f_n\|_p$$

$$\text{so } \left| \int_E \sum_{k=1}^n \varepsilon_k f_k^* f_n \right| \geq \frac{1}{3^n} \|f_n\|_p = n$$

$$\|f_n^*\|_q = 1 \text{ so } \|\varepsilon_n f_n^*\|_q = \frac{1}{3^n}$$

Thus  $\sum_{k=1}^{\infty} \epsilon_k f_k^* \rightarrow g$  in  $L^q$ .

(18.4)

$$\begin{aligned} |\int g \cdot f_n| &\geq |\int \sum_{k=1}^{\infty} \epsilon_k f_k^* f_n| - \int \sum_{k=n+1}^{\infty} \epsilon_k f_k^* \\ &\geq n - \|\sum_{k=n+1}^{\infty} \epsilon_k f_k^*\|_q \|f_n\|_p \\ &\geq n - \frac{1}{3^n} \|f_n\|_p \rightarrow 0. \quad \square \end{aligned}$$

Thm 18.9: Let  $1 < p < \infty$ . Then a sequence  $\{f_n\}$  (Banach-Alaoglu) has a weakly convergent subseq if it has a bounded subsequence.

i.e. "The unit ball is weakly compact".

Proof: See Royden 8.3.

Rem 18.10: This is a great result. If you want to show  $f_k \rightarrow f$  (say the minimum of a functional or sol of PDE)

Then can split as two easier results

- 1)  $f_k$  is bounded  $\Rightarrow \exists f$  st  $f_k \rightarrow f$ .
- 2) convergence is actually strong (no escaping mass).

Lecture 19 Fourier Series I: Hilbert spaces.

Def 19.1: A linear operator  $L: X \rightarrow Y$  on Banach spaces is said to be bounded if  $\exists C = C_L$  such that

$$\|Lx\|_Y \leq C_L \|x\|_X.$$

Lemma 19.2: A linear operator is continuous iff it is bounded.

Proof:  $\forall \varepsilon > 0$  take  $\delta = \frac{\varepsilon}{C_L}$  so

$$\|Lx - Ly\|_Y = \|L(x-y)\|_Y \leq C_L \|x-y\|_X \leq \varepsilon \quad \text{if } \|x-y\|_X < \delta.$$

Take  $\varepsilon = 1$ .  $\exists \delta$  s.t. if  $\|x-0\|_X < \delta$  then for  $\|x\|_X = \delta$

$$\|Lx\|_Y \leq 1 \leq \frac{1}{\delta} \|x\|_X, \quad \text{and both sides scale.}$$

Def 19.2: Two Banach spaces  $X_1, X_2$  are isomorphic if  $\exists$  a bounded linear isomorphism w/ bounded inverse.

19.i) Hilbert Spaces

Def 19.3: An inner product space  $H$  with

$$\langle \cdot, \cdot \rangle: H \times H \rightarrow \mathbb{R}$$

bilinear is a Hilbert space if the norm  $\|h\|_H = \sqrt{\langle h, h \rangle}$  makes it into a Banach space.

Ex 19.4:  $L^2(\mathbb{R})$  is a Hilbert space with

$$\langle f, g \rangle_{L^2} = \int_{\mathbb{R}} fg \, dx$$

Hölder's inequality specializes to Cauchy-Schwartz

$$|\langle f, g \rangle_{L^2}| \leq \|f\|_{L^2} \|g\|_{L^2}$$

Rem 19.5: In the case of a general inner product space Cauchy-Schwartz follows from general nonsense.

Def 19.6: A collection of vectors  $\{e_i\}$  is said to be an orthonormal set if  $\langle e_i, e_j \rangle = \delta_{ij}$ .

Def 19.7: An orthonormal set is said to be a basis if it is complete, i.e. the equivalent conditions hold

- 1) If  $\langle x, e_i \rangle = 0 \forall i$ , then  $x = 0$ .
- 2)  $\sum_{i=1}^{\infty} a_i e_i$  finite linear combinations are dense.

Proof: Suppose dense.  $\|x - \sum_{i=1}^N a_i e_i\| < \varepsilon$ , but  $0 = \langle x, \sum_{i=1}^N a_i e_i \rangle$   
 $= \langle x, \sum_{i=1}^N a_i e_i - x \rangle + \langle x, x \rangle$   
 $= \|x\|^2 - \varepsilon \|x\| \rightarrow \leftarrow$

Suppose complete, if not dense  $\exists y$  w/  
 $\|y - \sum_{i=1}^N a_i e_i\| > \delta \quad \forall a_i \in \mathbb{R}$  for some  $\delta$ .

Then  $y - \sum_{i=1}^{\infty} \langle y, e_i \rangle e_i$  has  $\langle y, e_i \rangle = 0 \quad \forall i \rightarrow \leftarrow$

Algorithm 19.8 (Gram-Schmidt)

Let  $a_i$  be a dense subset of  $H$ , for  $i=1, 2, \dots$

Set  $e_1 = a_1$   
 $\tilde{e}_{n+1} = a_{n+1} - \sum_{i=1}^n \langle a_{n+1}, e_i \rangle e_i$  then  $e_i = \frac{\tilde{e}_i}{\|\tilde{e}_i\|}$ .

Corollary 19.8: A separable Hilbert space has a (countable) orthonormal basis, hence

$$x = \sum_{i=1}^{\infty} \langle x, e_i \rangle e_i \quad \forall x \in H.$$

Proof: Existence follows from Gram-Schmidt on dense subspace

Note  $a_i = \langle x, e_i \rangle$  has  $\sum |a_i|^2 \leq \|x\|^2$

Then

$$0 \leq \|x - \sum_{i=1}^N a_i e_i\|^2 = \|x\|^2 + \sum_{i=1}^N |a_i|^2 - 2 \langle x, \sum_{i=1}^N a_i e_i \rangle$$

$$= \|x\|^2 - \sum_{i=1}^N |a_i|^2$$

so  $\sum_{i=1}^{\infty} \langle x, e_i \rangle e_i$  converges. By completeness,  $x - \text{this} = 0$ .

Corollary 19.10:  $\exists$  an isomorphism  $H \rightarrow \ell^2(\mathbb{N})$  for each basis (in fact an isometry)

Proof:  $x \rightarrow \{\langle x, e_i \rangle\}_{i=1}^{\infty}$



Thm 19.11 (Abstract Riesz Representation) Let  $H$  be a separable Hilbert space, then  $H \rightarrow H^*$  19.3  
 $x \mapsto \langle x, - \rangle$   
 is an isomorphism.

Proof: Obviously linear, and  $\sup_{\|f\|=1} \langle x, f \rangle = \|x\|$  so bounded, and obviously inj.

Let  $\varphi \in H^*$ . Set  $a_i = \varphi(e_i)$ . Then

$(\varphi - \langle \sum a_i e_i, - \rangle) e_j = \varphi(e_j) - a_j = 0$   
 so  $\varphi = \sum_{i=1}^{\infty} a_i e_i$ , hence surj. (isometry so inverse is bounded).

## 19.ii) Examples

Ex 19.12:  $\ell^2(\mathbb{N})$  or  $\ell^2(\mathbb{Z})$  with bases  $e_i = \{0, 0, 0, \dots, 1, 0, \dots\}$

Ex 19.13: The quantum mechanics Hilbert space is (usually)  $L^2(\mathbb{R}^3)$ .

Ex 19.14: More generally  $(X, \Sigma, \mu)$  any measure space  $L^2(X; \mu)$ .

Ex 19.15  $L^2[-1, 1]$  a basis is the Legendre polynomials  
 $1, x, \frac{1}{2}(3x^2-1), \frac{1}{2}(5x^3-3x), \frac{1}{8}(35x^4-30x^2+3), \dots$   
 (Gram-Schmidt on  $x^n$ ), density below.  $= \frac{1}{2^n n!} \left(\frac{d}{dx}\right)^n (x^2-1)^n$

Ex 19.16:  $L^2(\mathbb{R}, e^{-x^2/2})$

$$h_n(x) = (-1)^n e^{x^2} \left(\frac{d}{dx}\right)^n e^{-x^2} = (2x - \frac{d}{dx})^n \cdot 1.$$

(Hermite Polynomials)

$$= 1, 2x, 4x^2-2, 8x^3-12x, \dots$$

are orthonormal basis.

## 19.iii) Fourier Series

19.4

Def 19.17: an algebra of functions on  $K$  compact

- 1) separates points if  $\forall x, y \in K, \exists f$  w/  $f(x) \neq f(y)$
- 2) vanishes nowhere if  $\nexists x \in K$  w/  $f(x) = 0 \forall f$ .

Thm 19.18 (Stone-Weierstrass) Any algebra  $\mathcal{F} \subseteq C^0(K)$  that separates points and vanishes nowhere is dense in  $C^0(K)$ .

Upgrade 19.19: Consider  $\mathbb{C}$ -valued functions.

$L^2([a, b])$  works same w/  $\langle f, g \rangle = \int_a^b f \bar{g} dx$

Also  $L^2([a, b]) \cong L^2(S^1)$  as completion of  $f(a) = f(b)$  in  $C^0$ .

⇒ For next few lectures we work with  $L^2(S^1; \mathbb{C})$ ,  $\langle f, g \rangle = \frac{1}{2\pi} \int_0^{2\pi} f \bar{g} d\theta$ .

Thm 19.20:  $\{e^{in\theta}\}_{n=-\infty}^{\infty}$  for  $\theta \in [0, 2\pi]$  is an orthonormal basis for  $L^2(S^1; \mathbb{C})$ .

Proof:  $\therefore \frac{1}{2\pi} \int_0^{2\pi} |e^{in\theta}|^2 d\theta = \frac{1}{2\pi} \cdot 2\pi = 1$

$$\cdot \frac{1}{2\pi} \int_0^{2\pi} \langle e^{in\theta}, e^{im\theta} \rangle d\theta = \frac{1}{2\pi} \int_0^{2\pi} e^{i(n-m)\theta} d\theta = \frac{1}{i(n-m)} e^{i(n-m)\theta} \Big|_0^{2\pi} = 0$$

•  $\operatorname{Re}(e^{in\theta})$  and  $\operatorname{Im}(e^{in\theta})$  never simultaneously vanish

• ~~Since  $\operatorname{const} \in C^0$ , if  $\exists x, y$  then  $\exists x, y$  st  $f(x) \neq f(y) \neq 0$~~

$e^{in\theta}: S^1 \rightarrow S^1 \subseteq \mathbb{C}$  is already injective.

Def 19.21: The isomorphism / isometry

$$\mathcal{F}: L^2(S^1; \mathbb{C}) \longrightarrow L^2(\mathbb{Z}; \mathbb{C})$$

$$f \longmapsto \langle e^{in\theta}, f \rangle$$

is called the Fourier Series, and denoted

$$\mathcal{F}(f) = \hat{f}(n): \mathbb{Z} \rightarrow \mathbb{C}.$$

# Lecture 20 Fourier Series II: Derivatives and convolutions

20.1

Recall Fourier series gave an isomorphism

$$L^2(S^1; \mathbb{C}) \xrightarrow{\mathcal{F}} \ell^2(\mathbb{Z}; \mathbb{C})$$

$$f \mapsto \hat{f}(n) = \langle f, e^{in\theta} \rangle_{L^2}$$

$$\sum_{n=-\infty}^{\infty} c_n e^{in\theta} \longleftrightarrow \{c_n\}$$

In fact  
Thm 20.1 (Plancherel)  $\mathcal{F}$  is an isometry,

$$\|f\|_{L^2} = \sum_{n \in \mathbb{Z}} |c_n|^2 \quad \hat{f}(n) = c_n.$$

Ex 20.2 : For  $c_n = a_n + ib_n$   $c_n e^{in\theta} = (a_n + ib_n)(\cos(n\theta) + i\sin(n\theta))$

(Real Fourier Series)

And

$$= a_n \cos(n\theta) - b_n \sin(n\theta) + i(b_n \cos(n\theta) + a_n \sin(n\theta)).$$

$$\langle f + ig, \cos(n\theta) + i\sin(n\theta) \rangle_{\mathbb{C}} = \langle f, \cos \rangle + \langle g, \sin \rangle + i \langle f, -\sin \rangle + i \langle g, \cos \rangle.$$

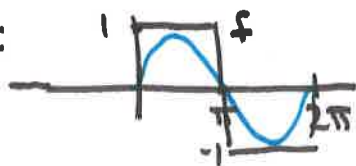
$\Rightarrow$  if  $f$  is real,

$$f = \text{Re} \left( \sum_n c_n e^{in\theta} \right) = \sum_n a_n \cos(n\theta) + b_n \sin(n\theta)$$

$$\text{where } a_n = \langle f, \cos(n\theta) \rangle_{L^2}$$

$$b_n = \langle f, -\sin(n\theta) \rangle_{L^2}$$

Ex 20.3 :



then  $a_n = 0 \quad \forall n$  b/c function is odd.

$$-b_n = \frac{1}{2\pi} \int_0^{2\pi} f(x) \sin(nx) dx$$

$$= \frac{1}{\pi} \int_0^{\pi} \sin(nx) dx$$

$$= \frac{1}{\pi n} [-\cos(nx)]_0^{\pi} = \begin{cases} \frac{2}{\pi n} & n \text{ even} \\ 0 & n \text{ odd} \end{cases}$$

$$f = \sum_{n \text{ odd}} (-1)^{\frac{n-1}{2}} \frac{2}{\pi n} \sin(nx).$$

$$= \sum_{n=1,3,5} \frac{4}{\pi n} \sin(nx).$$

## 20.i) The Derivative

20.2

Observe  $-i\partial_\theta e^{in\theta} = n e^{in\theta}$ .

Prop 20.4 : The Fourier series satisfies (for  $f \in C^\infty$ )

$$\mathcal{F}(-i\partial_\theta f) = n \hat{f}(n)$$

Proof : Let  $g = -i\partial_\theta f \in L^2$ .

$$g = \sum_{n=-\infty}^{\infty} d_n e^{in\theta} \quad \text{where}$$

$$f = \sum_{n=-\infty}^{\infty} c_n e^{in\theta} \quad \text{where} \quad c_n = \frac{1}{2\pi} \int_0^{2\pi} \langle f, e^{in\theta} \rangle d\theta$$

$$\begin{aligned} \text{And} \quad d_n &= \frac{1}{2\pi} \int_0^{2\pi} \langle -i\partial_\theta f, e^{in\theta} \rangle d\theta = \frac{1}{2\pi} \int_0^{2\pi} \langle \partial_\theta f, i e^{in\theta} \rangle d\theta \\ &= \frac{1}{2\pi} \int_0^{2\pi} \langle f, -i\partial_\theta e^{in\theta} \rangle d\theta \\ &= n \cdot c_n \end{aligned}$$

$$\text{Hence} \quad \sum_{n=-\infty}^{\infty} |n|^2 |c_n|^2 < \infty, \quad \text{and} \quad \left\| \sum_{n=-N}^N n c_n e^{in\theta} - (-i\partial_\theta f) \right\|_{L^2} \xrightarrow{N \rightarrow \infty} 0.$$

Corollary 20.5 :  $(-i\partial_\theta)^k f \in L^2$  if and only if  $\sum_{n=-\infty}^{\infty} |n|^{2k} |c_n|^2 < \infty$ .

Proof :  $\Rightarrow$  induction on above.

$$\Leftarrow \lim_{N \rightarrow \infty} \int_0^{2\pi} \left| \frac{f_N(\theta+h) - f_N(\theta)}{h} \right|^2 d\theta$$

$-i\partial_\theta f_N \rightarrow g$  in  $L^2$ . Claim  $g = -i\partial_\theta f$ . exchange limits + dominated convergence.

## 20.ii) The Convolution

Question 20.6: What is  $\mathcal{F}(fg)$ ?

Thm 20.7 : If  $f, g \in L^2$  with  $g \in \mathcal{O}'(S; \mathbb{C})$ . Then

$$\begin{aligned} \mathcal{F}(fg) &= \hat{f} * \hat{g} \\ &= \sum_{n \in \mathbb{Z}} \sum_{k \in \mathbb{Z}} \hat{f}(n-k) \hat{g}(k) \end{aligned}$$



Proof : First note  
 $(\hat{f} * \hat{g}) \in \ell^2$ ,  $\therefore$

20.3

$$\begin{aligned} \|\hat{f} * \hat{g}\|_{\ell^2} &= \sum_n \left| \sum_{k \in \mathbb{Z}} \hat{f}(n-k) \hat{g}(k) \right|^2 \\ &\leq \sum_n \left( \left( \sum_k \left| \frac{\hat{f}(n-k)}{(k^2+1)} \right|^2 \right)^{1/2} \left( \sum_k (k^2+1) |\hat{g}(k)|^2 \right)^{1/2} \right)^2 \\ &\leq \sum_n (\|\hat{g}\|_{L^2} + \|d\hat{g}\|_{L^2}) \sum_k \frac{|\hat{f}(n-k)|^2}{(k^2+1)^2} \\ &\leq \sum_k \sum_n \frac{|\hat{f}(n-k)|^2}{(k^2+1)^2} \leq (\|\hat{g}\|_{L^2} + \|d\hat{g}\|_{L^2}) (\|\hat{f}\|_{L^2}) \sum_k \frac{1}{k^4} \\ &< \infty. \end{aligned}$$

And

$$\begin{aligned} \langle f\hat{g}, e^{in\theta} \rangle &= \langle f \cdot \sum_{k \in \mathbb{Z}} \hat{g}_k e^{ik\theta}, e^{in\theta} \rangle + \langle f \cdot \sum_{k \geq n} \hat{g}_k e^{ik\theta}, e^{in\theta} \rangle \\ &= \langle f, \sum_k e^{i(n-k)\theta} \hat{g}_k \rangle + \\ &= \sum_{k \in \mathbb{Z}} \hat{g}_k \hat{f}_{n-k} + \\ \Rightarrow \langle f\hat{g}, e^{in\theta} \rangle &= \sum_{k \in \mathbb{Z}} \hat{g}(k) \hat{f}(n-k). \end{aligned}$$

$\leq \|f\|_{L^2} \|\sum_{k \in \mathbb{Z}} \hat{g}_k\|$   
 $\leq \|f\|_{L^2} \cdot \frac{1}{N^2} (\|\hat{g}\|_{L^2} + \|d\hat{g}\|_{L^2})$   
 $\square$

Thm 20.8 :  $\mathcal{F}(f * g) = \hat{f} \cdot \hat{g}$ ,  $2\pi$  when all are defined.

Proof :  $f * g = \int_0^{2\pi} f(\theta - \tau) g(\tau) d\tau$

$$\begin{aligned} \widehat{f * g}_n &= \frac{1}{2\pi} \int_0^{2\pi} \int_0^{2\pi} f(\theta - \tau) g(\tau) e^{-in(\theta + \tau - \tau)} d\tau d\theta \\ &= \frac{1}{2\pi} \int_0^{2\pi} \int_0^{2\pi} f(\theta - \tau) g(\tau) e^{-in\tau} e^{-in(\theta - \tau)} d\tau d\theta \end{aligned}$$

$$\begin{aligned} \theta' = \theta - \tau &= \frac{1}{2\pi} \int_0^{2\pi} \int_0^{2\pi} g(\tau) e^{-in\tau} f(\theta') e^{-in\theta'} d\theta' d\tau \quad \text{Fubini} \\ &= \int_0^{2\pi} \hat{g}(n) f(\theta') e^{-in\theta'} d\theta' = 2\pi \hat{f}(n) \hat{g}(n). \end{aligned}$$

Let  $\delta \in C^0(S^1)^*$  be s.t.  $\delta(f) = f(0)$ . Written

$$\delta(f) := \frac{1}{2\pi} \int_0^{2\pi} \delta(x) f(x) dx.$$

Then

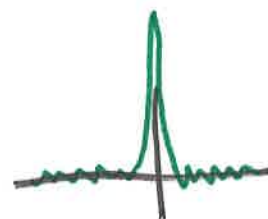
$$f(x) = \int_0^{2\pi} f * \delta(x) = \frac{1}{2\pi} \int_0^{2\pi} \delta(y) f(x-y) dy = f(x).$$

Intuitively,

$$\begin{aligned} \widehat{f}(\delta) &= \frac{1}{2\pi} \int_0^{2\pi} \delta(x) e^{inx} dx \\ &= \frac{1}{2\pi} \forall n, \text{ and } \|\delta(x)\|_{L^2} = \infty \notin L^2 \end{aligned}$$



Def 20.9 : The Dirichlet Kernel for  $N \in \mathbb{N}$

$$D_N(x) = \frac{1}{2\pi} \sum_{-N}^N e^{inx}$$


Thm 20.10 : 1)  $\lim_{N \rightarrow \infty} f * D_N = f$  in  $L^2$   $\forall f \in L^2$

(Approx Identity)

2)  $f * D_N \in C^\infty$  and  $\| \partial^m_{(f * D_N)} \|_{L^2} \leq N^m \|f\|_{L^2}$

Proof : 1)  $\widehat{f * D_N} = \chi_{[-N, N]} \widehat{f}$  so  $\|f * D_N - f\|_{L^2} = \left\| \sum_{|n| > N} c_n e^{inx} \right\| \rightarrow 0$ .

$$\begin{aligned} 2) \left\| \partial^m_{(f * D_N)} \right\|_{L^2}^2 &= \left| \sum |n|^{2m} \chi_{[-N, N]} \widehat{f} \right|^2 \\ &\leq N^{2m} \left| \sum |n|^{2m} \widehat{f}_n \right|^2 \leq N^{2m} \|f\|_{L^2}^2. \end{aligned}$$

and  $\partial^m f \in L^2 \forall m \Rightarrow f$  is smooth.

## Lecture 21: Convergence of Fourier Series (Fourier Series III)

21.1

Question 21.1: If  $f \in L^2$  then  $f_N = \sum_{|n| \leq N} \hat{f}(n) e^{in\theta} = f * D_N$  converges to  $f$  in  $L^2$ .

Under what conditions does  $f_N \rightarrow f$  pointwise? uniformly?

Ex 21.2: There exists an  $f \in C^0(S')$  such that  $f_N(0) \rightarrow \infty$ .

(Fejér)  $f(x) = \sum_{k=1}^{\infty} \sin((2^{k^3}+1)\frac{x}{2}) \cdot \frac{1}{k^2}$

Then  $f(x)$  is a  $\sup$  norm-limit of  $C^0$ , so  $C^0$ , but

$$\begin{aligned} \lim_{N \rightarrow \infty} f_N(0) &= \sum_{n=-\infty}^{\infty} a_n \\ &= \sum_{k=1}^{\infty} \frac{2}{2\pi} \sum_{n=-\infty}^{\infty} \frac{1}{k^2} \int_0^{\pi} \sin((2^{k^3}+1)\frac{\theta}{2}) \cos(n\theta) d\theta \\ &= \frac{1}{\pi} \sum_{n=1}^{\infty} \sum_{k=1}^{\infty} \frac{1}{k^2} \cdot \lambda_{n, 2^{k^3}+1} = \frac{1}{\pi} \sum_{k=1}^{\infty} \frac{\sigma_k}{k^2} \end{aligned}$$

Can show  $\sigma_k = \sum_{n=1}^N \lambda_{n, 2^{k^3}+1}$  has  $\sigma_k \geq \frac{1}{2} \log(N)$

$$\Rightarrow \text{For } N \geq 2^{k^3}-1, \quad f_N(0) \geq \frac{1}{k^2} \cdot \frac{1}{2} \log(2^{k^3}+1) \geq \frac{k}{L}.$$

### 21.i) Sobolev Spaces

Since  $\widehat{-i\partial_{\theta} f} = n \hat{f}(n)$  we say  
 $\partial_{\theta}^k f \in L^2 \iff n^k \hat{f} \in L^2$

Define 21.3 the (continuous) resp (discrete) Sobolev norm of regularity  $k$  by

$$\begin{aligned} \|f\|_{L^{k,2}} &= \left( \int_0^{2\pi} |f|^2 + |\partial f|^2 + |\partial^2 f|^2 + \dots + |\partial^k f|^2 d\theta \right)^{1/2} \\ \|\hat{f}\|_{\ell^{k,2}} &= \left( \sum_{n=-\infty}^{\infty} (1 + |n|^2 + |n|^4 + \dots + |n|^{2k}) |\hat{f}(n)|^2 \right)^{1/2}. \end{aligned}$$

Def 21.4

Define the Sobolev spaces

$$L^{k,2}(S'; \mathbb{C}), \quad \ell^{k,2}(\mathbb{Z}; \mathbb{C})$$

are the completions of  $C^\infty$ , finite seq in the above norms.

Thus  $f \in L^{k,2}$  iff it has  $k$ -derivatives in  $L^2$

\* this needs "weak derivatives" to be made precise

Corollary 21.5 : The following diagram commutes

$$\begin{array}{ccc}
 L^2(S'; \mathbb{C}) & \xleftrightarrow{\mathcal{F}} & L^2(\mathbb{Z}; \mathbb{C}) \\
 \uparrow & & \uparrow \\
 L^{1,2}(S'; \mathbb{C}) & \xleftrightarrow{\mathcal{F}} & L^{1,2}(\mathbb{Z}; \mathbb{C}) \\
 \uparrow & & \uparrow \\
 L^{2,2}(S'; \mathbb{C}) & \xleftrightarrow{\mathcal{F}} & L^{2,2}(\mathbb{Z}; \mathbb{C}) \\
 \uparrow & & \uparrow \\
 \vdots & & \vdots
 \end{array}$$

and each horizontal arrow is an isometry in the  $L^{k,2}$ ,  $l^{k,2}$  norms.

Proof: inclusions commute w/  $\mathcal{F}$  so commuting is obvious.

As before

$$\begin{aligned}
 \int |f|^2 + |\partial f|^2 dx &= \sum_{n \in \mathbb{Z}} |\hat{f}(n)|^2 + |\widehat{\partial f}(n)|^2 \\
 &= \sum_{n \in \mathbb{Z}} (1+n^2) |\hat{f}(n)|^2
 \end{aligned}$$

so  $\mathcal{F}$  lands in  $L^{1,2}$  precisely when  $|\partial f| \in L^2$ .

Caution 21.6 :  $L^{k,2} \subseteq L^{k-1,2}$  is a (bounded) inclusion,

$$\|f\|_{L^{k-1,2}} \leq \|f\|_{L^{k,2}}$$

of a dense Banach space. So  $L^{k,2}$  is closed in its own norm but dense in the  $L^{k-1,2}$ -norm.



Thm 21.7: If  $f \in L^{s,2}(S'; \mathbb{C})$ , then  $f_N \rightarrow f$  uniformly.

Proof: It suffices to show  $\sup_{S'} |f| \leq C \|f\|_{L^{s,2}}$

Then,  $f_N \rightarrow f$  in  $L^{s,2}$  so  $\sup_S |f_N - f| \leq C \|f_N - f\|_{L^{s,2}} \rightarrow 0$ .  
 ( $\hat{f}_N \rightarrow \hat{f}$  in  $L^{s,2}$  and isometry)

And  $\leq \left| \sum \hat{f}(n) e^{in\theta} \right|$

$$\sup_{S'} |f| \leq \sum_{n \in \mathbb{Z}} |\hat{f}(n)|$$

$$\leq \sum_{n \in \mathbb{Z}} \frac{|\hat{f}(n)|}{\sqrt{1+n^2}} \sqrt{(n^2+1)}$$

$$\leq \left( \sum_{n \in \mathbb{Z}} |\hat{f}(n)|^2 (n^2+1) \right)^{1/2} \left( \sum_{n \in \mathbb{Z}} \frac{1}{n^2+1} \right)^{1/2}$$

$$\leq C \|\hat{f}\|_{L^{s,2}} = C \|f\|_{L^{s,2}}.$$

Corollary 21.8: If  $f \in L^{k,2}(S'; \mathbb{C})$  then

$$\sup |f_N - f| + \sup |df_N - df| + \dots + \sup |d^{k-1} f_N - d^{k-1} f| \rightarrow 0.$$

Rem 21.9:  $L^{s,2}$  makes sense for non-integer  $s$ , which gives a notion of fractional derivatives

$$\left( \frac{d}{ds} \right)^s f = (\mathcal{F}^{-1} \circ \ln^s \circ \mathcal{F}) f.$$

Actually the proof shows any  $L^{s,2}$  for  $s > \frac{1}{2}$  is okay.

## 21.iii) Fourier transform of $L^p$

21.4

Recall since  $S'$  is compact

$$L^2 \subseteq L^p \quad 1 \leq p \leq 2$$

$\parallel$

$$L^2 \subseteq L^{p'} \quad p' \geq 2.$$

Thm 21.10 (Hausdorff-Young) Let  $1 \leq p \leq 2$ , and  $q$  be the Hölder conjugate.  
 $\frac{1}{p} + \frac{1}{q} = 1.$

$$\text{Then } \|\mathcal{F}(f)\|_{L^q(\mathbb{Z})} \leq \|f\|_{L^p(S'; \mathbb{C})}$$

In particular,  
 $\mathcal{F}: L^p(S'; \mathbb{C}) \rightarrow L^q(\mathbb{Z}; \mathbb{C})$   
is bounded.

Proof: For  $p=1$ ,  $q=\infty$ .

$$\|\hat{f}\|_{\infty} = \sup_n \left| \int \langle f, e^{inx} \rangle \right| \leq \|f\|_{L^1}.$$

$p=2$  is obvious. Interpolation or Young Convolution for general.

Ex 21.11:  $\{1, 1, \dots\} = \mathcal{F}(\delta)$  in  $\ell^\infty$  but  $\delta \notin L^1$  so not surj.

$$\|D_N\|_{L^1} \approx \log N.$$

Thm 21.11: If  $f \in L^p(S'; \mathbb{C})$  then for  $1 < p < \infty$ ,  
 $\|f_N - f\|_{L^p} \rightarrow 0.$

Proof: "Convergence of Fourier Series in  $L^p$  Space" J. Miao, various Harmonic Analysis texts.

# Lecture 22 | Fourier Series and PDEs I: Fredholm and Compact Operators (22.1)

Main Idea 22.1:  $\mathcal{F}$  turns differentiation into multiplication

$$\begin{array}{ccc} L^{2,2}(S^1; \mathbb{C}) & \xleftrightarrow{\mathcal{F}} & \ell^{2,2}(\mathbb{Z}; \mathbb{C}) \\ \downarrow -\partial_x^2 & & \downarrow \ell^2 \\ L^2(S^1; \mathbb{C}) & \xleftrightarrow{\mathcal{F}} & \ell^2(\mathbb{Z}; \mathbb{C}) \end{array}$$

Commutates. Eg.

$$L = \sum \alpha_j (-i\partial_x)^j f \longleftrightarrow (\sum \alpha_j n^j) \cdot \hat{f} \\ = (\alpha_N n^N + \alpha_{N-1} n^{N-1} + \dots)$$

$\Rightarrow$  Can solve  $Lf = g$  by dividing Fourier transform

$$f = \mathcal{F}^{-1} \circ \frac{1}{|2|n^N + \alpha|2|n^{N-1} + \dots + \alpha_0} \cdot \mathcal{F}g := P_g$$

• As  $|n| \rightarrow \infty$ , only leading order matters

• Write  $L = \underbrace{(|2|n^N + \alpha|2|n^{N-1} + \dots + \alpha_0)}_{\text{leading order}} \underbrace{\quad}_{\text{lower order}}$

$$\underbrace{\underbrace{\text{leading order}}_{\text{invertible}} + \underbrace{\text{lower order}}_{\text{Compact / f.d.}}}_{\text{Fredholm.}}$$

Question 22.2: Given  $f \in X \subseteq L^2(S^1; \mathbb{C})$ , when can we solve

$$Lu = f$$

and what space is  $u$  in?

## 22.1) : Compact Operators

(22-2)

Def 22.3 : Let  $H_1, H_2$  be Hilbert spaces. An operator

$K: H_1 \rightarrow H_2$   
is said to be compact if

$\{x_n\} \subseteq X$  bounded  $\Rightarrow \{Kx_n\}$  has a convergent subsequence.

i.e. the image of the closed unit has compact closure.

Lemma 22.4 : If  $K$  has finite rank, then  $K$  is compact.

Proof : Let  $E \subseteq H_2$  be  $\text{Im}(K)$  and choose  $E \cong \mathbb{R}^N$  some  $N$ .

If  $\{x_n\}$  bounded by  $M$ ,

$$\|Kx_n\| \leq \|K\| \|x_n\| \leq \|K\| \|M\|.$$

So  $\overline{\text{Im}(B_1)}$  is closed and bounded.

Lemma 22.5 : If  $K_n \rightarrow K$  i.e.  $\forall \varepsilon > 0 \exists N$  st  $\|(K - K_n)x\| \leq \varepsilon \|x\|$   
 $\forall n \geq N$   
and  $K_n$  is finite rank, then  $K$  is compact.

Proof : Let  $\{x_j\}$  be a bounded sequence

Let  $\varepsilon > 0$ , choose  $N$  so  $\|K - K_N\| < \varepsilon/3$ , and for that  $N$ , let  $J$  be large so  $\|K_N x_j - K_N x_k\| < \varepsilon/3$  for  $j, k \geq J$ . Then for  $j, k$

$$\begin{aligned} \|Kx_j - Kx_k\| &\leq \|Kx_j - K_N x_j + K_N x_j - K_N x_k + K_N x_k - Kx_k\| \\ &\leq \|Kx_j - K_N x_j\| + \|K_N x_j - K_N x_k\| + \|K_N x_k - Kx_k\| \\ &\leq \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3}. \end{aligned}$$

Rem 22.6 : This is iff (but not in Banach spaces!)



Lemma 22.7 : The unit ball  $B \in H$  is compact iff  $H$  is finite dimensional. (22.3)

Proof :  $\Leftarrow$  is Heine-Borel

$\Rightarrow$  if  $e_i$  is an infinite orthonormal basis  $\|e_i - e_j\| = \sqrt{2}$  so no conv. subsequence.

Thm 22.8 (Rellich's Lemma) The inclusion  
 $L^{k+1,2}(S'; \mathbb{C}) \subseteq L^{k,2}(S'; \mathbb{C})$

is compact.

Proof : Suffices to prove for  $L^{1,2} \subseteq L^2$ .

For each  $N$  let



$a_1, a_2, \dots \rightarrow x_1$  conv subsequence for  $|e| \leq 1 \in \mathbb{R}^3$   
 $b_1, b_2 \rightarrow x_2$  subsubseq for  $|e| \leq 2$ .

Claim diagonal is Cauchy. 1) For enough  $N$ ,  $\|y_n - y_m\| \leq \epsilon/2$  on  $|e| \leq N$   
 $\{y_n\}$   $= \sum_{i=1}^N |y_n - y_m|^2$

$$2) \sum_{|e| \geq N} |y_n - y_m|^2 \leq \frac{1}{N} \sum_{|e| \geq N} |e|^2 |y_n - y_m|^2 \leq \frac{2}{N}.$$

## 22.ii: Fredholm Operators

Def 22.9 : A bounded operator  $L: H_1 \rightarrow H_2$  is said to have closed range if  $L(H_1) \subseteq H_2$  is a closed subspace.

Lemma 22.10 (closed range estimate)  $L: H_1 \rightarrow H_2$  has closed range iff  $\exists C$  s.t.

$$\|x\|_{H_1} \leq C \|Lx\|_{H_2} \quad x \perp \text{Ker } L.$$

Proof :  $\Rightarrow$  If  $Lx_i \rightarrow y$  then  $\|x_i - x_j\| \leq C \|Lx_i - Lx_j\| \rightarrow 0$   
 so  $x_i \rightarrow x$  and by continuity  $Lx = y$ .

$\Leftarrow$  Suppose  $\|x_i\| = 1$  w/  $\|Lx_i\| \rightarrow 0$ ; then  $\text{Im}(B) \subseteq \text{Ker}(L)^\perp$  is closed, and contains 0, so  $\exists x \in B$  w/  $Lx = 0 \rightarrow \Leftarrow \square$

Def 22.11 : An operator  $L: H_1 \rightarrow H_2$  is said to be Fredholm if

- $L$  has f.d. kernel
- $L$  has closed range
- $L$  has f.d. cokernel.

Rem 22.12 : Closed range is key!  $H_2 / \text{Im}(L) \cong H_2 / \overline{\text{Im}(L)} \approx \text{Im}(L)^\perp$

eg  $L^2 / L^{1,2}$  is a bit hard to parse

Thm 22.13 : An operator  $L: H_1 \rightarrow H_2$  is Fredholm iff  
 $\exists P: H_2 \rightarrow H_1$  st  
 $LP = \text{Id} + K_1$        $PL = \text{Id} + K_2$   
 For  $K_i: H_i \rightarrow H_i$  compact.

Proof :  $\Rightarrow$  Suppose  $L$  Fredholm. Let  $P$  be inverse of  
 $\text{Ker}(L)^\perp \xrightarrow{P} \text{Im}(L)$  (bounded by open mapping thm)

Then  $PL = \text{Id} - \pi_{\text{Ker } L}$  which is compact b/c f.d. finite rank  
 $\forall LP = \text{Id} - \pi_{\text{Im } L}$  "

$\Leftarrow$   $0 = PL(x_i) = x_i - Kx_i$  for  $\|x_i\|=1$  in  $\text{Ker } L$ .  
 $\|x_i\| \leq \|Kx_i\|$  but converges of  $x_i$  so  
 $x_i$  converges  $\rightarrow B \cap \text{Ker } L$  is compact, so f.d.

For closed range

$$\|x\| \leq \|PLx + Kx\| \leq C\|Lx\| + \|Kx\|$$

Then  $LP(y_i) = y_i + Ky_i$   
 so  $y_i = Ky_i$  on  $\text{Im } L^\perp$  same.

Now suppose  $x \perp \text{Ker } L$  If  $Lx_i \rightarrow y$  then

$$\|x_i - x_j\| \leq C(\|L(x_i) - Lx_j\| + \|Kx_i - Kx_j\|) \text{ on subseq.}$$

Corollary <sup>22.14</sup> : Fredholm implies  $\exists C, K$  compact so  
 $\|x\|_{H_1} \leq C(\|Lx\|_{H_2} + \|Kx\|_{H_1})$

## Lecture 23 Fourier Series and PDEs II: Elliptic Equations

23.1

Recall the original goal was to study solutions of

$$Lu = f$$

For differential operators  $L$ .

Question 23.1 : For what  $f$  can this be solved?

What is  $\text{Ker}(L) \subseteq L^{k,2}$ ?

Question 23.2 : For  $f \in L^{k,2}$ , what space does  $u$  lie in?

Recall  $\Delta = -\partial_\theta^2 = (-i\partial_\theta)^2$

We will consider four examples

i)  $L = \Delta + 1$

ii)  $L = \Delta$

iii)  $L = \sum_{j=0}^N \alpha_j (-i\partial_\theta)^j \quad \alpha_j \in \mathbb{C}$

iv)  $L = \Delta + V(\theta)$  (time-ind Schrödinger)

Def 23.3 : An operator  $L = \sum_{j=0}^N \alpha_j(x) (-i\partial_\theta)^j$  is called a differential operator for  $\alpha_j : S^1 \rightarrow \mathbb{C} \in C^\infty$ . It is said to be elliptic if  $|\alpha_N(x)| > 0$ .

Observe that an  $N^{\text{th}}$  order operator is a bounded map

$$L : L^{k+N,2}(S^1; \mathbb{C}) \rightarrow L^{k,2}(S^1; \mathbb{C})$$

since  $\|Lu\|_{L^2} \leq \left\| \sum_{j=0}^N \alpha_j(x) (-i\partial_\theta)^j u \right\|_{L^2} \leq \sum_{j=0}^N \|\partial_\theta^j u\|_2 \leq \|u\|_{L^{N,2}}$

same for  $L^{k,2}$ .

23.i)  $L = \Delta + 1$

$$L^{2,2}(S^1; \mathbb{C}) \xrightarrow{\mathcal{F}} L^{2,2}(S^1; \mathbb{C})$$

$$\begin{array}{ccc} \Delta+1 \downarrow & & \downarrow |e|^2+1 \\ L^2(S^1; \mathbb{C}) & \xrightarrow{\mathcal{F}} & L^2(S^1; \mathbb{C}) \end{array}$$

Prop 23.4 :  $\Delta + 1 : L^{2,2}(S^1; \mathbb{C}) \rightarrow L^2(S^1; \mathbb{C})$  is an isomorphism

Proof : Suffices to show  $|e|^2 + 1 : L^{2,2} \rightarrow L^2$  is, but this obviously has inverse  $\frac{1}{|e|^2 + 1}$ .



To be pedantic,

23.2

$$\begin{aligned} \|\frac{1}{(1+|x|^2)^{k/2}} a\|_{L^{k,2}} &= \sum_{\ell \in \mathbb{Z}} (1+|\ell|^2+|\ell|^4) \frac{|a_\ell|^2}{(1+|\ell|^2)^2} \\ &= \sum_{\ell \in \mathbb{Z}} \frac{1+|\ell|^2+|\ell|^4}{1+2|\ell|^2+|\ell|^4} |a_\ell|^2 \leq C \|a\|_{L^2}^2. \end{aligned}$$

Corollary 23.5: 1)  $\Delta+1 : L^{k+2,2}(S'; \mathbb{C}) \rightarrow L^{k,2}(S'; \mathbb{C})$  is an isomorphism  $\forall k \in \mathbb{N}$ .

2)  $\Delta+1$  (a fortiori) is Fredholm  $\forall k$ ,

3)  $\Delta+1$  satisfies

$$\|u\|_{L^{k,2}} \leq C_k \|(\Delta+1)u\|_{L^{k,2}} \quad \text{for } C_k \text{ constants.}$$

4)  $Lu = f$  has a solution  $u, \forall f \in L^{k,2}$  and  $u \in L^{k+2,2}$ .

Proof: Only 1) needs proof

$$\begin{aligned} (1+|\ell|^2 + \dots + |\ell|^{2k} + |\ell|^{2k+2} + |\ell|^{2k+4}) &\sim 1+|\ell|^{2k+4} \\ &\sim (1+|\ell|^{2k})(1+|\ell|^2)^2 \end{aligned}$$

Remark 23.6: Actually the same result holds for  $s \in \mathbb{R}$ .

23.ii)  $L = \Delta$

Def 23.7: The (formal) adjoint of an operator  $L$  is that  $L^*$  such that

$$\langle Lu, v \rangle_{L^2} = \langle u, L^* v \rangle_{L^2}$$

$\forall u, v \in \mathcal{C}^\infty$ .  $L$  is said to be formally self-adjoint if  $L = L^*$ .

Ex 23.8: 
$$\begin{aligned} \langle Lu, v \rangle &= \int \sum (\alpha_j(x) \partial_j) \bar{\partial}_j u, v \rangle \\ &= \int \langle u, (-i \partial_j)^{n_j} \bar{\alpha}_j(x) v \rangle \\ &= \bar{\alpha}_j(x) (-i \partial_j)^{N-1} + \bar{\alpha}_{N-1} (-i \partial_j)^{N-1} + (\partial \alpha_N) (-i \partial_j)^{N-1} \end{aligned}$$



Ex 23.9  $\Delta$  is formally self adjoint.

[23.3]

$$\int \langle -\partial_{\bar{z}}^2 u, v \rangle = \int \langle -\partial_{\bar{z}} u, -\partial_{\bar{z}} v \rangle = \int \langle u, -\partial_{\bar{z}}^2 v \rangle.$$

Thm 23.10 : The operator

$$\Delta : L^{k+2,2}(S'; \mathbb{C}) \rightarrow L^{k,2}(S'; \mathbb{C})$$

is Fredholm. It has 1-dimensional kernel and cokernel, given by the constants.

Proof 1: On Fourier space  $\Delta = |\ell|^2$ . Set  $P = \begin{pmatrix} \frac{1}{|\ell|^2} & \ell \neq 0 \\ 0 & \ell = 0 \end{pmatrix}$ .

$$\text{Then } P\Delta = \text{Id} - \pi_0$$

$$\Delta P = \text{Id} - \pi_0$$

and  $\pi_0$  has rank 1, so compact.

Therefore,  $\Delta$  is Fredholm. Clearly constants are both the kernel and cokernel.

~~Proof 2 (of cokernel)~~ Corollary 23.11 (Elliptic estimates)

$$\|u\|_{L^{2,2}} \leq \|\Delta u\|_{L^2} + \|\pi_0 u\|_{L^2} \leq \|\Delta u\|_{L^2} + \|u\|_{L^2}$$

$$\|u\|_{L^{k+2,2}} \leq \|\Delta u\|_{L^{k,2}} + \|u\|_{L^{k,2}}$$

Corollary 23.12 (Elliptic regularity). If  $\Delta u = f$  w/  $f \in C^\infty$  (in particular  $0$ ) then  $u \in L^2 \Rightarrow u \in L^{2,2} \Rightarrow u \in L^{4,2} \dots \Rightarrow u \in C^\infty$ .

Proof :  $\|u\|_{L^{k+2,2}} \leq \|\Delta u\|_{L^{k,2}} + \|u\|_{L^{k,2}}$  same for  $f$ .

Remark 23.13 : "u is 2 degrees more regular than itself"  
This is called "elliptic bootstrapping".

Proof 2 (of cokernel) : If  $v \in \text{Coker}$ , then

$$0 = \langle \Delta u, v \rangle = \langle u, \Delta v \rangle \text{ for all } u \Rightarrow \Delta v = 0 \text{ and so } v \in L^{2,2}.$$

Therefore  $v \in \ker(\Delta)$ , so constant.

### iii) General Const Coefficient

(23.4)

Thm 23.14:  $L = \sum_{j=1}^N a_j (-i\partial)^j$  is Fredholm, it satisfies

$$\|u\|_{L^{k+2,2}} \leq C_k \|Lu\|_{L^{k,2}} + \|u\|_{L^{k,2}}$$

and elliptic regularity. It has ~~ker~~  $\dim \ker = \dim \operatorname{Coker} = \left\{ \begin{array}{l} \# \text{ integer roots} \\ \text{of } \sum a_j \lambda^j \neq 0 \end{array} \right\}$

Proof:  $\mathcal{F}^{-1} L \mathcal{F} = a_0 + a_1 i + a_2 i^2 + \dots + a_N i^N$ .

Rem 23.15: Agrees w/ ODE theory that  $N^{\text{th}}$  order has  $\leq N$  solutions.

### iv) The time-independent Schrödinger Equation

Say  $V(x) \in C^\infty$  w/  $V \geq 0$ .

$$\Delta + V \leftrightarrow |\xi|^2 + \hat{V}^* \quad \leftarrow \text{ew.}$$

Thm 23.16:  $\Delta + V$  is Fredholm and formally self adjoint.  
 $\exists C_k$  such that

$$\|u\|_{L^{k+2,2}} \leq C_k \|(\Delta + V)u\|_{L^{k,2}} + \|u\|_{L^{k,2}}$$

hold.

Proof:  $\|u\|_{L^{k+2,2}} \leq \|\Delta u\|_{L^{k+2,2}} + \|u\|_{L^{k,2}}$   
 $\leq \|(\Delta + V - V)u\|_{L^{k,2}} + \|u\|_{L^{k,2}}$   
 $\leq \|(\Delta + V)u\|_{L^{k,2}} + \|u\|_{L^{k,2}} \quad C_k \sim \|V\|_{C^k}$

Thm 23.17: In fact,  $\operatorname{Ker}(\Delta + V) = \{0\}$ .

~~It's not, it's not~~. Can show  $\|u\|_{L^2} \leq \|\Delta + V\|_{L^2}$  then  $\|u\|_{L^2} \leq C\|\Delta + V\|_{L^2}$   
 If not  $\exists u_n$  st  $\|u_n\|_{L^2} = 1$ , but  $\|\Delta + V\|_{L^2} \leq \frac{1}{n}$ .

Then 1) ~~It's not, it's not~~  $\langle (\Delta + V)u, u \rangle \leq \frac{1}{\sqrt{n}} \|u_n\|_{L^2}^2 + \sqrt{n} \|\Delta + V\|_{L^2}^2 \rightarrow 0$   
 $\| \nabla u_n \|_{L^2}^2 + \langle u, Vu \rangle_{L^2} \rightarrow 0$ .

2)  $u_n \in L^2$ , so  $\|u_n\|_{L^{2,2}} \leq \frac{1}{n} + \|u_n\|_{L^2} < \infty$ .  
 $\leq C_{u_2}$  bounded

$\Rightarrow u_{n_k} \in L^{1,2}$  convergent, but  $\|Vu_{n_k}\| \rightarrow 0$  so  $u_\infty = \text{const}$ ,  
 $\rightarrow u_\infty$  since  $\|u_n\|=1$ ,

but  $\langle u, Vu \rangle = \frac{1}{8\pi} \int v > 0. \rightarrow \leftarrow$

$\nabla u_\infty = \frac{1}{2\pi}$ .  
 $\square$ .



## Lecture 24 Fourier Series and PDEs III: the heat equation and 24.1 heat kernel.

### Question 24.1

Suppose that  $f(x) \in C^\infty(S^1; \mathbb{R})$  is a smooth heat distribution at time  $t=0$ . What is the distribution of heat  $u(x,t)$  for all  $t$ ?



Def 24.2: the initial value problem for the heat equation is to solve

$$\begin{cases} (\partial_t + \Delta_x) u(x,t) = 0 & = g(x,t) \text{ heat source} \\ u(x,0) = f(x). \end{cases}$$

for  $t \geq 0$ .

Ex 24.3: Suppose that  $f(x) = \cos(nx)$ . Then

$$\Delta_x f = -\partial_x^2 f = n^2 f \quad \text{so}$$

$$(\partial_t + \Delta_x) e^{-n^2 t} \cos(nx) = (-n^2 + n^2) e^{-n^2 t} \cos(nx) = 0.$$

is a solution.

Thm 24.4 (general solution) The heat evolution preserves Fourier modes and if

$$f(x) = \sum_{k \in \mathbb{Z}} a_k \cos(kx) + b_k \sin(kx)$$

then

$$u(x,t) = \sum_{k \in \mathbb{Z}} e^{-k^2 t} (a_k \cos(kx) + b_k \sin(kx))$$

solves

$$\begin{cases} (\partial_t + \Delta_x) u(x,t) = 0 \\ u(x,0) = f(x). \end{cases}$$



Lemma 24.5: If  $f(x) \in L^2(S; \mathbb{R})$  then  $u(x,t) \in C^\infty(S; \mathbb{R})$  for each  $t > 0$ . [24.7]

Proof: It suffices to show that  $\sum |u_k|^2 |k|^{2s} < \infty$  for all  $s$ .

But  $u_k = \frac{e^{-k^2 t}}{(a_k + ib_k)^{1/2}}$  so  $\sum (|a_k|^2 + |b_k|^2) |k|^{2s} e^{-k^2 t} \leq C \sum |a_k|^2 + |b_k|^2 \leq C \|f\|_{L^2}^2$

because  $|k|^{2s} e^{-k^2 t} < 1$  for  $k$  large.

Proof (of thm 24.4)

By the above  $\|u\|_{L^2,2} \neq \infty$ , so

$$\Delta_x \left( \sum_{|k|=1}^N e^{-k^2 t} (a_k \cos(kx) + b_k \sin(kx)) \right) \xrightarrow{u_N} \Delta u.$$

For each  $t \in [t_0, t_1]$  the bound is uniform,

$$\partial_t \left( \sum_{|k|=1}^N e^{-k^2 t} (a_k \cos(kx) + b_k \sin(kx)) \right) \rightarrow \partial_t u \text{ uniformly}$$

(use large  $s$ , so bringing down  $k^2$  doesn't matter).

Then by taking limits

$$(\partial_t + \Delta_x) u = (\partial_t + \Delta_x) u_N = 0 \quad \text{by previous example. } \square$$

obviously  $u(x,0) = f$

Corollary 24.6:  $u(x,t) \in C^\infty((t_0, \infty) \times S; \mathbb{R})$  for any  $t_0$ .

Proof: Lemma 24.5 shows  $x$  derivatives exist.

For  $t$  derivatives,  $k^{2s} u_N \rightarrow u$  and

$$k^{2s} u_N' \rightarrow v \text{ uniformly}$$

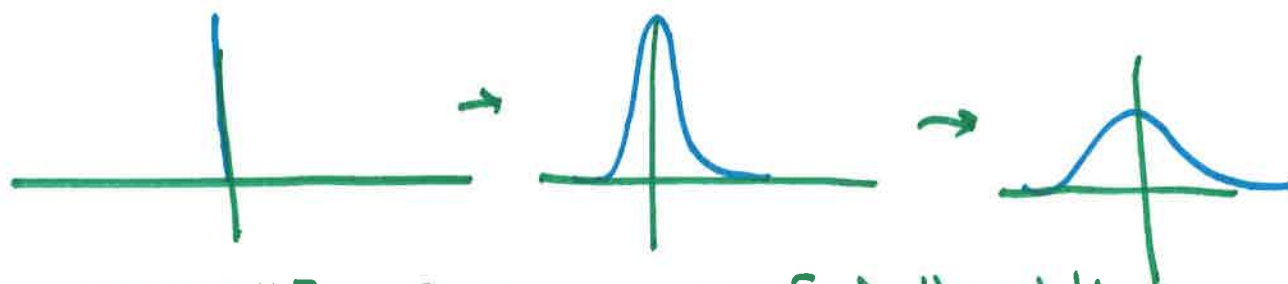
$$\text{for any } s, \text{ thus } \partial_t^s u = \partial_t^s u_N = k^{2s} \cdot u_N.$$

Mixed partials same. □

## 24.ii) The idea of a fundamental solution

24.3

Consider a pinprick of heat



Observation 24.7: Suppose we can find the solution of  $\begin{cases} (\partial_t + \Delta_x) u(x,t) = 0 \\ u(x,0) = \delta_x. \end{cases}$

Then  $f(x) \approx \sum \delta(x - \frac{j}{n}) f(\frac{j}{n}) \rightarrow \int_0^{2\pi} \delta(x-y) f(y) dy.$

the heat equation is Linear so

$$u_t(x,t) = \sum_{j=1}^N K(x - \frac{j}{n}, t) \cdot f(\frac{j}{n}). \rightarrow \int_0^{2\pi} K(x-y, t) f(y) dy.$$

Def 24.8: the solution  $K(x-y, t) : S^1 \times S^1 \times \mathbb{R}^{\geq 0} \rightarrow \mathbb{R}$  is called the heat kernel or fundamental solution.

Thm 24.9: There exists a heat kernel  $K(x,t)$  such that

$$(\partial_t + \Delta_x) K(x,t) = 0 \quad \forall t > 0$$

and as  $t \rightarrow 0$ ,

$$\int_0^{2\pi} K(x,t) dx \rightarrow 1 \quad \text{as } t \rightarrow 0$$

$$\int_{\delta}^{2\pi-\delta} K(x,t) dx \leq \epsilon, \quad \forall \epsilon > 0, \exists t \text{ such that}$$

$$\int_0^{2\pi} K(x,t) f(y) dy \rightarrow f(\frac{x}{n}) \text{ uniformly as } t \rightarrow 0.$$

Proof: Recall the Dirichlet Kernel

$$D_N(x) = \sum_{j=-N}^N e^{in\theta}.$$

then is  $\hat{D}_N = \{ \delta_{k,0} \}_{k \in \mathbb{Z}}$ .

Thus set

$$K(x,t) = \sum_{k \in \mathbb{Z}} e^{-k^2 t} \cos(kx).$$

By the same logic as before,  $K(x,t) \in C^\infty((0,\infty) \times S^1; \mathbb{R})$  and  $(\partial_t + \Delta_x) K(x,t) = 0$ .

Lemma 24.10 (Poisson Summation)

proved next week.

$$\sum_{k \in \mathbb{Z}} f(x+k) = \sum_{n \in \mathbb{Z}} e^{in\theta} \hat{f}(n)$$

In particular, since  $\int_{\mathbb{R}} \mathcal{F}(e^{-ax^2}) = e^{-x^2/a}$ , the following results

$$K(x,t) = \sum_{k \in \mathbb{Z}} e^{-k^2 t} \cos(kx) = \sum_{k \in \mathbb{Z}} \frac{1}{\sqrt{4\pi t}} e^{-\frac{(x-k)^2}{4t}}$$

~~bullet~~ bullet points 1), 2) obvious from Fourier, real space side resp.  
For 3) know  $D_N$  satisfies this, and  $K_N \rightarrow D_N$  uniformly.  $\square$

Rem 24.11: Kernels are used as models for the linear parabolic theory underlying "flow" equations e.g. Mean Curvature flow, Ricci flow, Yang-Mills flow.

It's not a coincidence that  $\int K(x,y,t) f(y) dy \xrightarrow{t \rightarrow \infty}$

$$\lim_{t \rightarrow \infty} \text{Tr}(f \mapsto \int K(x,y,t) f(y)) = 1 = \dim \text{Ker } \Delta$$

This is used in Getzler's famous Heat kernel proof of the Atiyah-Singer Index theorem.



# Lecture 25 Fourier Transforms I: Schwartz functions + distributions.

Question 25.1: What is the analogue of Fourier series on  $\mathbb{R}$ ?

Recall functions on  $\mathbb{R}$  are more subtle b/c we must control the behavior as  $|x| \rightarrow \infty$ .

On  $[-\pi, \pi]$ , write  $f(x) = \sum_k a_k e^{ikx}$  periodic functions  
 $[-\pi, \pi]$   $f(x) = \sum_k a_k e^{i \frac{k}{T} x}$  for  $\frac{k}{T} = 0, \frac{1}{T}, \frac{2}{T}, \dots$

Note the discretization of the series gets



finer as  $T \rightarrow \infty$ . Write  $\xi_k = \frac{k}{T}$  then

$$g(\xi_k) = \frac{1}{2\pi} \int_{-\pi/2}^{\pi/2} f(x) e^{i \xi_k x} dx$$

$$f(x) = \frac{1}{\pi T} \sum g(\xi_k) e^{i x \xi_k} = \sum g(\xi_k) e^{i x (\xi_{k+1} - \xi_k)} \\ \rightarrow \frac{1}{2\pi} \int_{-\pi/2}^{\pi/2} e^{i x \xi} g(\xi) d\xi$$

In limit,  $g = \hat{f}$  becomes a function of continuous variable  $\xi$ .

Rem 25.2: Fourier series has nice properties

1) Countable basis

2)  $-i\partial_x \leftrightarrow k$

3)  $\|f(\omega)\|_{L^2} = \|\hat{f}\|_{L^2}$

$\Leftrightarrow$

Transform or Spectral

X

✓

✓

X

✓

X

Observation 25.3: It is no longer clear  $\hat{f}$  is integrable, or  $\mathcal{F}, \mathcal{F}^{-1}$  make sense etc.

Def 25.34: Given  $f$  in some function space  $X$ , define the Fourier transform

by  $\hat{f} = \mathcal{F}(f) = \frac{1}{2\pi} \int_{\mathbb{R}} f(x) e^{i \xi x} dx$

and the inverse transform by

$$g = \mathcal{F}^{-1}(\hat{f}) = \frac{1}{2\pi} \int_{\mathbb{R}} \hat{f}(\xi) e^{-i \xi x} d\xi$$



## 25.i: Schwartz Functions

Def 25.5: A function is said to be Schwartz if for all  $\alpha \in \mathbb{N}$ ,  $N \in \mathbb{N}$

$$|x^N \partial_x^\alpha f| \leq C_{N,\alpha}$$

i.e. functions whose derivatives all decay faster than polynomially.

Ex 25.6:  $e^{-x^2}$  is Schwartz.

Prop 25.7:  $\mathcal{S}(\mathbb{R})$  the Schwartz functions satisfies

- 1) is a vector space and algebra
- 2)  $\partial_x, \int_{-\infty}^x : \mathcal{S}(\mathbb{R}) \rightarrow \mathcal{S}(\mathbb{R})$ , and  $e^{isx} : \mathcal{S}(\mathbb{R}) \rightarrow \mathcal{S}(\mathbb{R}) \quad \forall s \in \mathbb{R}$
- 3) ~~obviously~~  $\mathcal{S}(\mathbb{R}) \subset L^\infty(\mathbb{R}) \cap L^2(\mathbb{R}) \cap L^1(\mathbb{R})$ .

Proof: these are all obvious.

Warning 25.8:  $\mathcal{S}(\mathbb{R})$  is not a Banach or Hilbert space. The topology is a bit more complicated, and stronger.

$$\varphi_n \rightarrow \varphi \quad \text{iff} \quad |x^N \partial_x^\alpha (\varphi_n - \varphi)| \rightarrow 0 \quad \forall N, \alpha$$

It is what called a <sup>(Fischer)</sup> Fréchet space w a countable filtration of norms.

Lemma

Def 25.9: For each  $\varphi \in \mathcal{S}(\mathbb{R})$ ,  $\hat{\varphi}$  is defined and  $\hat{\varphi} \in L^\infty(\mathbb{R})$ .

$$\hat{\varphi}(\xi) = \frac{1}{2\pi} \int_{\mathbb{R}} f(x) e^{isx} dx \leq \frac{1}{2\pi} \int_{\mathbb{R}} |f| dx = \|f\|_1 < \infty.$$

So take sup.

## 25.ii) Properties of the Fourier transform

Prop 25.10: The Fourier transform on  $\mathcal{S}(\mathbb{R})$  obeys

- 1)  $\widehat{f(x+y)} = e^{-is y} \hat{f}(\xi)$ ,  $\widehat{e^{isx} f(x)} = \hat{f}(\xi + \zeta)$
- 2)  $\widehat{\partial_x^k f} = (-i\xi)^k \hat{f}$

$$3) \widehat{(ix)^k f(x)} = \partial_\xi^k \hat{f}(\xi).$$

$$4) \widehat{f * g} = \hat{f} \cdot \hat{g} \quad (\text{inverse hard to show directly.})$$

Proof:

$$1) \widehat{f(x+y)} = \int_{\mathbb{R}^n} f(x+y) e^{i\xi x} dx = \int_{\mathbb{R}^n} f(x) e^{i(x-y)\xi} dx = e^{-i\xi y} \hat{f}(\xi).$$

$$\begin{aligned} 2) \widehat{(\partial_x^k f)} &= \int_{\mathbb{R}^n} (\partial_x^k f) e^{i\xi x} dx \\ &= (-1)^k \int_{\mathbb{R}^n} f \partial_x^k (e^{i\xi x}) dx \quad \text{Int by parts.} \\ &= (-i\xi)^k \int_{\mathbb{R}^n} f e^{i\xi x} dx \end{aligned}$$

$$3) \frac{\partial}{\partial \xi} \hat{f} = \lim_{h \rightarrow 0} \int_{\mathbb{R}} f(x) \frac{e^{+i(\xi+h)x} - e^{i\xi x}}{h} dx$$

$$= \int_{\mathbb{R}} f(x) (ix) e^{i\xi x} dx$$

$$= \widehat{(ix)f(x)} \quad \text{and induct.}$$

↓ dominated convergence

4) same as series.

Corollary 25.11 :  $\mathcal{F}: \mathcal{D}(\mathbb{R}) \rightarrow \mathcal{D}(\mathbb{R})$   
 $\mathcal{F}^{-1}: \mathcal{D}(\mathbb{R}) \rightarrow \mathcal{D}(\mathbb{R})$

Proof: As before, if  $\varphi \in \mathcal{D}(\mathbb{R})$ ,  $|\hat{\varphi}| < \infty$ .

and  $|\xi^N \partial_\xi^k \hat{\varphi}| = |\widehat{x^k \partial_x^N \varphi}| < \infty$  b/c  $x^k \partial_x^N \varphi \in \mathcal{D}(\mathbb{R})$

The same properties hold w/  $\mathcal{F}^{-1}$  but w/  $(-1)^{N+k}$ .  $\square$

Thm 25.12 :  $\mathcal{F} : \mathcal{S}(\mathbb{R}) \rightarrow \mathcal{S}(\mathbb{R})$  is an isomorphism with inverse  $\mathcal{F}^{-1}$ .

Proof : 
$$\mathcal{F}^{-1} \circ \mathcal{F} (x^N \partial_x^k \varphi) = \mathcal{F}^{-1} (\partial_y^N \mathcal{F}^k \mathcal{F}(\varphi))$$
$$= x^N \partial_x^k \mathcal{F}^{-1} \mathcal{F}(\varphi).$$

Thus  $T = \mathcal{F}^{-1} \circ \mathcal{F} : \mathcal{S} \rightarrow \mathcal{S}$  commutes w/  $x, \partial_x$ .

Lemma 25.13 : If  $[T, x], [T, \partial_x] = 0$  on  $\mathcal{S}$ , then  $T = c \text{Id.}$  for  $c \in \mathbb{R}$ .

Proof: Step 1 : If  $\varphi(y) = 0$  then  $T\varphi(y) = 0$ .

$\varphi = (x-y)\varphi_1$  w/  $\varphi_1 \in \mathcal{S}$  by Taylor's theorem.

Thus 
$$T\varphi(y) = T(x-y\varphi_1)(y)$$
$$= [(x-y)T\varphi_1](y) = 0.$$

Step 2 : Fix  $y \in \mathbb{R}$ ,  $g$  st  $g(y) = 1$  st  $c(y) = (Tg)(y)$ .

Then  $Tf(y) = c(y)f(y)$ . To see this,

$\varphi_2 = f(y) - f(y)g(x)$  so  $\varphi_2(y) = 0$ ,

so  $0 = T\varphi_2 = Tf(y) - f(y)(Tg)(y)$ 
$$= Tf(y) - f(y)c(y) \text{ so } Tf(y) = c(y)f(y) \quad \forall y.$$

Step 3 : Take  $f \neq 0$  eg  $e^{-x^2}$ , then  $c(y) = \frac{Tf}{f} \in \mathcal{S}(\mathbb{R})$ .

Step 4 :  $c(y)(\partial_x f)_y = T(\partial_x f)_y = \partial_x T f(y)$ 
$$= \partial_x [c \cdot f](y).$$
$$= (\partial_x c)_y \cdot f(y) + c(y) \partial_x f(y)$$

thus  $(\partial_x c)(y) = 0 \quad \forall y \Rightarrow c$  is const.

Step 5 : By normalizing conventions, can take  $c=1$ .



# Lecture 26 Fourier Transforms II: Distributions and Plancherel.

[26.1]

Recall that if  $E \subseteq \mathbb{R}^n$  is compact

$$L^\infty(E) \subseteq \dots \subseteq L^4(E) \subseteq L^3(E) \subseteq L^2(E) \subseteq L^1(E)$$

$\Downarrow$  \* dual

$$\dots \supseteq L^{4/3}(E) \supseteq L^{3/2}(E) \supseteq L^2(E) \supseteq L^1(E)$$

Thus in general  $X \subseteq Y \Rightarrow Y^* \subseteq X^*$  (all functionals bounded on  $Y$  are automatically bounded on  $X$ , but maybe not).

Def 26.1: The space of tempered distributions, denoted  $\mathcal{S}'(\mathbb{R}) = \mathcal{S}(\mathbb{R})^*$  is  $\mathcal{S}'(\mathbb{R}) = \{ \phi: \mathcal{S}(\mathbb{R}) \rightarrow \mathbb{R} \mid \phi \text{ is bounded in the sense } \exists C, N, K \text{ s.t. } |\phi(\varphi)| \leq C \sum_{\substack{K, n \\ \leq K, N}} \sup_{\mathbb{R}} |x^n \partial_x^k \varphi| \}$

Ex 26.2:  $\alpha = |x|^N$ . Then

$\alpha \mapsto \int_{\mathbb{R}} |x|^N \cdot \varphi dx = \langle \alpha, \varphi \rangle$  is a tempered distribution as

$$\int_{\mathbb{R}} |x|^N \varphi = \int_{\mathbb{R}} |x|^N \varphi \leq \sup_{\mathbb{R}} |x|^N |\varphi| \text{ so } N \leq N.$$

Ex 26.3:  $|x|^n \partial_x^k = k! \beta$

$$\varphi \mapsto \int_{\mathbb{R}} |x|^n \partial_x^k \varphi dx \text{ for same reason}$$

$\Rightarrow$  functions w/ growth slower than some polynomial.

Ex 26.4: Let  $\delta \in \mathcal{S}'(\mathbb{R})$  be the distribution given by

$$\langle \delta, \varphi \rangle = \int_{\mathbb{R}} \delta(x) \varphi(x) dx = \varphi(0).$$



Def 26.5:  $\delta$  is called the Dirac delta "function" or distribution.

26.7

### 26.ii) Operators on distributions

Suppose  $A: \mathcal{D}(\mathbb{R}) \rightarrow \mathcal{D}(\mathbb{R})$  is a linear (continuous) operator.

Def 26.6: The formal adjoint is the operator on distributions

$A^*: \mathcal{D}(\mathbb{R}) \rightarrow \mathcal{D}(\mathbb{R})$  given by

$$\langle A^* u, \varphi \rangle := \langle u, A\varphi \rangle$$

Lemma 26.7:  $\forall u \in \mathcal{D}'(\mathbb{R}), \exists \varphi_n \in \mathcal{D}(\mathbb{R})$  s.t.

$$\int_{\mathbb{R}} \langle \varphi_n, \varphi \rangle dx = \langle \varphi_n, \varphi \rangle \longrightarrow u(\varphi) \text{ for all } \varphi.$$

Proof: Take  $\varphi_n(x) = \int K_{1/n}(x-y) u(y) dy$   
 $= \langle K_{1/n}(x-y), u \rangle$   
and apply approximate identity stuff.

Def 26.8:  $A^*$  is said to extend  $B$  if

$$\langle A^* u, \varphi \rangle = \langle u, A\varphi \rangle = \langle Bu, \varphi \rangle$$

for all  $u, \varphi \in \mathcal{D}'(\mathbb{R})$

Ex 26.9: The distributional derivative of  $u$  is

$$\langle \partial_x u, \varphi \rangle := \langle u, -\partial_x \varphi \rangle.$$

Obviously, if  $u \in \mathcal{D}(\mathbb{R})$  this holds by integration by parts, so the extension is valid.

Ex 26.10:  $H(x) = \begin{cases} 1 & x \geq 0 \\ 0 & x < 0 \end{cases}$ .  $-\delta = \partial_x H$ . Since  $\forall \varphi$ ,

$$\langle H, \partial_x \varphi \rangle = \int_0^\infty \partial_x \varphi = \varphi(\infty) - \varphi(0) = -\varphi(0)$$

### 26.iii) The Plancherel Theorem

(26.iii)

Let  $A: \mathcal{L}(\mathbb{R}) \rightarrow \mathcal{L}(\mathbb{R})$  be the Fourier transform. Thus to extend to  $\mathcal{L}'(\mathbb{R})$  we need a  $B: \mathcal{L}(\mathbb{R}) \rightarrow \mathcal{L}(\mathbb{R})$  satisfying

$$\int_{\mathbb{R}} (B\psi)(x) \varphi(x) dx = \langle B\psi, \varphi \rangle = \langle \psi, A\varphi \rangle = \int_{\mathbb{R}} \psi(x) \hat{\varphi}(x) dx$$

Thm 26.11 (Plancherel):  $B = \mathcal{F}$ . i.e., the Fourier transform is self-adjoint on  $S$ . i.e.

$$\int_{\mathbb{R}} \hat{\psi} \varphi dx = \int_{\mathbb{R}} \psi \hat{\varphi} dx$$

Proof :

$$\begin{aligned} \int_{\mathbb{R}} \hat{\psi} \varphi dx &= \int_{\mathbb{R}} \int_{\mathbb{R}} \psi(\xi) e^{i\xi x} d\xi \varphi(x) dx \\ &= \int_{\mathbb{R}} \int_{\mathbb{R}} \psi(\xi) e^{i\xi x} \varphi(x) dx d\xi \quad (\text{Fubini}) \\ &= \int_{\mathbb{R}} \psi(\xi) \hat{\varphi}(\xi) d\xi. \end{aligned}$$

Corollary 26.12:  $\mathcal{F}$  extends to a map  $\mathcal{F}: \mathcal{L}'(\mathbb{R}^n) \rightarrow \mathcal{L}'(\mathbb{R}^n)$  as does  $\mathcal{F}^{-1}$ , and they remain inverses.

Proof: Follows directly, and extension of id is obviously id.

Corollary 26.13:  $\mathcal{F}$  extends to an isomorphism on  $L^2$ , in fact an isometry.

So

$$\begin{array}{ccccc} \mathcal{L}(\mathbb{R}) & \hookrightarrow & L^2(\mathbb{R}) & \hookrightarrow & \mathcal{L}'(\mathbb{R}) \\ \downarrow \mathcal{F} & & \downarrow \mathcal{F} & & \downarrow \mathcal{F} \\ \mathcal{L}(\mathbb{R}) & \hookrightarrow & L^2(\mathbb{R}) & \hookrightarrow & \mathcal{L}'(\mathbb{R}) \end{array}$$

commutes. The Plancherel Formula  $\|f\|_2 = \|\hat{f}\|_2$  holds.

Proof :  $\mathcal{F}(f)$  is defined as a distribution, so suffices to show 26.4  
 $\|\mathcal{F}(f)\|_{L^2} < \infty$ . Take  $\psi = \hat{\varphi}$ . Then

$$\hat{\psi} = \mathcal{F}(\hat{\varphi}) = \int \hat{\varphi} e^{-i\xi x} dx = \mathcal{F}^{-1}(\hat{\varphi})$$

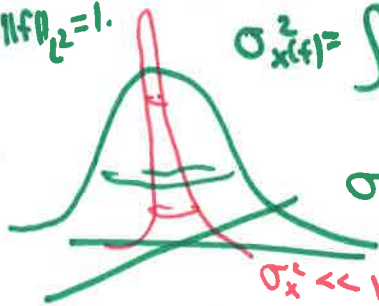
so  $\hat{\psi} = \mathcal{F}\mathcal{F}^{-1}(\hat{\varphi}) = \hat{\varphi}$ . Conclude

$$\int |\hat{\varphi}|^2 dx = \int |\hat{\psi}|^2 dx$$

□.

### 26.iv) The Uncertainty Principle

Let  $f \in L^2$ . Then the variance of  $|f|$  is  
 $\sigma_x^2(f) = \int_{-\infty}^{\infty} x^2 |f|^2 dx - \left( \int_{-\infty}^{\infty} x |f|^2 dx \right)^2$  ← mean



$$\sigma_x^2 \gg 1 \quad \sigma_y^2 = \sigma_x^2(\hat{f}).$$

### Thm 26.14 (Heisenberg Uncertainty Principle)

$$\sigma_x \sigma_y \geq \frac{1}{2}.$$

Proof : may assume mean 0 by translation.

Let  $\varphi = xf$   $\hat{\psi} = \xi \hat{f}$ . so  $\psi = -i \partial_x f$ .

By Cauchy Schwartz,

$$|\langle \varphi, \psi \rangle|^2 \leq \left( \int x^2 f^2 \right) \left( \int \xi^2 \hat{f}^2 \right) = \sigma_x^2 \sigma_y^2.$$

But  $|\langle \varphi, \psi \rangle|^2 \geq |\operatorname{Im} \langle \varphi, \psi \rangle|^2 = \left| \frac{\langle \varphi, \psi \rangle - \overline{\langle \varphi, \psi \rangle}}{2i} \right|^2$

$$= \left| \frac{1}{2i} \right|^2 = \frac{1}{4}.$$

And  $-\langle \varphi, \psi \rangle + \langle \psi, \varphi \rangle = \int x \bar{f} (-i \partial_x f) - x f (i \partial_x \bar{f}) dx$   
 $= \int x \bar{f} (-\partial_x f) + i \partial_x (xf) \bar{f} dx$   
 $= \int x \bar{f} (-\partial_x f) + i f \bar{f} + i x \partial_x f \bar{f} dx = i \int |f|^2 = i$



## Lecture 27 Fourier Transforms II: PDEs and Convolutions

(27.1)

Recall that on  $L^2(S'; \mathbb{C})$ ,  $\mathcal{F}$  turned Differential operators into multiplication.

Question 27.1: How do we extend the analysis on  $S'$  to  $\mathbb{R}$ ?

How do we solve e.g.

$$\Delta u = f \in L^2(\mathbb{R})$$

$$(\Delta + 1)u = f$$

$$Lu = \sum \alpha_j(x) \partial_x^j u = f \in L^2(\mathbb{R})$$

or parabolic 
$$\begin{cases} (\partial_t + \Delta_x) u(x, t) = g \\ u(x, 0) = f \end{cases}$$

and can we extend the results of e.g. elliptic regularity?

### Observation 27.2

$$\begin{array}{ccc} L^{2,2}(\mathbb{R}) & \xleftrightarrow{\widehat{\mathcal{F}}} & L^2(\mathbb{R}, (1+|\xi|^2)^2 d\xi) \\ \downarrow L & & \downarrow \widehat{L} \\ L^2(\mathbb{R}) & \xleftrightarrow{\mathcal{F}} & L^2(\mathbb{R}) \end{array}$$

Note e.g.  $\partial_x f \in L^2 \iff \int |\xi f|^2 d\xi = \int |f|^2 \underbrace{|\xi|^2}_{\leq \infty} d\xi < \infty$ .

Thus to solve, one must study the "inverse operator"

$$\widehat{f} \longrightarrow \frac{1}{\alpha_0 + \alpha_1 \xi + \dots + \alpha_N \xi^N}$$

(for  $\alpha_j \in \mathbb{C}$  constant coefficient).

Def 27.3: Let  $K(z) := \mathcal{F}\left(\frac{1}{\alpha_0 + \dots + \alpha_N \xi^N}\right)$ .

Then if  $Lu = f$ ,  $\Rightarrow u = \int_{\mathbb{R}} K(x-y) f(y) dy = \mathcal{F}^{-1}\left(\frac{1}{\alpha_0 + \dots + \alpha_N \xi^N} \cdot \widehat{f}\right)$ .

$K(z)$  is called by various authors:

- 1) the Fundamental solution
- 2) the Green's function (of  $L$ )
- 3) the integral kernel
- 4) the Schwartz kernel.



Rem 27.4 :  $\mathcal{F}(\delta)(\varphi) = \langle \delta, \mathcal{F}\varphi \rangle = \hat{\varphi}(0) = \int_{\mathbb{R}} \varphi(x) e^0 dx$   
 $= \int_{\mathbb{R}} \langle \varphi(x), 1 \rangle dx$

$$\begin{aligned}\hat{\mathcal{F}}(\delta)(y) &= \langle \delta, \hat{\mathcal{F}}\varphi \rangle = \hat{\varphi}(y) = \int_{\mathbb{R}} \varphi(x) e^{iyx} dx \\ &= \int \langle \varphi(x), 1 \rangle dx\end{aligned}$$

so  $f(s) = 1$ . Thus

$$L K(\mathbb{Z}) = \mathbb{Z}$$

is an equivalent definition, similar to as with the heat Kernel.

Def 27.5: The operator  $u \mapsto K * u$  is called the  
1) Green's operator, or parametratrix. It is a special  
case of a convolution operator.

Convolution operators are a key aspect of the study of PDEs. See Math 173/205.

### 27.ii: Young's Inequality for Convolutions

One fundamental result is

Thm 27.6 (Young's Convolution)  $p, q, r \geq 1$ .

Suppose  $\frac{1}{p} + \frac{1}{q} = 1 + \frac{1}{r}$ . Then  $\exists C$  such that  $f, g \in L^p(\mathbb{R}), L^q(\mathbb{R})$

$$\|f * g\|_r \leq C \|f\|_p \|g\|_q.$$

Hence  $f * g \in L^1(\mathbb{R})$ . The same holds on  $\mathbb{R}^n$ .

Proof :

$$f * g = \int_{\mathbb{R}} f(x-y)g(y)dy$$

$$|f * g| \leq \int_{\mathbb{R}} |f(x-y)| |g(y)| dy$$

$$\leq \int_{\mathbb{R}^n} |f(x-y)|^{1+p/r-p/r} |g(y)|^{1+q/r-q/r} dy$$

$$\leq \int_{\mathbb{R}^n} (|f(x-y)|^p |g(y)|^q)^{1/r} |f(x-y)|^{1-p/r} |g(y)|^{1-1/r}$$

$$\leq \left( \int_{\mathbb{R}} |f(x-y)|^p |g(y)|^q \right)^{1/r} \int \| |f(x-y)|^{\frac{r-p}{r}} \|_{\frac{pr}{r-p}} \| |g(y)|^{\frac{r-q}{r}} \|_{\frac{qr}{r-q}}$$

Here, we used if  $\frac{1}{p^*} + \frac{1}{q^*} + \frac{1}{r^*} = 1$

27.3

$$\int fgh \leq \|f\|_p \|g\|_q \|h\|_r$$

multi-Hölder, (induction) applied for  $r^* = r$ ,  $p^* = \frac{pr}{r-p}$ ,  $q^* = \frac{qr}{r-q}$ .

$$\leq \left( \int |f|^p |g|^q \right)^{1/r} \|f\|_p^{r-p} \|g\|_q^{r-q}$$

translation invariance of Lebesgue!

$$\|f+g\|_r^r = \int_{\mathbb{R}^n} |f+g|^r dx$$

$$\leq \int_{\mathbb{R}^n} \left[ \int_{\mathbb{R}^n} |f(x-y)|^p |g(y)|^q dy \right] \|f\|_p^{r-p} \|g\|_q^{r-q} dx$$

$$\leq \|f\|_p^{r-p} \|g\|_q^{r-q} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} |g(y)|^q |f(x-y)|^p dx dy \quad (\text{Fubini})$$

$$\leq \|f\|_p^{r-p} \|g\|_q^{r-q} \int_{\mathbb{R}^n} g(y) \|f\|_p^p = \|f\|_p^{r-p+p} \|g\|_q^{r-q+q}$$

Ex 27.7: On  $\mathbb{R}^n$   $n \geq 3$

•  $K(x-y)$  for  $\Delta$  is  $\frac{1}{|x-y|^{n-2}}$  so  $\int |K(\frac{x}{r})| dv = \int \frac{1}{r^{n-2}} r^{n-1} dr$

$\Rightarrow$  the Green's function of  $\Delta$  just barely fails to be  $L^1$ !

•  $K(x-y)$  for  $\Delta + \beta^2 = \frac{e^{-\beta|x-y|}}{|x-y|^{n-2}}$  same (but only at 0).

Rem 27.7: This is part of why studying integral kernels is difficult, cannot just apply Young.

### 27.iii) The Fourier transform of $L^p$

[27.4]

Recall  $\mathcal{F}: L^2 \rightarrow L^2$  is an isometry.

Question 27.9: What is the Fourier transform of  $L^p$  for  $1 \leq p < \infty$ ?

Ex 27.10  $\mathcal{F}(1) = \delta$  and  $1 \in L^\infty$  but  $\delta \in L^p$  for any  $p$ .

Thm (Hausdorff-Young) <sup>27.11</sup> If  $1 \leq p \leq 2$ ,  $\frac{1}{p} + \frac{1}{q} = 1$ , then

$$\|\hat{f}\|_{L^q} \leq C \|f\|_{L^p}$$

Corollary 27.12: For  $1 \leq p \leq 2$ ,

$$\mathcal{F}: L^p(\mathbb{R}) \longrightarrow L^q(\mathbb{R})$$

is continuous and injective. (It is not surjective) <sup>this is extremely difficult.</sup>

Proof: Suppose that  $q = 2k$  for  $k \in \mathbb{N}$ ,  $p = \frac{2k}{2k-1} = 2, \frac{4}{3}, \frac{6}{5}, \dots$

$$\begin{aligned} \|\hat{f}\|_{L^q} &= \|\hat{f}\|_{L^{2k}} = \|(\hat{f})^k\|_{L^2}^{1/k} \\ &= \|\hat{f} \cdot \hat{f} \cdots \hat{f}\|_{L^2}^{1/k} \\ &= \|f * f * f * \cdots\|_{L^2}^{1/k} \end{aligned}$$

$$\text{If } k=2, \quad 1 + \frac{1}{2} = \frac{3}{2} = 2\left(\frac{3}{4}\right) = 2\left(\frac{1}{4}\right):$$

$$\leq \|f\|_{L^p} \|f\|_{L^p}^{1/2}$$

For general  $k$ , iterated Young convolution does same

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